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Making the Case for the Multiverse

A Radical New Theory of Aging and Disease

How Uncertainty Breaks the Brain

Escaping the Infinite Scroll

We're addicted to social media. How do we get clean?



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From the Editor

- **Our love-hate relationship with social media**
  Regardless of whether social media addiction is real, these platforms hack the human brain, often to our detriment

Our love-hate relationship with social media

Regardless of whether social media addiction is real, these platforms hack the human brain, often to our detriment

By Jeanna Bryner edited by Jeanna Bryner



Scientific American

Two opposing ideas can be true at once. Something can be helpful and harmful, good and bad, annoying and endearing. That's the premise of dialectical behavioral therapy (DBT), and it's also, I think, partly why we are having such a tough time navigating social media. We genuinely benefit from the "infinite scroll of doom." We are social beings, and our very survival has always depended on connections. A lone human ancestor would have been easy pickings for a hyena on the African savanna, and without cooperative hunting or food gathering, our forebears simply wouldn't have made it. Even today human infants are helpless without at least one connection. We're built for bonding, so much so that the brain rewards us for it, releasing bonding chemicals such as oxytocin when we connect. Maybe, then, it's no wonder we are drawn to

social media platforms, where connections can be forged in minutes.

Here's where things get complicated. As associate editor Allison Parshall points out in our cover story, social media isn't quite the same as real-world meetups or handwritten letters (I'm now reading Virginia Evans's 2025 debut novel, *The Correspondent*, whose theme touches on the lost art of letter writing). Platforms such as X, Facebook and Instagram run on algorithms that hack the brain's mesolimbic pathway, whose neurons rely on dopamine, a type of "feel-good" chemical that can drive motivation and reward. "I've come to almost understand algorithms as being heat-seeking missiles that identify psychological vulnerability," one psychologist told Allison.

That quote really hit home. After talking with Allison in the initial stages of her reporting, I realized I'm apparently vulnerable to sad medical stories, and my Instagram algorithm picked that up right away. How could I *not* read a post describing a recent disease diagnosis or the loss of a child that my feed was serving up?

On top of these heat-seeking missiles, the platforms play to humans' (particularly teens') need to be accepted and admired by peers. Witness the kid constantly checking a new social post to see how many "likes" it's received; many of us adults do the same thing. To me, the draw feels automatic and irresistible.

But how irresistible? The first step in 12-step addiction-treatment programs involves admitting you are powerless to resist your vices, including the urge to dive headfirst into social media so deeply you can no longer manage other parts of your life. For Meg, who is part of an Internet-addiction program and who spoke to Allison for this feature, her social media habit kept her up all night and left her unable to attend to her kids in the morning. "I would wake up at 10 A.M. just feeling such shame, but I couldn't stop it," Meg said. "I did it over and over."

You don’t have to cross that threshold for social media to take a toll on your mental well-being. Like with many things we love that are also bad for us, some people find a balance, one where they get more of the good—reconnecting with old friends, a creative outlet—than the bad.

People whose lives spiral when social media enters the picture might need to abstain, although doctors are finding that for behavioral addictions abstinence might not be feasible (screens are everywhere). Instead they are shifting to harm reduction. This kind of recovery can involve DBT, a therapy that focuses on mindfulness and emotional regulation. Gradual progress is the key because a healthy relationship with technology won’t develop overnight.

When I’m not lying awake worrying about what screen time is doing to my own kids’ social and cognitive development, I’ll admit this whole detour into human nature is kind of fascinating. I think you will find our feature on the subject fascinating as well.

Jeanna Bryner is editor in chief of *Scientific American*. Previously she was editor in chief of Live Science and, prior to that, an editor at Scholastic's *Science World* magazine. Bryner has an English degree from Salisbury University, a master’s degree in biogeochemistry and environmental sciences from the University of Maryland and a graduate science journalism degree from New York University. She has worked as a biologist in Florida, where she monitored wetlands and did field surveys for endangered species, including the gorgeous Florida Scrub Jay. She also received an ocean sciences journalism fellowship from the Woods Hole Oceanographic Institution. She is a firm believer that science is for everyone and that just about everything can be viewed through the lens of science.

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https://www.scientificamerican.com/article/our-love-hate-relationship-with-social-media

Letters

- **Readers respond to the June 2026 issue**
  Letters to the editors for the June 2026 issue of Scientific American

Readers respond to the June 2026 issue

Letters to the editors for the June 2026 issue of Scientific American

By Aaron Shattuck edited by Aaron Shattuck



Scientific American

POLE POSITION

"Early Plates," by Marissa Grunes [Advances], states that Earth's magnetic north pole is "technically" now in the Southern Hemisphere. Can readers get an explanation of this?

ALEX SMITH *LITTELTON, MASS.*

GRUNES REPLIES: *The terms "north" and "south," as applied to magnetic poles, are somewhat arbitrary designations. Imagine a bar magnet: Magnetic force flows out of what we call its "north" pole. The force bends around the magnet (forming a magnetic field) and flows back into the magnet through its "south" pole. This is why opposite poles attract and similar poles repel each other.*

In the case of Earth, since the last magnetic reversal about 780,000 years ago, lines of magnetic force flow out from the Antarctic, bend around our planet and reenter Earth in the Arctic. So our world's "north" magnetic pole is in the Antarctic, and its "south" magnetic pole is in the Arctic.

Magnetic compasses arose from the discovery that free-floating magnetic materials align themselves along the north-south axis of the globe. The poles on a bar magnet received their names from the direction in which they would point. But remember that opposite poles attract! So the pole of a magnet that's called "north" may point toward the cardinal direction of geographic north, but the pole that's pulling it is, magnetically speaking, "south."

To avoid confusion, we've continued to use the geographic designations of "north magnetic pole" for the pole in the Arctic and "south magnetic pole" for the pole in the Antarctic. But technically, the planet's "south" magnetic pole is in the Northern Hemisphere, and vice versa.

SAFEGUARDING THE MOON

In the June issue, I was especially fascinated by the articles about the Artemis II moon mission and the space dreams it has inspired ["Back to the Moon" report]. But I was troubled by how little attention was paid to the U.S.'s history of irresponsible resource extraction, which is especially concerning now that billionaires and others are obviously advocating for the rapid exploitation of whatever the moon may hold. I would love to see substantial guardrails erected to protect the moon in a way that we've so far failed to protect our home planet.

For instance, "Eyes in the Sky," by Joseph Howlett, reports that researchers hope to make the moon a site for a new gravitational-wave detector. I believe that goal can be reached only if mining is

severely restricted or even prohibited on the moon because, as demonstrated by the Laser Interferometer Gravitational-Wave Observatory (LIGO), even minuscule tremors interfere with measurements. Similarly, Howlett describes how the lunar farside is an ideal location for a highly sensitive radio telescope because radio communications from Earth can't reach it. But that promise can be realized only if that portion of the lunar sky is kept free of a maze of communications satellites.

MICHAEL KEARNS *COLUMBUS, OHIO*

PIECE OF HISTORY

"*Tetris* Paradox," by Jack Murtagh [Advances; September], refers to the shapes used in the game as its "pieces." These shapes are, in fact, called tetrominoes. The shapes were first introduced 30 years before *Tetris* by the late mathematician Solomon Golomb, and within a couple of years they were popularized in *Scientific American*! So it's unfortunate to see them called *Tetris* pieces.

I have a book with MIT Press exactly about this: *Soviet Blocks*, a legal and geopolitical history of *Tetris*.

JULIEN MAILLAND *PROFESSOR OF MEDIA LAW & MANAGEMENT, INDIANA UNIVERSITY BLOOMINGTON*

AUTISM AND CONTEXT

"Visualizing the Spectrum," by Allison Parshall and Amanda Montañez [Graphic Science; April], shows how autism spectrum disorder has many dimensions rather than being a single linear spectrum. It's encouraging to see that autism research has begun, over the past decade, to recognize that complexity and the fact that divergence in quite a few of those dimensions does not represent a "disease" that needs treatment.

As a neurodivergent individual myself, I note that there's still one set of dimensions the field is missing: context. The article represents 39 traits from the Autism Symptom Dimensions Questionnaire, and my answers to some of these questions, especially those relating to interpersonal relationships, can differ markedly depending on the situation. For example, a couple of the questions lump family and friends into one group, but my answers differ depending on whether I'm interacting with my spouse, other family, friends or unfamiliar people and also depend on whether those interactions are in a group setting versus one-to-one.

"JIM" *VIA E-MAIL*

ORBITAL DATA CENTERS

Partly inspired by "Battle for the Night Sky," by Jonathan O'Callaghan [September], our online Discussions space asked readers, "Should we build data centers in space?" The following responses were edited for space and clarity.

I can't believe it would be less costly to put a data center in space rather than on the ground. And there are other factors to consider. The most common objections to data centers on Earth focus on electricity consumption and cooling. Neither concern goes away in orbit.

I think the objections to massive data-center construction here on Earth's surface are well founded. Regulation is badly needed here. And if I'm wrong and space-based data centers turn out to be economically feasible, regulation will be needed there, too.

GARY NOVOSIELSKI *VIA DISCUSSIONS*

There is no way to stop the advancement of our technology, whether it is good or not. Perhaps we have no realistic choice about building data centers somewhere. Then, considering the

environmental impact of these massive projects, space is a better place.

“NOBU” *VIA DISCUSSIONS*

Assuming they work, building, launching and operating space-worthy data centers would exact a stupendous environmental cost during production and after. Much smarter solutions abound. China is powering data centers, here on Earth, with renewable energy and placing them underwater to minimize climate and land-use impact. It apparently understands that however data centers are built, preservation of habitable environment has to come first, or there will be no one needing them.

DOROTA NIECHWIEJ CLEGG *VIA DISCUSSIONS*

ERRATUM

“Spying on the Universe,” by Jonathan O’Callaghan [September], should have said that astronomers met at Princeton University to discuss what they could do with telescopes from the National Reconnaissance Office in 2012.

Aaron Shattuck is managing copy editor at *Scientific American*.

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https://www.scientificamerican.com/article/readers-respond-to-the-june-2026-issue

Advances

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  Growing crystals “like alchemy,” researchers plotted out the ideal site within the lattice to build a nuclear clock
- **Sex-changing mouse clones may help fight to save endangered species**
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- **Lettuce engineered to produce meat proteins**
  Protein-packed leaves might be the key to better meat alternatives
- **How to spot an AI-generated face, according to science**
  Training people to pay attention to the right visual cues nearly doubled how accurately they could spot AI-generated faces
- **New JWST images reveal the Lion Nebula in all its glory**
  This dying star appears to have a face of gas and a mane of space dust
- **This beer-making waste may offer a new stream for cleaner water**
  The research offers yet another way the brewery industry might repurpose its myriad by-products

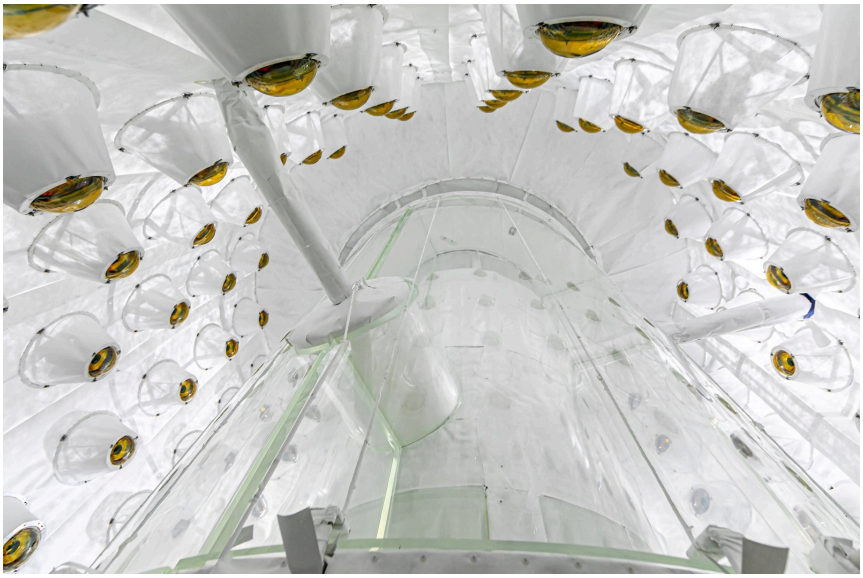
- **Plants move to get away from their decaying brethren**
  A new discovery reveals how plants protect themselves from microbes in rotting material
- **Math Puzzle: Baby bottle mix-up**
  Solve a new-parent dilemma with this math puzzle
- **Sperm with DNA damage might be the true cause of repeated miscarriages**
  New guidelines say testing sperm for DNA damage could help explain recurrent pregnancy losses
- **Honeybees deemphasize potential ‘misinformation’ in their waggle dances**
  Confident bees work harder to convince their peers of tasty treats

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Have we finally found dark matter?

The world's biggest dark matter search saw a particle it cannot explain

By Joseph Howlett edited by Lee Billings & Sarah Lewin Frasier



LUX-ZEPLIN hunts for dark matter using seven metric tons of liquid xenon buried in an abandoned mine.

Matthew Kapust/Sanford Underground Research Facility

On June 16, 2023, 39 seconds after 3:22 P.M. and one mile beneath the Black Hills of South Dakota, there was a sudden flash of light.

It happened in a dark, sealed tank of extremely cold liquid in a cavern where miners once chiseled gold from the walls. Caught by the tank's many cameras, the flash also sent an electric charge through hair-thin wires submerged in the liquid.

The tank is the centerpiece of the LUX-ZEPLIN (LZ) experiment, one of several ongoing attempts by physicists to snare and study dark matter, a shadowy *something* they suspect is everywhere—yet somehow seems nowhere to be found. They've sought it for the

past 40 years by placing detectors in some of Earth's most silent and secluded spaces, places where dark matter's whispers might register. But every time so far, they've come up empty, with every burst of light or surge of electricity having a more ordinary explanation.

This flash from three years ago is different. The charge and light together seem to indicate a violent collision between an unknown particle and the nucleus of one of the detector's atoms. The event bears almost no resemblance to any of the "backgrounds"—mundane sources of flashes that mimic dark matter—which the LZ collaboration's hundreds of scientists have spent their careers battling.

Yet the collision is also out of sync with the kind of dark matter signal they've been searching for. That is, it's a total mystery. But after four decades of nothing, a new mystery is welcome.

"I have been waiting for a positive result for, oh, my god, 40 years," says Katherine Freese, a theoretical physicist at the University of Texas at Austin, whose work helped inspire the search. "So you can bet I'm very, very excited by the LZ results."

The LZ collaboration announced the finding in a presentation today at the TeVPA conference in Japan. An accompanying paper has been submitted to *Physical Review Letters*. According to the project's best estimates, the flash's chances of being a statistical fluke are about half a percent (which is still far above the 0.00003 percent threshold used by particle physics experiments to hail a discovery).

The universe we see when we look out into space is almost impossible to countenance without dark matter. The stuff our telescopes can track is swirling too fast, forming too quickly into clumps that are too dense, to easily explain. The easiest explanation, in fact, is that, compared with the matter we can see,

there must be five times more stuff that's invisible and inert, still carrying gravity's weight but emitting no light. This dark matter is ubiquitous and plentiful, but we have no idea what it's made of.

Detectors like LZ are predicated on a guess: that dark matter is made of weakly interacting massive particles (WIMPs). These are hypothetical particles that almost always slip through ordinary matter like ghosts. But very rarely, they should interact with it, colliding at random with an atom. LZ and its competitors are racing to pack as many atoms as possible into the quietest places they can find to catch these collisions.

Still, traces of background radioactivity sometimes break the subterranean silence, with that activity piercing through LZ's mile of protective bedrock or even seeping out of the materials lining the detector's walls. But physicists can discern many of these events from their best guess at dark matter's signals and try to reduce the incidents as much as they can. That a single unexplained event can carry so much statistical significance shows how incredibly sensitive the technology has become. "We made our meter so that the needle would jump when it saw an event like this," says Knut Morå, a physicist at the University of Zurich and a member of the LZ collaboration. "Everything is working as intended."

The collision from June 2023 meets most of the team's WIMPy expectations—except that it's far too explosive. For LZ to find such a high-energy WIMP event, most standard theories say that it should have already seen dozens or hundreds of gentler ones—which it has not. So if what LZ saw *is* a WIMP, this particle must be exceptionally odd—interacting in a slightly different way with regular matter or somehow having been boosted to a higher speed.

"Given that we don't know what the dark matter is, it doesn't strike me as strange that it might not be exactly the thing we expect," says Hugh Lippincott, an associate professor at the University of California, Santa Barbara, and a member of the LZ collaboration.

Whether or not this signal will prove to be genuine is presently impossible to say. What's already certain is that it will spark a frenzy of papers seeking to explain how tweaks to various theoretical flavors of WIMPs could account for LZ's mysterious event. "They are potentially seeing other types of interactions which are perfectly reasonable," Freese says.

The collaboration itself also advises caution. "We shouldn't be comfortable claiming specific things about dark matter or about new physics based on one event," Morå says. Additionally, the team's scientists analyzed the data without "blinding" themselves to it, an approach to avoid bias that's considered the gold standard for particle physics. (They attempted an alternative approach called "salting" but stated that it was unsuccessful.)

Fortunately, more answers should arrive in the near future, based on a larger tranche of data that has been properly blinded. "This was 200 days of data. We actually have over 700 days blinded," says Richard Gaitskell, a professor at Brown University and co-spokesperson of LZ. "So that puts us in a strong position to study this."

Those data will either reveal this one event to be an anomaly or uncover more like it. In the latter case, two competing experiments, PandaX-4T and XENONnT, will help determine whether LZ has really seen WIMPs or some as-yet-unknown background.

For now, the LZ collaboration is trying its best to manage expectations. "You get these flashes of, like, 'Wait, could this really be it?'" Morå says. "But it would be damaging to the analysis to be in that mindset, and I don't think many people are." Lippincott agrees, albeit with a nod toward the extraordinary. "It is far from any of the known backgrounds," he says, "so it's hard not to wonder."

The mystery collision comes at a perilous time for the dark matter race. “It’s a good moment for nature to give us this gift,” says Juan Collar, a physicist at the University of Chicago, who is not part of the LZ collaboration. Detectors are becoming so large and so sensitive that they will soon be swamped with other ghostly particles called neutrinos that are generated in the sun and in Earth’s atmosphere. At that point, the race would have to cease—or at least to shift away from WIMPs to instead seek some other elusive quarry.

“Here’s your last window of opportunity to see a WIMP—if it’s not there, you have to close up shop at some point in the next 10 years,” Collar says. “This could be the beginning of something spectacular.”

A version of this article entitled “Flash of Darkness” was adapted for inclusion in the November 2026 issue of Scientific American.

Joseph Howlett is a senior reporter at *Scientific American* covering physics, math, astronomy, and more. He was previously a math staff writer at *Quanta Magazine* and holds a Ph.D. in particle physics from Columbia University.

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https://www.scientificamerican.com/article/have-we-finally-found-dark-matter

Mathematicians invented a knitting stitch that stops mistakes from spreading

With a little help from knot theory, scientists have invented a "robust knit" to fight laddering

By Emma Gometz edited by Sarah Lewin Frasier

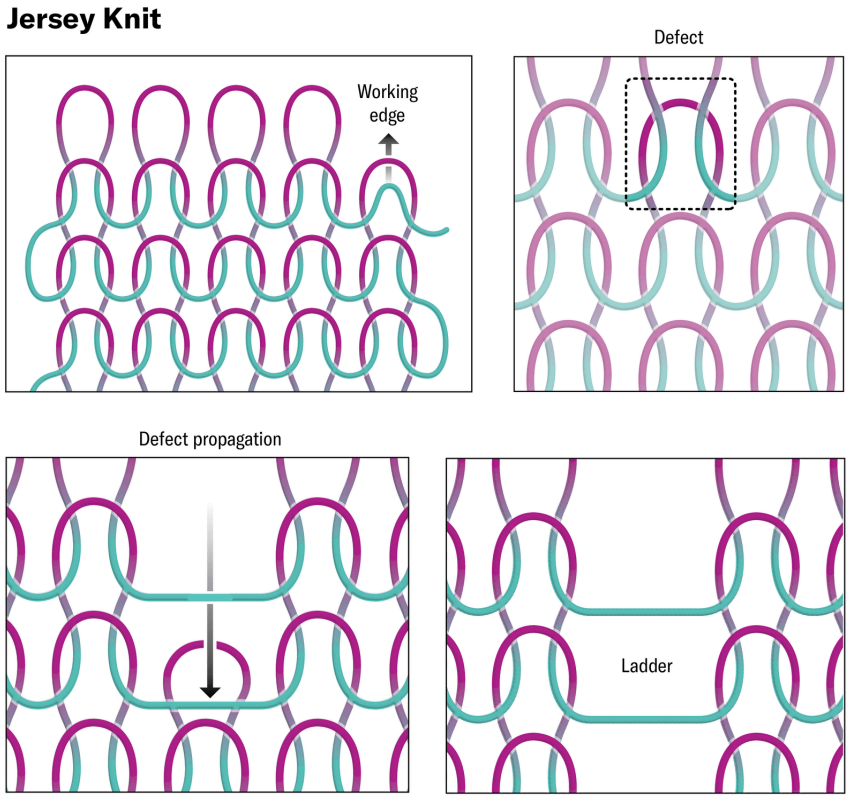


glamstock/Imazins/Getty Images

Despite its tranquil reputation, knitting is anything but boring. One stitch slip-up can cause a defect that spreads across the entire fabric. But does it have to be that way? In a recent paper in *Physical Review X*, a team of researchers mathematically investigated knitting to invent a new, more resilient stitch.

Knitting is a special form of fabrication in which the two ends of the yarn remain outside the fabric. In knitting, loops from the middle of an entangled yarn are slipped through one another so the yarn holds its position but isn't knotted together. The knitting process is portable and easy, but the structure also means a tug on one loose end can quickly unravel the whole project. It's also vulnerable to other quickly spreading errors.

Daisuke S. Shimamoto, a physicist at Ritsumeikan University in Japan, and his colleagues wrapped a three-dimensional model of a single jersey knit stitch—a simple beginner's stitch—around a doughnut shape called a torus, which preserves the pattern without including any edges at the top or bottom. It basically converts the knit pattern into one giant knot that can be analyzed with knot theory. The scientists projected multiple knit rows on the torus to reveal how defects from one row of stitches spread until the entire fabric unravels. The process is called defect propagation, and the researchers say it's one of the defining characteristics of knittable fabrics.

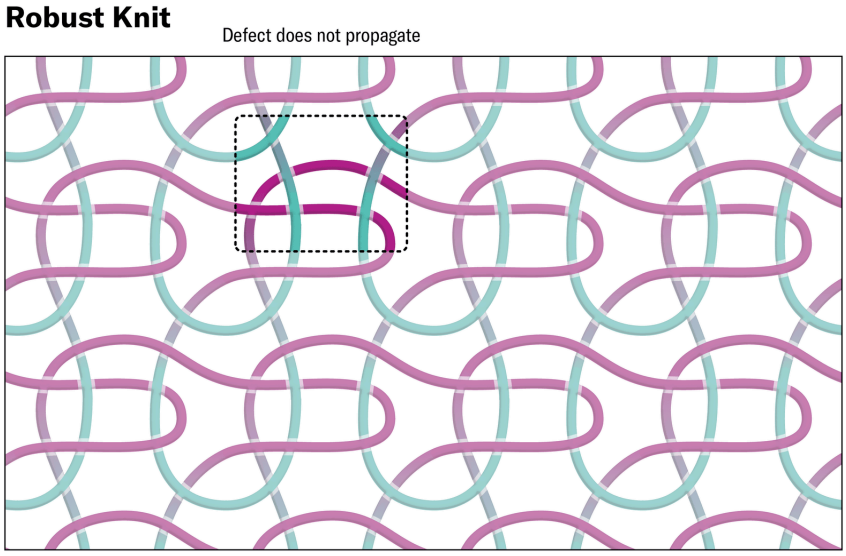


Amanda Montañez; Source: "Topological Defect Propagation to Classify Knitted Fabrics," by Daisuke S. Shimamoto et al., in *Physical Review X*, Vol. 16; July 14, 2026 (*reference*)

Researchers have mathematically described stitch defects in the past, says applied mathematician Alexander Omelchenko of Constructor University in Germany, but only as static objects. This study shows how defects move dynamically across a textile. "It's a very original approach," says Omelchenko, who wasn't involved in this study but originated the topological analysis method used.

One common type of defect propagation is “laddering,” in which a single missed loop slips down a column of stitches, unraveling the whole column along the way. “If one unit cell is disentangled, the cell just below that cell is disentangled, and it is repeated, and it propagates endlessly,” Shimamoto says.

After his topological analysis of the jersey knit, Shimamoto used moves from basic knot theory to design a knittable stitch in which defect propagation can still happen but not endlessly. This “robust knit” is a complex stitch with sweeping loops that cross over through two stitches rather than one, and it reduces propagated defects row by row until they stop. Defects expand only when there is more than one slipped stitch in a row.



Amanda Montañez; Source: “Topological Defect Propagation to Classify Knitted Fabrics,” by Daisuke S. Shimamoto et al., in *Physical Review X*, Vol. 16; July 14, 2026 (*reference*)

Omelchenko appreciates the study authors’ ability to translate math to real life but notes that fully evaluating the possibilities of new stitches by using torus diagrams would be impractically expensive to compute. Shimamoto hopes his work with the robust knit, as a proof of concept, can one day inspire stronger fabric design for manufacturing.

Besides a fascination with fabric, the two researchers have another thing in common: both consulted their mothers' knitting advice for their papers, and Shimamoto even gave his mother an author credit as a thank you for teaching him how to knit. According to Omelchenko, "It sometimes was a little bit more useful to talk with her rather than textile researchers."

Emma Gometz is a journalist and artist based in Queens, N.Y. Before becoming associate newsletter editor at *Scientific American*, Gometz was a digital producer for WNYC's *Science Friday.* Their favorite musical is *A Little Night Music.*

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https://www.scientificamerican.com/article/mathematicians-invented-a-knitting-stitch-that-stops-mistakes-from-spreading

Physicists pinpoint the perfect crystal site for an ultra-accurate nuclear clock

Growing crystals “like alchemy,” researchers plotted out the ideal site within the lattice to build a nuclear clock

By Jacek Krywko edited by Sarah Lewin Frasier



akinbostanci/Getty Images

Today’s best atomic clocks, which track time by measuring the way electrons consistently jump between energy states when exposed to particular laser light, could run for 13 billion years—the age of the universe—and lose less than one second of accuracy. They are incredibly precise but also incredibly delicate: they require an entire laboratory’s worth of vacuum chambers and shielding to prevent interference from stray electric or magnetic fields.

To make them hardier, scientists aim to go one layer deeper into the atom, from the electrons on the outside to the nucleus nestled within, and track how energy flips a single neutron between quantum states. According to Vienna University of Technology physicist Thorsten Schumm, such a switch would let the atom itself

shield the clock from much outside interference—and such atoms could be embedded in a crystal for even more protection. (Plus, this switch could make the clocks up to 10 times as precise.)

In a study published in *Science*, Schumm and his colleagues grew just such a crystal and mapped the four places atoms can sit inside it, finding the best position to measure when building the most precise clock ever.

In most atoms, the energy required to flip a neutron between states would need a gamma-ray laser, which scientists can't make yet. There's just one known exception: an isotope called thorium-229, which researchers showed in 2024 can be triggered with an ultraviolet laser instead.

"Thorium-229 is a quirk of nature. Its nuclear transition energy is remarkably low, making it accessible to high-purity laser light," explains Andrei Derevianko, a physicist at the University of Nevada, Reno, who was not involved in the study. Researchers sought a crystal structure that could house and protect thorium-229 atoms, removing the need for intensive vacuum chambers for the clock to function.

"Crystal growing is like alchemy or like cooking," Schumm says. "You try, and then you try something else, and then you keep trying." After 15 years of experiments, Schumm's group ended up with a recipe for solid calcium fluoride crystals, doped with little impurities of thorium-229, that produced transparent millimeter-size cubes. "In the end, it looks like a tiny little piece of glass, but there's tens of thousands of hours of learning how to do that," he says. The researchers realized, however, that if they measured thorium-229 in the wrong internal position, the crystal would have an uneven electric field that would harm the clock's timing.

For this latest study, to investigate the possible ways thorium atoms can be positioned in crystals, the scientists shined a custom-built

ultraviolet laser at one wavelength for 60 seconds, switched it off, and watched for five minutes as excited nuclei decayed back down to lower energy levels and emitted the light back out. Then the researchers altered the wavelength and did it again. It turned out thorium atoms can sit at four sites in the crystal lattice. Three return light at multiple wavelengths, which indicate an uneven electric field around them. The fourth responds to just one wavelength, a sign of an even field that made it perfect for the clock.

“The data in this paper are something we’ve all wanted to see for some time,” says Eric Hudson, a physicist at the University of California, Los Angeles, who was not involved in the study. “This is a very important result for the research on the solid-state nuclear clock,” adds Ekkehard Peik, head of the time and frequency department at the National Metrology Institute in Germany, who was also not involved in the study. Based on their discovery, Schumm and his colleagues have already built the first early prototypes of new clocks, and they have patented a route to building one that fits on a small chip. A different team in China, led by physicist Shiqian Ding of Tsinghua University, has also demonstrated a prototype nuclear clock; that one achieved comparable results with a more powerful laser aimed at a crystal containing a slightly lower concentration of thorium-229. Schumm thinks progress in the field is gaining momentum.

As for his own team’s effort: “I’m not going to give you a timescale on when we will beat the best atomic clocks,” Schumm says, “but we will be better [than our current prototypes] by at least three orders of magnitude by the end of the year.” For now, though, his team isn’t chasing superior precision. The priority, the researchers say, is miniaturizing the clocks—first going from lab-size to shoebox-size, Schumm says, a form that could be used in a data-center server rack to keep things such as banking transactions going when communications are lost with today’s satellite-based clocks.

Jacek Krywko is a freelance writer who covers space exploration and computer science.

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https://www.scientificamerican.com/article/physicists-pinpoint-the-perfect-crystal-site-for-an-ultra-accurate-nuclear-clock

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The fight to save endangered species may have a new weapon: sex-changing mouse clones

For the first time, scientists managed to turn cloned male mouse embryos into females, giving life to healthy and fertile individuals

By Gennaro Tomma edited by Sarah Lewin Frasier



A male donor mouse meets its female (*right*) and male (*left*) clones.

Takashi Ichiushi and Shogo Matoba

Scientists racing to save critically endangered species may soon have a new tool at their disposal: sex-changing clones. For the first time, researchers managed to turn cloned male mouse embryos into females, giving life to healthy and fertile animals. The technique could one day help scientists rescue endangered species with only a few surviving individuals.

“I think it’s a really neat foundational breakthrough in reproductive science,” says Ben Novak, lead scientist at Revive and Restore, a nonprofit that promotes biotechnology for wildlife conservation. “Being able to manipulate chromosome counts and chromosome

makeup will open up a new avenue of functional genomic science." Novak wasn't involved in the new study, which has not yet been peer-reviewed.

Inspired by the Okinawa rubble goby, a Japanese fish that can switch sex, researchers used a method based on the gene-editing tool CRISPR to eliminate the Y chromosome from male embryos, which typically carry both X and Y chromosomes. This process turned the embryos female with just a single X chromosome—typically notated as XO.

The scientists first successfully tested the tool, which they named Y-CUT, on male mouse embryos obtained through in vitro fertilization. Then they produced cloned male mouse embryos through a process called somatic cell nuclear transfer, in which a donor mouse's cell nucleus was transferred into an egg cell with its own nucleus removed. The team applied Y-CUT to the cloned cells and transferred them to surrogate mothers, which gave birth to healthy, fertile female clones. In other words, the researchers were able to produce both male and female clones from the same donor male. The cloned females carried the exact same genome as the donor male (except for the Y chromosome).

In a surreal twist, the donor male mouse had the chance to meet its female clones. And it got weirder: The cloned females mated with male clones, producing offspring with regular XX or XY chromosomes. That's why the researchers say the tool could be useful for rescuing endangered species with a few individuals left—scientists could create fertile breeding pairs using just one male donor.

"It certainly can have long-term future applications for extreme cases of species revival," Novak says. Still, as tantalizing as the prospect of cloning a creature out of extinction may be, the method is limited—and not just because it requires at least one male. For example, female rodents carrying XO chromosomes may be fertile,

but many other mammals, including horses and humans, can’t reproduce when they have this chromosome combination. It’s possible that supplementing the technique with other elements, such as future technologies that could add an X chromosome to XO embryos, might help overcome this problem.

Gennaro Tomma is a freelance journalist who covers science, with a focus on the natural world, biodiversity, conservation, climate change, environmental and science-related policies, and more. His work has appeared in the *New York Times, Science, National Geographic, New Scientist* and other outlets. Find more on his website: https://gennarotomma.it

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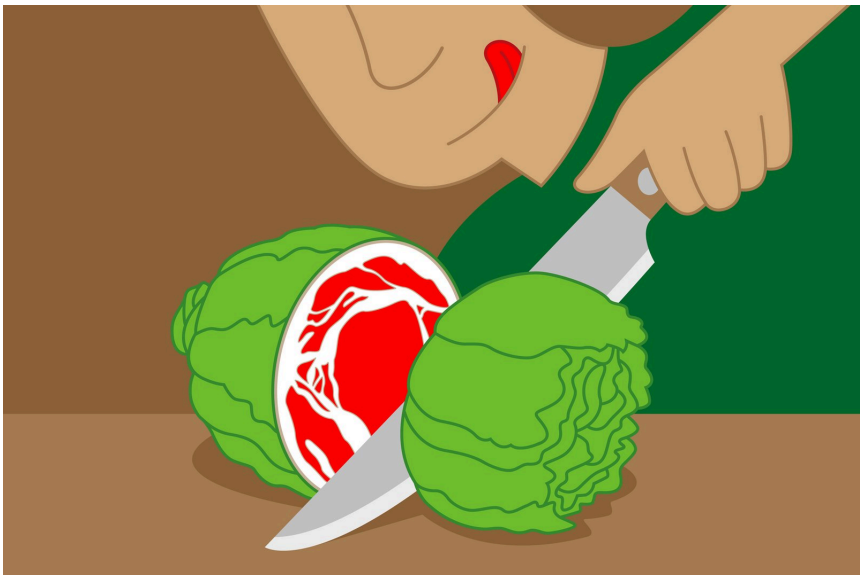
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Lettuce engineered to produce meat proteins

Protein-packed leaves might be the key to better meat alternatives

By Lila Magbilang edited by Sarah Lewin Frasier



Thomas Fuchs

Perhaps you enjoy a burger in a lettuce wrap once in a while—but what if greens were also crucial to the patty? That's the ultimate promise of new research showing lettuce can be engineered to produce myoglobin, a protein found in meat.

Myoglobin (and its better-known relative hemoglobin) contains the compound heme, the red pigment in blood. Adding heme can give plant-based meat alternatives a meaty flavor and the ability to "bleed." For example, Impossible Foods uses a compound called leghemoglobin (a heme protein derived from soybeans) to flavor its meat alternatives. But using myoglobin, which is found in animal tissue, could lead to vegetarian products that not only look and bleed like red meat but smell more like it, too.

In the study, published in *Frontiers in Plant Science*, the researchers describe how they got lettuce and tobacco plants to produce myoglobin for the first time, says lead author Alexia Groff, who worked on the project as a Ph.D. student at Imperial College London. The team engineered the plants' photosynthesizing chloroplasts—the cell structures that typically turn sunlight into food—to make them do the job.

In previous attempts to make myoglobin using nonanimal sources such as yeast, the protein-production directions were placed in the nucleus's genome, but Groff and her colleagues found that adding them to the chloroplast genome instead could yield more of it—perhaps producing myoglobin even more efficiently than animals do.

Right now yeast fermentation is the standard method of producing heme (as leghemoglobin), but the sterile environment and industrial equipment necessary for manufacturing can be pricey. Plants are much easier to grow. "Row-crop protein production would likely be faster than fermentation-derived production," says Adam Leman, a principal scientist at the Good Food Institute, who was not involved in the study. "If existing crop-processing equipment can be used, costs for isolating and purifying a myoglobin-rich fraction could be quite low."

Lettuce is especially promising: the medium itself is edible, meaning lettuce-produced myoglobin could move more easily through the regulatory steps that can slow down the path from laboratory to market. A more streamlined process could also mean less money spent in production—and far less energy use.

The meat industry is a voracious energy consumer: 60 percent of agricultural energy use comes from farming and transporting animal products. Making meat substitutes less resource-intensive is a step toward more sustainable food systems—even if it means making meat proteins in a vegetable.

“In science, there were things that we used to think were impossible that now are an everyday occurrence,” Groff says. “I’m very excited for what the future of this sector is going to look like.”

Lila Magbilang is a California-based freelance science journalist who covers environmental and biological sciences.

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https://www.scientificamerican.com/article/lettuce-engineered-to-produce-meat-proteins

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How to spot an AI-generated face, according to science

Training people to pay attention to the right visual cues nearly doubled how accurately they could spot AI-generated faces

By Sam Macdonald edited by Claire Cameron & Sarah Lewin Frasier



The top three AI-generated faces are balanced, generic and conventional—traits recognizable with training that distinguish them from the bottom three human faces.

StyleGan3 (*AI faces*); Flickr (*human faces*)

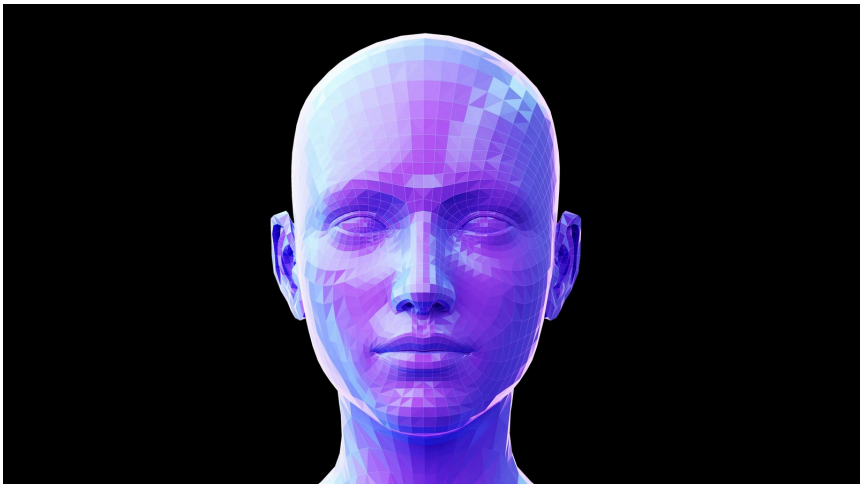
It used to be easy to tell when a face had been generated by artificial intelligence. Whether it was a distinctive uncanny sheen, impossibly smooth skin, eyes that didn't quite make sense or a conspicuous third ear, older AI models' attempts at human faces were simple to notice and easy to dismiss. That's just not true anymore.

Now AI image generators can produce portraits so convincing that even careful observers struggle to distinguish fact from figment. That's why apps including Zoom and Tinder let their users submit biometric identification, such as retinal scans, to help prove that a

real person exists behind a profile picture. But a recent study in the *Proceedings of the National Academy of Sciences USA* suggests you can train your brain to get better at spotting fakes.

Past attempts to teach people to identify AI faces have focused on training viewers to look for visual glitches or statistical fingerprints left behind by a particular image generator, such as a wonky ear or an eye with two pupils. The problem is that those clues can disappear with a software update or even the use of a different prompt. “The AI is getting too good,” study lead author Amy Dawel, a clinical and cognitive psychologist at the Australian National University, said in a press release. “And fraudsters may avoid using pictures with obvious flaws anyway.” The result is an endless technological arms race humanity seems destined to lose.

Instead the researchers taught the participants to recognize broader patterns in how AI systems generate images. Current AI image generators are themselves trained on datasets composed of millions of photos. When prompted to generate a face, they don’t copy specific faces but instead compose a new one that is based in part on the mathematical patterns shared across the faces in those datasets—which allow the AI to construct a “typical” human face.



Sebastian Kaulitzki/Science Photo Library via Getty Images

This means AI-generated faces often drift toward statistical averages. They’re not so much overly unrealistic as a little too

balanced, a little too generic and a little too conventional. Individually, none of these traits is necessarily suspicious. But together, the whole is blander than the sum of its parts—a subtle banality that humans can often implicitly sense.

“It is particularly interesting that the global-impressions-of-faces training approach used in the current study resulted in such substantial improvements,” says Sophie Nightingale, a senior lecturer in psychology at Lancaster University in England. The researchers found that even relatively short training sessions helped to boost people’s accuracy in identifying the AI.

Compared with real faces, AI-generated faces tend to be more symmetrical, more proportional and more attractive—while also being less expressive, less distinctive and significantly less memorable. When the researchers taught participants to look for these six markers instead of fleeting artifacts such as malformed ears or mismatched jewelry, their accuracy in spotting an AI face almost doubled, from about 40 to 80 percent success on a comparison task.

In other words, AI gravitates toward the center; individual humans do not. Our faces are shaped by countless small deviations from the norm—the subtle asymmetries, distinctive features and unique expressions that make us memorable. Those imperfections are not flaws. They are our signature.

Sam Macdonald is an assistant professor of mathematics at Augustana University and was a 2026 AAAS Mass Media fellow at *Scientific American*. He enjoys rock climbing, strategic hammock placement and the axiom of choice.

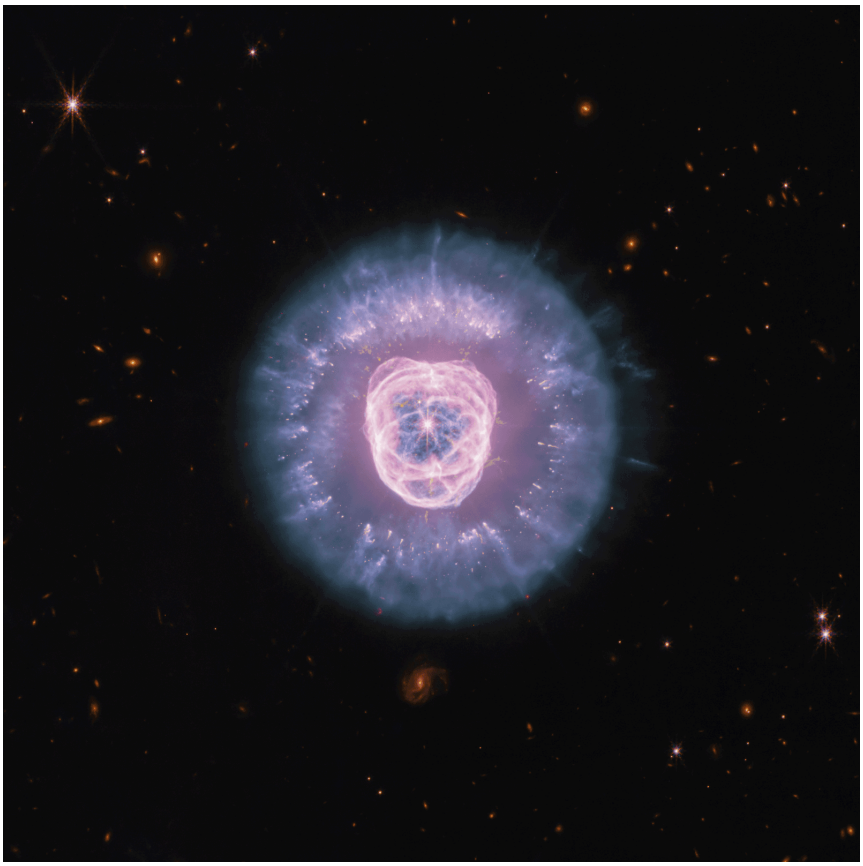
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https://www.scientificamerican.com/article/how-to-spot-an-ai-generated-face-according-to-science

New images reveal the Lion Nebula in all its majestic glory

This dying star appears to have a face of gas and a mane of space dust

By Mary Randolph edited by Claire Cameron & Sarah Lewin Frasier



The Lion Nebula was first captured by the Hubble Space Telescope in 2000. The new images from JWST capture the planetary nebula's lionlike nose and mane in far greater detail.

NASA

Imagine a lion with the nose of a dying star, a mane of space dust and a habitat 10,000 light-years away from Earth. On its infrared galactic safari, NASA's James Webb Space Telescope just captured such a beast on camera.

The planetary nebula NGC 2392, nicknamed the Lion Nebula for its resemblance to the animal, was previously photographed in 2000 by the Hubble Space Telescope. Now astronomers have captured an even clearer view.

The Lion Nebula, like other planetary nebulae, is a glowing shell that surrounds a dying star in the form of a white dwarf at its center. The white dwarf emits radiation and produces a giant gas bubble—the lion's face—that destroys dust in its path. The border, or "mane," of the planetary nebula is a halo of dust surrounding the white dwarf.

The new images reveal the ionized gas and clumps of dust surrounding the white dwarf more clearly than the older Hubble pictures. The structure formed by the gas and dust, visible in the new observations as a set of complex rings and shells, is an astronomical phenomenon researchers still don't totally understand.

The star will eventually die, destroying the lion and dispersing all the gas and dust into interstellar space. Astronomers estimate that its final roar will occur around 10,000 years from now.

Mary Randolph is a news intern at *Scientific American* and is based in New York City. Her work has been featured by *Smithsonian Magazine*, Pew Research Center, the Local News Initiative and various local newsrooms.

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https://www.scientificamerican.com/article/new-jwst-images-reveal-the-lion-nebula-in-all-its-glory

This beer-making waste may offer a new stream for cleaner water

The research offers yet another way the brewery industry might repurpose its myriad by-products

By Mary Randolph edited by Sarah Lewin Frasier



Beer moves down a brewery conveyor belt.

industryview/Getty Images

Scientists still haven't cracked how to turn water into wine—but they might have figured out how to turn beer into clean water. From the literal ashes of breweries' by-products, researchers made a charcoal-like substance that clings to bacteria, and they are using it to test a new way of filtering water.

In a recent study, scientists heated up a common waste product from beer making called spent malt rootlets, which are little sprouts that grow from barley during the malting process, to produce biochar, a porous material created from burned organic matter. They found that when they mixed this substance with sand and sent water through it, the biochar was able to remove around five times

as much *Escherichia coli* bacteria as a filtration system that used only sand.

Sending water through sand filtered bacteria like a colander: by physically stopping microbes from moving through. But when the team used a sand mix that was 10 percent biochar, the biochar actively glommed on to slow microbes' passage.

Groundwater is constantly moving through soil. There it can become contaminated with agricultural waste or fecal matter—potentially infecting any humans who drink it. Adding biochar to certain soils could offer a public health solution and "a sustainable approach for water remediation and the protection of groundwater," says Ioannis Manariotis, senior author on the study and an environmental engineer at the University of Patras in Greece.

Manariotis and his team have also tested whether waste from olive oil and coffee production can have similar water-purifying effects. Now that the researchers have found the beer by-product can filter out bacteria in laboratory conditions, he says, they plan to test its efficacy under more unpredictable real-life conditions. The results were published in the journal *Biochar*.

Using brewery waste for water filtration in this way is a "proof of concept," says Alcina Johnson Sudagar, an independent chemistry researcher who studies other beer-making by-products and was not involved in the study. But the repurposing of by-products from this industry for other applications is already common, she says. Leftover brewery grain is frequently used to make animal feed, and other research suggests beer-making by-products can be used for everything from sunscreen to electricity. The brewing process can have up to 11 steps, and each step produces organic waste materials with different chemical makeups—potentially ideal for different uses.

Sudagar says she hopes Manariotis and his team continue to study biochar. Her most pressing question is whether the material can be used more than once for filtration. If it can trap bacteria only once, it quickly becomes waste again, negating its value as a consistent filtration system or sustainable brewery by-product.

As the global beer industry grows, these new and innovative ways of using the by-products are crucial to sustainability, Sudagar adds. “It’s important to continue because the waste produced will keep increasing, and you would need different applications,” she says. “You cannot just send it to one place and say that we are done with it.”

Mary Randolph is a news intern at *Scientific American* and is based in New York City. Her work has been featured by *Smithsonian Magazine*, Pew Research Center, the Local News Initiative and various local newsrooms.

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https://www.scientificamerican.com/article/this-beer-making-waste-may-offer-a-new-stream-for-cleaner-water

Plants move to get away from their decaying brethren

A new discovery reveals how plants protect themselves from microbes in rotting material

By Joanna Thompson edited by Sarah Lewin Frasier



Thomas Fuchs

Plants seem stationary: they don’t skip, fly or walk around. But that doesn’t mean they don’t move. The organisms bend their roots and shoots in response to stimuli such as water, light, salt and gravity. These responsive movements, called tropisms, are crucial for plants’ survival.

Now researchers have discovered a new way plants move, which they call “saprotropism”: the bending of roots to avoid dead or rotting plant tissue. The results were published in *Science*.

The discovery began with a failed experiment and an unexpected observation. Yuzhou Zhang, a biologist at Northwest A&F University in China, was studying how plants move in response to

gravity in a laboratory in Austria when one of his agar plates became contaminated with airborne microbes. Just before he threw it away, he noticed something striking: almost every single one of the plants' roots in the plate was bent away from the spoilage. The consistency "immediately made me wonder whether this was more than a random response," Zhang says.

His team decided to investigate further. The researchers grew the common research plant *Arabidopsis thaliana* in containers, some of which were dotted with patches of rotting apple. The plants that encountered "decay zones" bent their roots to avoid the bad apple, whereas the ones in containers lacking decay grew straight through. Repeating the experiment with other plants, including tomatoes and wheat, yielded similar results—but an encounter with putrefying animal tissues had no effect. Zhang and his colleagues theorize that this reaction might help plants avoid pathogens, similar to the way unpleasant smells help animals steer clear of rotten food.

Saprotropism seems to operate differently than most other plant tropisms. "It does not follow the classical mechanism," Zhang says. For example, gravitropism, or movement in response to gravity, and phototropism—in response to light—occur when plants redistribute the hormone auxin in their stems or roots, causing them to change their growth direction.

In contrast, saprotropism relies on a hormone called abscisic acid (ABA). When the plant roots encounter an area acidified by decaying plant matter—a chemical gradient distinct from that of animal matter—ABA builds up on the side facing the rot, encouraging growth in the other direction.

The work's scope is impressive, says Chris Wolverton, a plant biologist at Ohio Wesleyan University, who was not involved in the study. He adds, however, that "there's still a lot of open space in the signaling mechanism for more detail to get fleshed out." At what

point, for instance, does decaying plant tissue break down enough to become nutritious compost?

Zhang’s team hopes to fill those gaps in future studies. Ultimately the research could help illuminate how plants sense and interact with the world around them.

Joanna Thompson is an insect enthusiast and former *Scientific American* intern. She is based in New York City. Follow Thompson on X (formerly Twitter) @jojofoshosho0

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Math Puzzle: Baby bottle mix-up

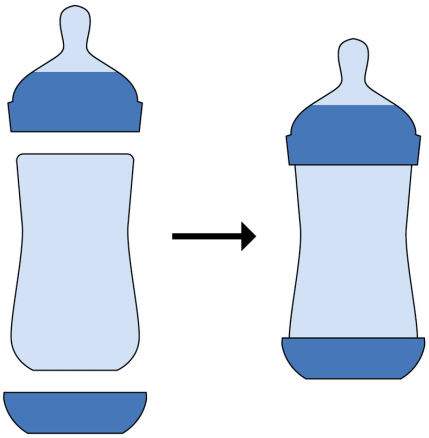
Solve a new-parent dilemma with this math puzzle

By Benjamin E. Grossman-Ponemon edited by Sarah Lewin Frasier



Ross MacDonald

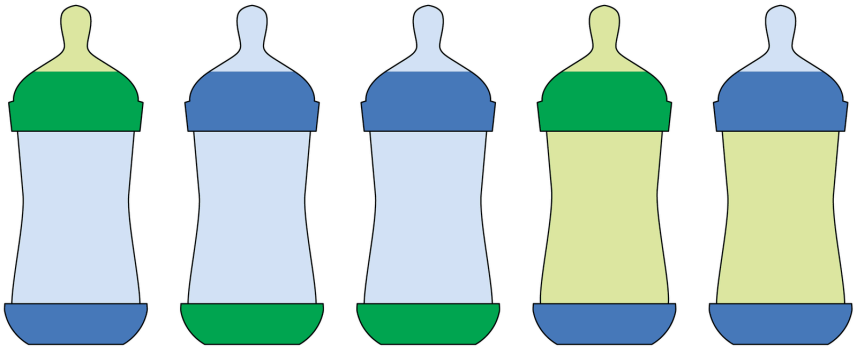
This problem is the result of many hours of cleaning baby bottles for two daughters. Suppose your baby uses a certain type of bottle that is composed of three pieces (say, top, middle and bottom). The bottles come in two colors (say, blue and green). You have three blue bottles and two green bottles, for a total of five bottles. Your baby uses all five bottles in a single day. Because you're perpetually tired, you do not disassemble and wash the bottles until the end of the day. After the bottles have dried overnight, you reassemble them and begin your next day. Such is the life of a new parent.



Amanda Montañez

When you go to reassemble the bottles, you notice that you can exchange parts among them so that some are “mixed up.” For example, you can put a green top on a blue middle and blue bottom. Naturally, you try your hand at swapping parts so there is no complete bottle of a single color—in other words, you find a “derangement” of the bottles.

Here is one such derangement:



Amanda Montañez

Suppose you then buy some new bottles in a third color (say, pink). What is the minimum number of pink bottles required to ensure that no derangement can occur?

View puzzle with solution at this link.

Benjamin E. Grossman-Ponemon is an assistant professor in the department of physics and engineering at John Carroll University in Ohio. In addition to his love of puzzles, Grossman-Ponemon has research interests in computational mechanics, with particular focus on the problems of algorithms for systems of wearable inertial measurement units and the physics of fracture. His

teaching interests include numerical methods for engineers, dynamics, and signals and systems. You can reach him at bgrossmanponemon@jcu.edu.

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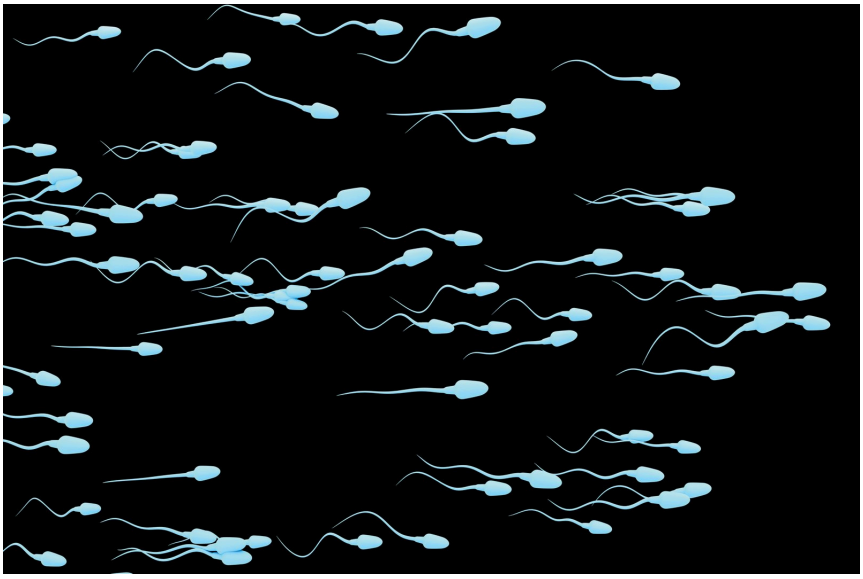
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Damaged sperm might be the true cause of repeated miscarriages

New guidelines say testing sperm for DNA damage could help explain recurrent pregnancy losses

By Sarah Elizabeth Richards edited by Claire Cameron & Sarah Lewin Frasier



Sperm cells.

feellife/Getty Images

In the summer of 2024 fertility doctor Carrie Bedient sat across from a young couple and braced herself for a difficult conversation. The man and woman, both 32 years old, were trying to have a baby and had been seeking medical help at her clinic for the previous five years. But multiple rounds of in vitro fertilization (IVF) treatment had failed. By the time Bedient met with them to discuss the next steps, they had been through failed embryo transfers, repeated miscarriages and a kind of early pregnancy loss known as a chemical pregnancy.

Bedient, a reproductive endocrinologist at the Fertility Center of Las Vegas, had already ordered the usual battery of tests, including

blood work, sperm count and motility measurements, and uterine anatomy exams. So she suggested what was then a more controversial idea: testing for DNA damage in the man's sperm. He said yes.

By that point a growing body of research had linked DNA fragmentation in sperm to an increased risk of miscarriage. But despite tests for fragmentation being available for some 15 years, they weren't commonly done—not least because doctors weren't sure what people could do with the information. It was also unclear what caused the DNA damage in the first place; possible culprits ranged from environmental toxins to advanced paternal age to lifestyle factors such as drug use and smoking. Plus, tests could cost more than $500 and weren't always covered by insurance.

In this man's case, the test results revealed a moderate level of DNA damage. Bedient gave him some frank advice on how to rehabilitate his sperm. "Hey, I need you to clean up everything you can," she recalls telling him. He vowed to stop drinking alcohol, stop smoking, start exercising and take vitamin supplements—and it worked.

Six months later the couple made another batch of embryos. "He completely turned everything around, and they've now got a beautiful baby boy," Bedient says.

When a man starts infertility treatment, he typically undergoes a workup that analyzes the volume and concentration of his semen, as well as the shape of his sperm and their ability to move. Yet the association of damaged sperm with miscarriage—in addition to newer research linking it to lower IVF live birth rates and poorer early embryo development—is pushing doctors to delve deeper into sperm health.

This year, in its first update in 14 years, the American Society for Reproductive Medicine (ASRM) released new treatment guidelines

for recurrent miscarriage, noting that testing for sperm DNA fragmentation “may be considered” in cases of multiple pregnancy losses. The guidelines’ authors stop short of outright recommending the tests; instead they say more research is needed to determine whether treatments for reducing sperm DNA fragmentation improve pregnancy outcomes.

But it is still an important inclusion, says Clarisa Gracia, a reproductive endocrinologist at the University of Pennsylvania and chair of the ASRM practice committee that develops clinical guidance for fertility doctors.

“Our summary statement was hedging, but at least it spoke to the fact that the male evaluation is also important,” Gracia says. “If a couple is trying to get pregnant, I don’t think we should leave out the male, because we are learning more and more that male health affects reproductive outcomes as well.” The so-called male factor accounts for 20 percent of infertility cases and contributes to an additional 30 to 40 percent of cases in combination with the female factor, according to the American Urological Association.

Fertility doctors didn’t even suspect that sperm played a critical role in miscarriage until just a few decades ago. Stories circulated in clinics about women who were former patients getting pregnant and giving birth after a divorce and finding new partners.

“I think anecdotally people had an inkling that there was something up related to the male in terms of miscarriages,” says Denny Sakkas, chief scientific officer at Boston IVF and IVI RMA North America, part of a global reproductive medicine group. He conducted research in the 1990s that looked at the DNA of sperm from men with fertility problems and found little breaks in the molecule’s structure.

“DNA is like an instruction booklet. If there are a lot of spelling mistakes in the DNA instruction, you might get a pregnancy,”

Sakkas says. “But when the pregnancy gets to a certain stage where a particular spelling mistake will impact, say, the development of the heart tissue, the pregnancy won’t continue.”

Sakkas predicts that consensus will shift toward more proactively testing sperm in the case of repeat miscarriage, especially as companies offer increasingly sophisticated and lower-cost DNA testing. And doctors have more treatment options, including a new generation of sperm-sorting machines that can select the healthiest sperm to inject directly into an egg.

In the meantime, recognizing men’s role in miscarriage can help to dismantle the stigma around pregnancy loss, says Brennan Peterson, a professor of marriage and family therapy at Chapman University.

“There is a lot of shame, depression and secrecy,” Peterson says. “So ASRM putting [new guidelines] out there hopefully will help the public understand that the pain of infertility and miscarriage is not related solely to women’s issues.” Avenues are also open for men to explore.

Sarah Elizabeth Richards is a San Diego, Calif.–based health and science journalist who writes frequently about reproductive medicine. Her writing has appeared in *Wired*, *National Geographic*, the *Atlantic*, *Men’s Health* and the *New York Times*, among other publications.

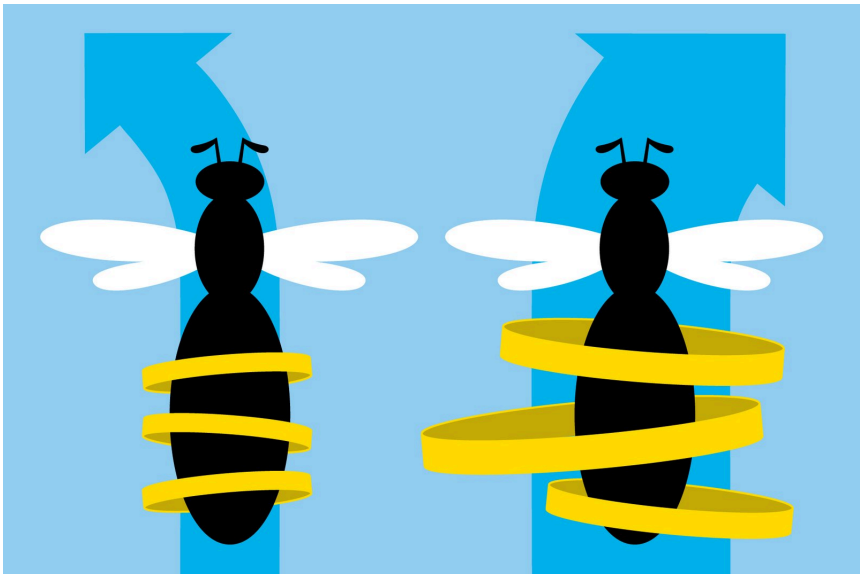
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Honeybees deemphasize potential ‘misinformation’ in their waggle dances

Confident bees work harder to convince their peers of tasty treats

By Marta Hill edited by Sarah Lewin Frasier



Thomas Fuchs

When a bee wants to tell its hive mates how to find a scrumptious field of flowers, it wiggles its rear end. A complicated combination of figure eight–like movements known as a waggle dance tells nearby bees what direction to fly in and how far to go. But what happens if a waggling bee is unsure of its directions?

A study published in *Behavioral Ecology and Sociobiology* earlier this year shows that bees amplify their communication based on how certain they are that their information is true. It seems the more confident a bee is in what it’s conveying, the more emphatically it communicates with the hive, says study lead author Karmi Oxman, a behavioral ecologist at Hebrew University of Jerusalem.

“Confidence is sort of a human-y word, but it’s useful in this context,” Oxman says. Confidence reflects the communicator’s belief in a message’s relevance or reliability, “and the dance is a self-report of the situation in the field.” This extra information also prevents the bee from accidentally misleading the hive.

The experiment started with one bee flying to a specialized feeder. This first bee, also known as the focal bee, was always met with a sugar solution. The researchers then manipulated the scenario by changing what happened after the focal bee returned to the hive and gave directions to the feeder. In the “honest” scenario, the feeder remained full, and all the bees following the directions got a sugary treat. In the “liar” scenario, the feeder was found empty. And in the “unverified” scenario, all the bees were caught before they could return to the hive and report what happened.

When given the chance to communicate the feeder’s location to the hive again in a later experimental stage, the focal bee in the honest scenario tried harder to convince its fellows to visit the sugar source by performing more waggle dance circuits—suggesting the bee could assess the reliability of its own signal. Unverified and “liar” bees did not show that same increase; in fact, unverified bees decreased their recruitment efforts.

“For me, the big takeaway is that bees are not little robots,” says James C. Nieh, a behavioral ecologist at the University of California, San Diego, who was not involved in the study. “They’re getting information from their environment, and that [information] also includes how successful they are in communicating to their nest mates.”

Both Oxman and Nieh say they wonder what other types of communication beyond what we can see—such as chemical signals or vibrations—could be playing a role in waggle dances. Further research into how bees communicate can help with efforts to “maintain the world in a way that they can still thrive,” Nieh says.

Marta Hill was the spring 2026 multimedia intern at *Scientific American*. Her work has appeared in *Smithsonian Magazine*, the Transmitter, *Eos* and elsewhere.

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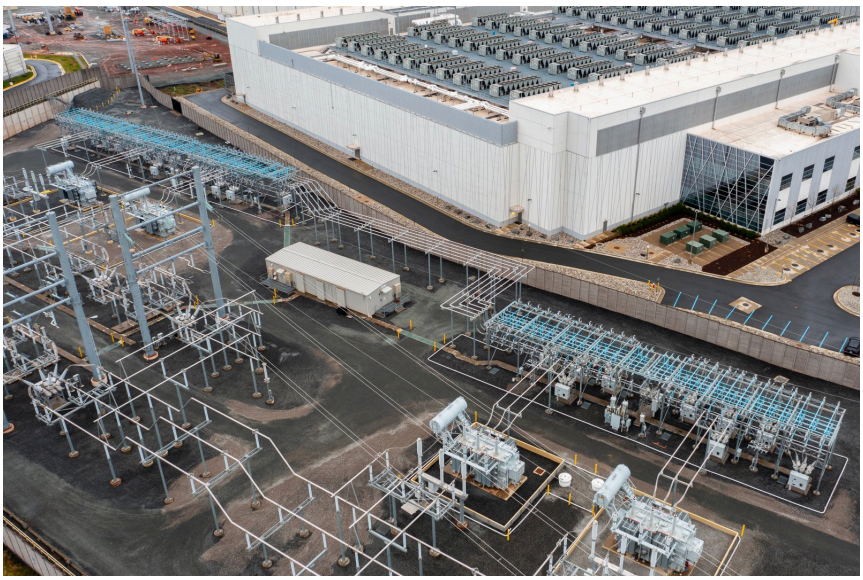
Industry

- **Why data centers are reviving solid-state transformers**
  Solid-state transformer designs have been around for decades. Now data centers may finally give the chip-filled devices their killer application

AI’s power boom is giving the transformer a high-tech makeover

Solid-state transformer designs have been around for decades. Now data centers may finally give the chip-filled devices their killer application

By Alex Pasternack edited by Eric Sullivan



As AI data centers demand ever more electricity, engineers are rethinking the bulky equipment that feeds them. Here a substation sits beside a data center in Ashburn, Va.

Bloomberg/Getty Images

On a humid morning in June, Ramadan Elmoudi nervously watched grid current flow into a machine his colleagues had spent years building and kept his expectations low. Years earlier, a similar test by another group of researchers left their machine smoking, two days in a row. This time nothing burned. “I was stunned,” says Elmoudi, a senior research engineer at the New York Power Authority.

The humming metal box, installed at a research substation in the wooded hills of western Massachusetts, looked like a large

refrigerator, indistinguishable from much of the electrical gear around it. But inside was a solid-state transformer, or SST—an array of power electronics that was meant to do the work of several pieces of that gear—and hopefully much more. Now the researchers, from a laboratory at North Carolina State University, were pushing electricity from a working power line through an electric-vehicle charging station and into a car. As far as they knew, it was the first test of a megawatt-rated SST on a live grid. “It might sound like this is old technology,” says Oscar Andrés Montes, one of the graduate students who designed and built the unit. “But it’s not something that people have done.”

Every transformer on the grid—from the cylinders perched on utility lines to the big spiky devices at substations—trades voltage for current at a fixed ratio. For a century, that was enough to satisfy a slowly growing demand. AI data centers have scrambled that math. Nvidia, Google, Microsoft, and other tech heavyweights have collaborated to sketch a new electrical architecture for them, with a different kind of transformer facing the grid, and a small industry has formed to build it. Whether that is worth paying for is still open because the device it would replace can be near 99 percent efficient, lasts for decades and has been doing its job since before anyone alive was born.

The first practical transformer was switched on about 15 miles down the road, in 1886, when William Stanley lit stores and offices lining Great Barrington’s Main Street. The tech has barely changed since: current passes between coils wrapped around magnetized steel cores, stepping voltage up for transmission and back down where it meets a customer. Typically custom-designed using specialized steel, some are so massive that transporting them can require moving power lines. And they are increasingly hard to get: Amid a scramble to expand the U.S. grid, surging demand and materials constraints have stretched lead times for higher-voltage units to years.

The initial concept for the solid-state transformer, which miniaturizes the magnetic core and surrounds it with power semiconductors, was patented in 1968 but largely remained a concept until the 2010s. Switches made from wide-bandgap semiconductors such as silicon carbide could handle far higher voltages than silicon, and the EV industry's appetite for these materials began driving the cost down just as a hungrier customer arrived.

"If we are going to have data centers, we want them to be as efficient as possible," says Srdjan Lukic, a professor of electrical and computer engineering at North Carolina State whose lab, at the university's FREEDM Systems Center, built the Lenox, Mass., prototype. Solid-state conversion would mean smaller footprints, fewer conversion stages and less copper. First seeded nearly two decades ago with federal funding, the center has become a global brain trust for solid-state electronics and spun off companies such as DG Matrix, a Raleigh, N.C., start-up that says it's been selling the technology to data centers.

"There's a lot of pushback" against data centers, including in North Carolina, Lukic says. The faceless, hulking boxes have become a proxy for the whole argument about AI—its water use, its demands on power and land, the question of who benefits from it all. "If we can reuse the floor space and existing land and grid connections that have already been allocated to data centers, we can do more with less."

NYPA has provided some funding for the Lenox prototype as it tries to meet rising load forecasts for data centers, EV chargers, and more in a state where power is among the costliest in the country. New York households now pay roughly 70 percent more per kilowatt-hour than in 2019, against a national rise of under 40 percent. The most recent long-term assessment by NERC, the grid-reliability watchdog, projects summer peak demand rising 224 gigawatts over the next decade, a forecast 69 percent higher than

the one it published a year earlier, with data centers accounting for most of the increase. “There are some ambitious companies—they’re going to use AI in everything,” Elmoudi says, “and we are not ready yet for that.”

Some data centers are now being built at gigawatt scale—roughly the power demand of a small city, says Anuja Ratnayake, emerging technology executive at the Electric Power Research Institute. And then there are larger ones, such as the campus in Pike County, Ohio, where OpenAI said in August it agreed to ultimately take eight gigawatts of capacity over a six-year buildout. That “is bigger than one of the largest cities that we serve.”

A conventional transformer steps voltage up and down, but it does not actively shape or route the power passing through it. A solid-state transformer offers control. Grid current passes through silicon carbide transistors that switch at high frequency before sending it through a small conventional transformer, preserving the essential electrical isolation between the grid and everything downstream.

On the far side, another bank of electronics reassembles the power as alternating current or direct current at the voltage the load wants. Because the switching happens thousands of times a second, the device can hold voltage steady, absorb a disturbance or operate in reverse, pushing power backward from batteries toward the grid.

Moving key stages from AC to DC also shrinks the footprint and can cut losses by a few percentage points, which sounds trivial until you scale it.

This is where SSTs are most useful, says Haroon Inam, DG Matrix’s CEO and co-founder. They aren’t likely to solve the legacy transformer shortage anytime soon or replace the large AC-to-AC transformers used to send power over long distances. Their value is closer to the load, at the edges of the grid. “The real problem,” he says, “is that the grid, which is based on centralized

generation and unidirectional power flow, has a constraint in providing a large amount of power at the end of transmission or distribution feeders.”



DG Matrix’s Flex unit is designed to do the work of several pieces of conventional power equipment in one semiconductor-studded box.
DG Matrix, via @Karamijt Singh, Chief Delivery Officer of DG Matrix

Launched in 2022, DG Matrix put its first pilot in the field, a charging system in North Carolina, in April 2024. The next month Nvidia called. “When the data-center people realized that they had a limitation in using more power in data-center racks, they looked at the EV industry to say, ‘What has already been commoditized and built up?’ ” Inam says.

The limitation is physical. Nvidia’s road map calls for pumping a megawatt into a single rack. Data centers typically run on AC power, and at existing distribution voltages, the current that would be required would demand impossibly thick cabling—“something that resembles a roadway of copper,” Inam says—and add more heat. “You can’t physically fit that in.”

So Nvidia and some of its largest customers proposed a new design, skipping several conversions and distributing power inside the building not through typical AC but through DC, the native

form the chips want anyway. In June, Nvidia, Google and Microsoft published a draft specification through the Open Compute Project for the solid-state transformer that would slot into that architecture, converting the grid's medium-voltage AC—thousands of volts—straight to 800 volts DC, a number inherited from the supply chain built for EV fast charging. Although the spec stops short of requiring an SST, more than 80 manufacturers and infrastructure providers are now building to meet those requirements, according to Nvidia.

While DG Matrix develops a medium-voltage device, one that's capable of connecting to the grid, it's selling a smaller SST designed to sit on the customer's side of the meter, downstream of the utility's transformer. For developers managing a mix of AC and DC loads, Inam says the device collapses a number of machines—including a rectifier, an inverter, backup power and energy management—into a single unit four feet square and 12 feet high. The company has logged more than 100,000 noncontinuous run hours in the lab and in the field, with 10 of its Flex units "at multiple sites," he says.

More savings, Inam says, come from pooling power sources. With multiple ports, DG Matrix's units can link grid power, batteries and on-site generation into a microgrid, redirecting power to server racks or other parts of the system and allowing data centers to start operating sooner, without a grid connection. Ultimately what customers want is the "most tokens for the cheapest amount of power and the cheapest infrastructure buildout." The tokenomics has fueled a surge of investment. Since late 2025 DG Matrix has raised $60 million; Singapore-based Amperesand, $80 million; and California's Heron Power, $140 million. Eaton bought a solid-state transformer company outright; giants such as Schneider Electric, Delta Electronics and Hitachi Energy are all circling.

The larger hope goes beyond efficiency. Paired with on-site energy storage, SSTs could turn data centers from blunt-force loads into

more flexible participants in the grid. Murali Baggu, who leads the grid-integration program at the National Laboratory of the Rockies, says technical challenges remain and standards are still in flux. But he wrote in an e-mail that “these advanced interfaces can transform data centers from potentially disruptive loads into active contributors to grid stability.”

The problem, Ratnayake says, is not only how much power data centers want but how abruptly they swing between consuming it and dropping it. On the morning of July 22, a transmission line fault in Ashburn, Va., the heart of “Data Center Alley,” prompted data centers to switch to backup power, pulling roughly 3,800 megawatts off the PJM grid, which serves 13 states and Washington, D.C., causing voltage to spike and lights to flicker as far away as Chicago. The synchronized GPU workloads of AI training runs can also whipsaw power demand by hundreds of megawatts in seconds, swings that “pose a significant threat to the stability of the utility grid,” Nvidia said in a blog post last year. And that risk was “making grid interconnection a primary bottleneck for AI scaling.”

A device that can route power can also respond to what the grid requires. Paired with battery storage and control software, a solid-state transformer can combine the work of multiple devices to act as a buffer or shock absorber, helping to decouple the chaotic appetite of AI chips from the utility’s need for stability.

That flexibility, whether enabled by SSTs or other controls, could lower costs and speed up grid connections. A 2025 Duke University study found that curtailing large loads such as data centers by just 0.5 percent of annual electricity use could make room for nearly 100 gigawatts of new load without new power generation. More operators are now exploring “non-firm interconnection,” accepting less than perfect service in exchange for connecting years sooner.

EPRI is developing flexibility frameworks with partners such as the Open Compute Project and Nvidia while a parallel effort validates hardware, including SSTs, against strict grid codes. Data-center operators, Ratnayake says, need to start asking, “How do I design my data center differently?”

Utilities are also experimenting with deals meant to keep the costs of new data centers from spilling onto everyone else’s bills. In northern Indiana, an approved agreement with Amazon is expected to use revenue from new data centers to fund grid upgrades and return credits to other customers. “That’s the kind of creative thinking we need,” Ratnayake says, “to ensure this revolution won’t create an unnecessary burden on the rest of society.”

The tech needs far more testing, too. Fault currents, which can fry components and cause dangerous arc flashes, are one challenge, requiring new protection measures and solid-state circuit breakers. As power passes through the miniaturized magnetic core, the insulation that electrically isolates the medium-voltage grid from everything downstream must survive hundreds of thousands of thermal cycles without once failing. “That’s what everybody’s worried about, whether they say it or not,” Inam says. “And then you want it certifiable.” Semiconductors also fail in ways iron does not, and a box with thousands of switching components has thousands of things that can go wrong on a Tuesday.

Still, amid the commercial rush, few will reveal what they have learned from field tests and pilots. “The deployments are never public, and the results are never in the public domain for these demos,” Lukic says. “So we don’t really know as a community where the technology is, what the problems are.”

Federal programs are pushing from the other end. Oak Ridge National Laboratory is designing solid-state power substations; ARPA-E is funding research into converters for high-voltage DC corridors. Johan Enslin, an ARPA-E program director, envisions a

grid in which power can be converted, directed and managed automatically—much like the way data are routed around the Internet—rather than merely stepped up or down. The data-center buildout is “an opportunity to now say, ‘Hey, we need this stuff,’” he says. “What I do worry sometimes is that we’re just going to build the same old stuff we’ve done the past 50 years for substations, for transformers.”

Lukic, whose lab continues to scale up its medium-voltage unit, assumed the killer application for SSTs would be EVs. Instead the race to force more power into one kind of semiconductor is driving demand for another. That could ease pressure on conventional transformers, unlock new efficiencies and make it easier to add renewables to the grid. For the booming data-center buildout, that could be a “happy accident”—that is, if the new machines prove as dependable as old-fashioned iron and steel.

Alex Pasternack is founding editor of Motherboard, Vice Media's tech-culture website, and a contributing editor at *Fast Company*, where he covers technology and science.

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https://www.scientificamerican.com/article/why-data-centers-are-reviving-solid-state-transformers

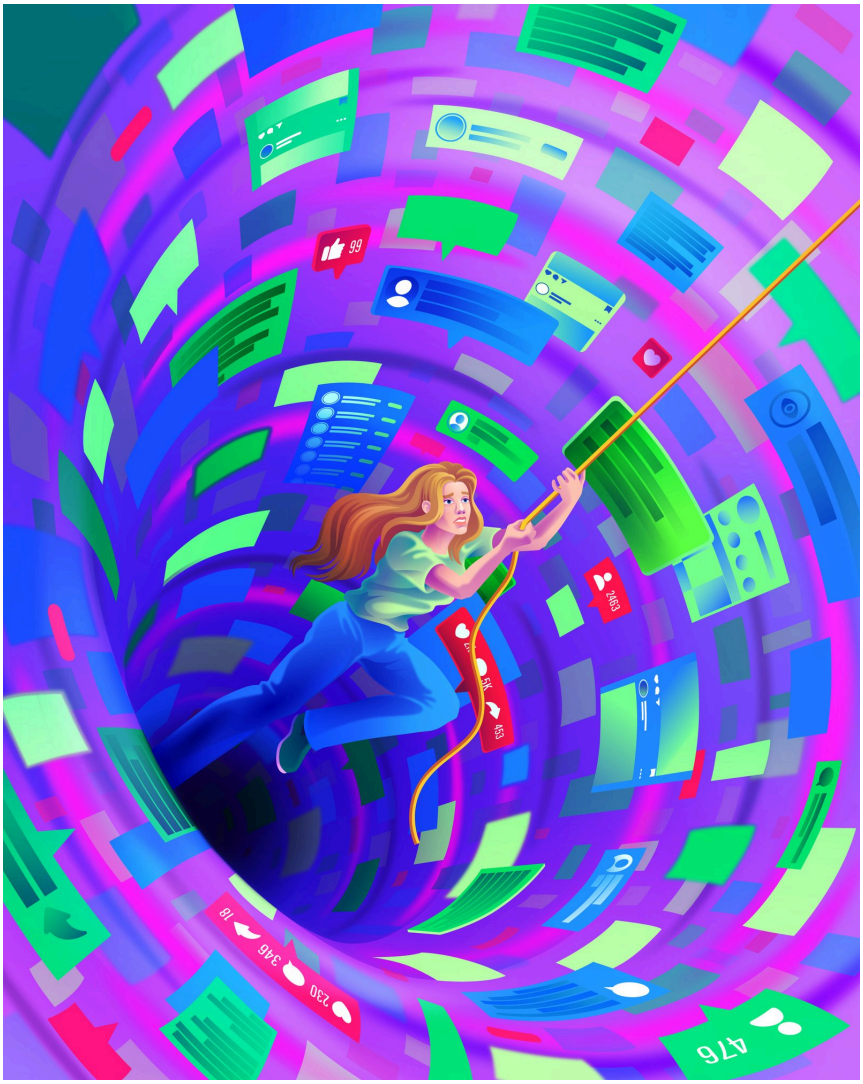
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We’re addicted to social media. How do we get sober?

How doctors, scientists and “anonymous” programs are working to free people from their phones

By Allison Parshall edited by Seth Fletcher



Ari Liloan

In 2022, when Tom was 32, he went to extreme lengths to defeat his tech addiction. He flew from his home in Michigan to a town outside Stockholm and gave up his devices for a three-month, self-

directed “dopamine detox.” He did cold plunges, worked out and hiked—all without looking at his phone.

It was a grand display of his willpower over a technology that had taken over his life. What had started as regular media binges in college had spiraled out of control. His sleep schedule and relationships deteriorated, his obsessive-compulsive disorder (OCD) worsened, and he had begun to think of suicide. “I was just constantly in a battle for my sobriety,” he says. “Literally any imaginable thing, I tried it. And nothing worked.”

The Swedish detox didn’t work, either. Now that he’d felt the serenity of sobriety, he was sure that he could return to social media and video games in moderation once the three months were up. Why would he ever choose to go back? “But then of course, I found myself, as we say, ‘on my back’: on YouTube or on the Dead by Daylight video game, just bingeing, bingeing, unable to stop,” he says.

Cover story Allison Parshall talks about what motivated her to write this story.

Eventually Tom found something that helped him get sober. At his most desperate point in 2025, he found two 12-step programs: Internet and Technology Addicts Anonymous (ITAA) and Media Addicts Anonymous. “Once I joined 12-steps, what happened was I learned how to let go,” he says. The advice he found in recovery programs’ “Big Book” allowed him to stop white-knuckling his sobriety, calm his OCD symptoms and take back his life.

I first attended an ITAA meeting in Manhattan on a sunny Saturday afternoon in March. I listened, intrigued and uncomfortable, as I compared people’s stories of compulsive screen use with my own. That week I had averaged nearly seven hours a day on my phone, outside of work. That was fine, I thought. I’d gotten really into a television show that winter and spent a lot of time chatting and creating with other fans on social media. Sure, I’d often go on

TikTok for a “quick” scroll and get spat out hours later, hating myself for it. Sure, the first thing I’d do in the morning was turn on a podcast. But I didn’t have a problem. Right?

The first of the 12 steps is admitting that your relationship with your vice has become unmanageable. This is true in “anonymous” programs for addictions to alcohol, opioids and gambling and to technology such as social media. If our society as a whole could follow the Big Book, this is the stage we’d be at right now: collectively, we are beginning to admit that we have a problem.

> “I would wake up at 10 A.M. just feeling such shame, but I couldn’t stop it.” —Meg *ITAA Member*

In March a landmark ruling in a California state court deemed social media addictive and, for the first time, held tech companies accountable for those harms. Meta, facing lawsuits from nearly every state in the U.S., agreed to a massive $18-billion settlement in August that requires it to change how kids can interact with Instagram and Facebook. The list of countries that have banned these apps for children under a certain age—Australia, Turkey, Indonesia, the U.K.—continues to grow.

“We’ve clearly crossed a Rubicon,” says lawyer Matthew P. Bergman, founder of the Social Media Victims Law Center. And although scientists remain split on whether we should call all of this unhealthy social media use “addiction,” their research is starting to reveal who is most at risk of harm and how we can regain control of our digital lives.

How can we enforce a balance between the good and bad of online connection? In other words, how do we get sober? “I think this is the question of our time,” says psychology researcher Lina Christin Brockmeier of the University of Basel.

To Meg, a 56-year-old ITAA member from Wisconsin, Facebook isn't that different from alcohol. She has been sober from alcohol since November 2, 1996, but when Facebook became available a decade later, there were signs that it could become her next addiction. Before her wedding in 2008, her family and friends warned her not to join yet. "They actually came to me together and said, 'Meg, you have to wait until after the wedding to join this,'" she says. "'We don't want you to put down your wedding decorations!'"

By 2018 she was solidly hooked. She posted about every aspect of her life, obsessing over the font and color of the text adorning her photographs, staying up all night and sleeping until 10 A.M. This left her husband to get their two children with special behavioral needs ready for school alone, which she saw as her responsibility as a stay-at-home mom. "I would wake up at 10 just feeling such shame, but I couldn't stop it. I did it over and over," she says. Decades after putting down alcohol, "this just felt like the same kind of addiction."

Psychiatrists call these "behavioral" addictions. They don't involve chemical dependence, but they can still get people hooked. In Washington, D.C., ITAA shares a meeting space with 12-step recovery programs called Sex and Love Addicts Anonymous, Debtors Anonymous and Overeaters Anonymous. With the exception of gambling disorder, behavioral addictions have received slim recognition in psychiatry's diagnostic manuals—although doctors and researchers often still treat them as legitimate and potentially devastating disorders.

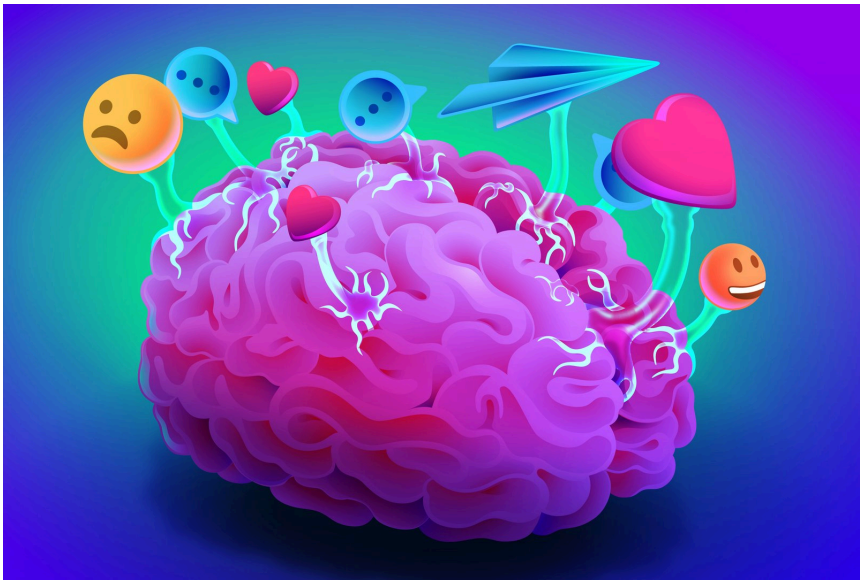
Even in the Internet's infancy, there were signs it could become the site of a new kind of behavioral addiction. On a 1997 Internet addiction questionnaire, respondents are asked if they have "spent at least 3 h on the Net at least twice." Internet addiction, however, never became an officially diagnosable condition. Researchers found it too imprecise: you get addicted not to the Internet itself but

to the things you do on it. But now, as everything online morphs into algorithmic social media, the case for concern has crystallized into something quite specific. “I don’t think ‘Internet addiction’ goes anywhere,” says Luke Clark, a psychologist at the University of British Columbia. “But when we start considering social media addiction, it becomes a tractable question.”

Clark, who studies gambling addiction, sees modern social media as engrossing in the same way as slot machines. Their main similarity is a variable reward system: every swipe could net you a reward or a disappointment, and that payoff occurs at unpredictable intervals. The rewards also come almost instantaneously, can happen in quick succession, and don’t contain any natural cues to stop because of the “infinite scroll” feature, he adds. These features of slot machines and For You pages pique the interest of neural circuitry originating deep in the brain, called the mesolimbic pathway. Here brain pathways tuned to salient information such as reward and uncertainty pass messages using the chemical dopamine that make us feel good.

We evolved this system so that we will seek out information important for survival, and activating it is “the shared feature that all addictive drugs have,” says Paul Kenny, an addiction neuroscientist at the Mount Sinai Icahn School of Medicine. Although different drugs or addictive behaviors cause different patterns of activation across the whole brain, all of that activity ultimately funnels down to this deep-brain reward system, he explains. In a world flooded with hedonic products and possibilities, from cigarettes to junk food to bite-size media, it’s easy to throw this reward system out of balance.

Researchers often characterize addiction with six core aspects. It involves a *salient* behavior or drug that causes *mood modification* effects, which you develop *tolerance* to over time, leading to a cycle of *withdrawal* and craving-induced *relapse* that can cause *conflict* in your life.



Ari Liloan

This pattern of escalation is the hallmark of addiction for doctors and clinicians. Just using something excessively shouldn't count—otherwise, you'd risk diagnosing people with healthy high enthusiasm. "These are people who spend as much time gaming as they do working in their life, but they're not addicted to games," says Mark Griffiths, a behavioral-addictions researcher retired from Nottingham Trent University in England. But our everyday definition of addiction is much more lax. We may use the word to hyperbolically refer to anything we do more than we probably should, like saying "I'm so addicted to *Love Island*," says Angela Lee, a communications researcher at Stanford University.

In the slew of social media addiction lawsuits brought against tech companies, including Meta and Google, plaintiffs' lawyers are generally using a more colloquial version of the term "addiction." "When I started doing this work [in 2021], we used the word but understood it to be a bad habit," says Bergman, who leads the Social Media Victims Law Center. Still, Bergman says he's come around to believing that these products cause widespread addiction in the more severe, clinical sense, too.

Yet these lawsuits don't rest on proving that social media is addictive like cigarettes or gambling. Instead plaintiffs' lawyers

have focused on what tech companies knew about their products' capacity to hook kids and how they proceeded regardless. State attorneys general alleged that Meta covered up internal studies on the impact of Instagram on teenagers' mental health. In court, lawyers read out loud from internal documents saying that "teens are hooked despite how it makes them feel. Instagram is addictive." They also quoted other Meta documents stating that "problematic use" should be used instead of "addiction" because of the "legal risk" involved with that word. Instagram "is a drug," wrote one Meta employee in a work chat uncovered in court. The reply: "We're basically pushers."

Meta avoids the term "social media addiction" because medical authorities do not recognize it as a diagnosis, a spokesperson told me. They added that the company is continually working to reduce problematic use. As part of the $18-billion settlement, Meta has agreed to turn off the Likes feature for teen accounts, as well as cap screen time at two hours a day, disable notifications during school hours and give teens the option of switching to a nonpersonalized feed. The company is explicitly incentivized to lobby its peers, Google and TikTok, to do the same.

The trials have focused on children and teens not only because they are particularly vulnerable to harm but also because they are a valuable demographic of users for social media companies. Researchers at Harvard University estimated that Instagram alone generated $4 billion in revenue from users age 13 to 17 in 2022—and $802 million from users 12 and younger, despite them not technically being allowed on the platform. One internal study from Meta revealed in the court cases was entitled "Long Term Retention: The Young Ones Are the Best Ones." In the words of a 2019 Meta presentation: "Teens talk of Instagram in terms of an 'addict's narrative' spending too much time indulging in a compulsive behavior that they know is negative but feel powerless to resist."

A few years ago most of the patients at Stanford University's youth addiction clinic were there for substance-use concerns. That has changed, says the clinic's director, psychiatrist Bradley Zicherman: "Here we are in 2026, and I'd say about half of my referrals are actually for screen addiction."

Other teen treatment centers have been flooded with tech-addicted patients, too. "The main thing that I was seeing was a lot of really young, bright people being sucked into this way of being that was really dampening their development as human beings," says psychologist Nicholas Kardaras, who has run a tech-addiction program in Austin called Omega Recovery since 2018.

Adolescence is an exquisitely sensitive time in the brain's development. "The brain is so sensitive to social cues, to the sense of belonging, to rewards, to peers," says neuroscientist Eva Telzer, who co-directs a tech and teen development center at the University of North Carolina at Chapel Hill. In the past few years she and her colleagues have gotten a close look at how problematic social media use shapes the developing brain. For a 2024 study, she and her colleagues spent two years following 103 sixth and seventh graders on the cusp of puberty. They assessed students' addictionlike social media use at each check-in, then scanned their brains while they completed a task that simulates receiving a social punishment or reward.

> "The fuller your life becomes, the less you want to escape it."
> —Nicholas Kardaras *Omega Recovery & New York Center for Living*

Telzer expected the kids' brains to become increasingly activated by social feedback over the course of puberty. In those who used social media less addictively, that's what happened. But in those who used it more addictively, social reward activity peaked *before* puberty's onset. It then decreased over time in a pattern that mimics increasing tolerance to a drug. "That reward becomes less and less

potent over time," Telzer says. "So you might need more and more of it to get that boost."

In another, forthcoming study, Telzer and her colleagues found signs of tech withdrawal in the brains of late adolescents. Forty-five college students with heavy social media use volunteered to have their phones locked in Telzer's laboratory for 24 hours. Before and after, they lay in a brain scanner and watched 10 minutes of TikTok videos. Their brain states changed markedly. After, networks that process emotions and internal states were activated longer, and a network that supports executive functioning was activated for less time. "It basically resembles what we would see in substance-use withdrawals," Telzer says.

But the parallel has one intriguing difference. When the researchers ran the study again and asked about mental well-being, students reported feeling better in the phone's absence. They had higher self-esteem, lower anxiety and fewer negative feelings overall, unlike what you would see in withdrawal from addictive substances or behaviors. "Kids now report, in pretty equal measure, both positive and negative [emotions] from their time on social media," says psychologist Mitch Prinstein, a collaborator of Telzer's at U.N.C. Chapel Hill. "They like the stimulation, but then they feel more lonely. They enjoy being entertained, rather than bored, but then they feel bad about the amount of time they spent on there that felt out of their control."

This year in Austria and four other countries, a massive experiment showed just how much appetite young people may have for change. A program funded by Austria's Federal Ministry of Education asked students to decrease or outright eliminate their social media use for three weeks. "We thought that maybe 1,000 or 2,000 [students] would take part," says Oliver Scheibenbogen, a clinical psychologist at Sigmund Freud Private University in Vienna. Instead more than 45,000 students participated in this "Handy Experiment" ("Handy" is a colloquial term for a cell phone in

Germany). Researchers gathered data on students' mental well-being before and after the intervention in March. Scheibenbogen reports that the kids who cut out social media entirely showed modest increases in mental well-being that started wearing off five weeks later. Most students stuck with it for the full three weeks. "It makes me very proud of the students," he says. At the same time, he says, "I see this as a cry for help from young people."

Chris joined his first ITAA meeting at a very low point in his life in February 2024. "I've been in bottoms like that before, but this one was particularly bad," he says. He would stay up until dawn watching and scrolling on multiple devices; he would stream TV shows holding up his phone in the shower. He was 35, and he'd lost about a decade of his life to the deep unhappiness and anxiety of active addiction. "Something had just *really* broken inside me and said, you've got to try something extreme," Chris recalls. (We're using a pseudonym to protect his privacy.) So when he binged a show that featured a 12-step recovery program, he decided to seek one out himself. In that first online meeting, he bawled his eyes out. Then he got to work.

Whereas 12-step programs typically aim for total sobriety from the problem behavior or substance, tech-addiction fellowships have to be more specific in their definition of "sober." They have participants identify their "top," "middle" and "bottom" lines—online behaviors that are healthy, risky or flat-out dangerous, respectively. Bottom lines are often described as "the behaviors that once you start, you can't stop, and once you stop, you can't stay stopped." In fellowships such as ITAA, bottom lines are what you try to cut out. Chris's are: streaming TV shows by himself, scrolling social media, playing video games, watching porn, joining chat rooms and using dating apps.

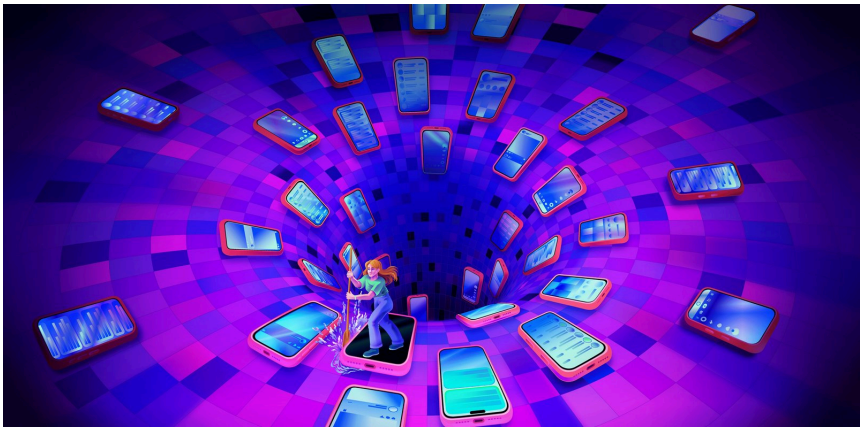
Twelve-step recovery programs are free and accessible, making them a good (and often the only) option for many people. But doctors and therapists who treat drug and behavioral addictions

have been steadily shifting away from sobriety to emphasize gradual progress and a reduction of harm instead. “I don’t think abstinence is a realistic goal for a lot of people,” says Joshua Grubbs, who studies behavioral addictions at the University of New Mexico. He notes that, especially when it comes to behavioral addictions such as those to sex or technology, focusing on sobriety can cause fixation and a cycle of shame.

Instead modes of therapy that help people reduce black-and-white thinking, such as dialectical behavioral therapy (DBT), are beginning to show promise in treating addictions. DBT, plus a four-week device “detox,” is Kardaras’s go-to for treating people of all ages in his practice. This type of therapy “is really good for tech addiction because the core principles are distress tolerance, emotional regulation and mindfulness,” he says—all things that are often absent or eroded for people with tech addictions.

Most of the work of recovery is about treating whatever is causing people to escape their lives into endless scrolling and stimulation, Kardaras says. “Look at what’s making the person feel empty,” he says. “Whether it’s substances or screen time, it’s usually an inside job.” For Chris, the source was religious trauma. Growing up in a tightly controlled religious community, he was forbidden from watching many kinds of movies and TV shows. Leaving the faith “kind of sent my life tailspinning,” he says. But a new world of entertainment that he could use to distract and numb himself was suddenly available to him.

In recovery, he has developed a relationship with his past self. “Now I can look back and be like, ‘Okay, 19-year-old Chris, I also like you. I want to have you as part of my narrative,” he says. “That’s what sobriety has enabled me to do: sit with myself, be with myself, and be okay with being a messy, messy human. And I think because of that, I just feel so much more alive in the world.”



Ari Liloan

Chris calls this shift a “deep inner replacement” of the role of media in his life. Kardaras aims for something similar with his patients. “That’s what recovery looks like: having a full, meaningful life, where you recreationally might use your digital device but not compulsively use it,” he says.

Building this life is the gargantuan task facing people looking to fix the role of technology in their lives. Chris has started playing music again and going to open-mic nights. Meg took up voice lessons, despite always singing off key. Tom, who always felt like he was wearing a mask around other people, learned to show others his authentic self and found that people gravitate toward him in a way they didn’t before. As Kardaras says, “The fuller your life becomes, the less you want to escape it.”

Even though I have what I consider to be a pretty full life, the TikTok app sits there on my phone like a threat. Maintaining a full life can be hard. It requires energy I do not always have, especially for socialization, which I find exhausting. “I’ve come to almost understand algorithms as being heat-seeking missiles that identify psychological vulnerability,” Kardaras says. And so many of us have a vulnerability in common: the human need for connection, along with the various reasons that it might be easier to look for it on social media than to seek out the real deal. “It makes sense that people turn online when they’re coping with isolation,” says Jeffrey Hall, who researches relationships and technology at the University

of Kansas. “The problem is, by and large, that is not a terribly effective means of coping with loneliness.”

For one thing, social media has gotten far less social since the advent of engagement algorithms and infinite scroll. For another, relying on the Internet for company can hurt us in the long term. Our social skills and opportunities can atrophy. Maybe they never get built in the first place. “Social behavior is like an exercise,” Hall says. “If you get out of the habit of doing it, you’ll become less effective at doing it.”

Hall doesn’t think social media is the root of our current epidemic of loneliness. The amount of time we spend alone has been increasing since at least 1965, a recent study found. In a 2017 experiment, he found that social media mostly displaces time we would have spent watching other kinds of media. But it certainly may exacerbate our existing loneliness and tendency toward isolation, he says.

Ashley Gearhardt, who studies and treats food addiction at the University of Michigan, points out that social media fulfills our base need for socialization in the same way that ultraprocessed products fulfill our need for food. “It looks like something we need to survive, like social affiliation or calories,” Gearhardt says. But it has been deconstructed and reassembled into something optimized for maximum consumption and the producer’s profit, she says. It’s also hardest to resist in the times we need that willpower the most —don’t go to the grocery store while hungry, and don’t scroll while lonely.

For this reason, Gearhardt emphasizes compassion. “You are in a system that is fundamentally mismatched with your biology,” she says. “This isn’t because you just lack willpower, and ‘if you’d only tried hard enough ...’ Fundamentally the system is rigged to make it hard to make healthy choices.”

Most people are probably not addicted to social media in the way Chris, Meg and Tom experience it. Tom describes himself as uniquely susceptible to this kind of stimulation, something that Kenny, the Mount Sinai neuroscientist, says may have some credence. “I do wonder if certain individuals may have a dopamine system—and other systems in the brain—wired in such a way that they’re really vulnerable to this type of stimulus, this type of reward,” Kenny says. For the rest of us angsting over our screen time, the products may simply be working as intended: capturing our attention and keeping it with sleek design and endless feeds and becoming harder to resist at more challenging times of life.

I relate to Mathilde, a 26-year-old from San Francisco who lives in New York City and feels like she’s in a constant battle against her screen time. “Short-form videos have the ability to let you kind of turn your brain off for a little bit,” she says. She’s tried many screen-blocking apps, but after a while, they lose their authority, and she starts bypassing the blocks. Mathilde doesn’t know whether she’d call it an addiction, but as she scrolls, she feels her brain becoming more unreasonable. “I enter a headspace where I’m not able to think clearly.”

And during challenging periods in our lives, we may become more like Jack, a recent high school graduate from a rural town in upstate New York. “I had a really rough summer” before senior year started, Jack says. “I was averaging about 18 hours of screen time a day. Which is way more than I ever expected.” Like many people his age, he uses mainly Instagram, TikTok and Snapchat, he says: “On TikTok, it’s just a quick flick, flick, flick, flick, flick....”

Feeling like we’ve lost control over our behaviors is part of what makes it feel like an addiction. But California Institute of Technology neuroscientist Ian Anderson, who studies social media habits, says automatic behaviors can also just indicate an ingrained habit. If you want to cut down on your screen time, it may be more helpful to think of it as a habit you can shape rather than an

addiction that needs to be treated by a doctor. “Framing it as a habit not only offers a less pathologizing term but also offers research-based solutions for how to change your behavior,” he says.

Habits are usually paired with physical or emotional cues, and disrupting those cues can disrupt the habit. If you tend to scroll at night, for example, bedtime has become a trigger; you’re better off leaving your phone outside of the bedroom. You can also change where the app is on your home screen, remove the shortcut entirely or turn off notifications, Anderson adds. Plus, replacing scrolling with a different activity you’d look forward to can also make it easier to stick with your new normal.

There’s also a big difference between active and passive screen time, researchers have found. Passive screen time, especially with personalized feeds, can be a mind-numbing time suck that makes you feel less in control. But there are many ways to be more active during the time we spend online. “We can actually change what we’re doing *while* we’re scrolling to make it a more values-aligned behavior,” says Alison Tuck, a digital health researcher at Washington University in St. Louis. She suggests setting intentions for screen time that align with your values: finding connection, gaining knowledge or expressing creativity, for example. Keeping these in mind while you scroll can keep you feeling more in control, she says.

“Every time you get on technology, you have two paths: one is going up to the top of the mountain to see the sunset, and the other is going down a rabbit hole,” says Larz May, founder of a teen-led digital-wellness organization called #HalfTheStory. She’s made it her life’s mission to give people, especially kids, the resources they need to avoid harmful rabbit holes; Jack pulled himself out of his social media binge by joining the organization’s Teen Tech Council. It’s an uphill battle to go from the “doom loops” of algorithmic feeds to “joy loops” that enrich our lives, May says: “We have to fight for the joy loops.”

> “The stronger the technology, the more protection we humans need.” —Aza Raskin, *Center for Humane Technology*

In Kardaras’s office at the New York Center for Living, a teen and young adult addiction-treatment center in Manhattan, I mentioned that I sometimes reach for the word “addiction” to describe my dissatisfaction with my own ballooning screen time, even though I am objectively not addicted—I’d just gone on a five-day vacation where I delighted in the excuse to not look at my phone at all.

“I don’t have any trouble stopping,” I said, a walking cliché.

He looked at me doubtfully. “You said there’s a dissatisfaction. What does that look like?”

I sputtered a little. I talked about finishing work mentally exhausted, telling myself I’ll just scroll for 15 minutes on TikTok to refresh myself before I make dinner—then getting up an hour later with even less energy than when I started. I know it will happen, but I do it anyway, and I can’t really stop once I’ve started. Yet at the same time, my online life is one of the most creative and sociable parts of my world; it often *does* refresh me. This screen contains some of the best parts of me, even as it threatens to bring out the worst.

“I don’t think that it’s absolutely a terrible way of being,” Kardaras said slowly, referring to my seven-hour-average daily screen time. “But this can be a rabbit hole.” If my life were a little bit different, he said, I might fall deeper in that hole. “We all have foundational tethers, things that keep us grounded: family supports, career things. But if you don’t have those things, then you’re kind of adrift.”

Kardaras continued, “That’s the part that they’re fishing for. They’re looking for the low-hanging fruit—the person that’s just kind of bored and scrolling and has a lot of self-loathing.”

Aza Raskin had a part in creating the slick social media feeds that engross us today. Microsoft patented infinite scroll in 2006; Raskin, then a tech developer, independently developed a similar version that same year. Now he lobbies against his creation and testifies in court to its addictive potential. “One of the best ways of subverting human free will is by appealing to the parts of ourselves that get addicted,” says Raskin, who co-founded the tech watchdog group Center for Humane Technology. Those vulnerabilities will always be there, and technology will only get better at finding them. “The stronger the technology, the more protection we humans need,” he says.

In our moment of reckoning, most of the current momentum is going toward enacting social media bans for kids younger than 16. Some experts believe this is a good start, but many see it as wholly inadequate. “It’s not changing that cesspool of what social media is like,” says S. Bryn Austin, who researches technology and youth well-being at the Harvard T. H. Chan School of Public Health. “It’s just saying, ‘Oh, for your 16th birthday, we’re going to throw you in!’ ” The new changes to Instagram following Meta’s settlement may have similar limitations because they apply only to teen accounts and have minimal impact on the algorithms themselves.

Austin and her colleagues have created sample legislation for the kinds of laws they would like to see instead. Such laws would require third-party audits of social media algorithms to identify any harmful patterns, such as serving eating-disorder content to young girls more than to other groups. “They shouldn’t have that bad content pushed at them when they’ve never asked for it,” says Nancy Costello, a law professor at Michigan State University who helped to develop this model legislation.

Because these laws don’t touch the content on the sites—only adjust how social media companies deliver them—they may not run afoul of the First Amendment or a statute called Section 230, which have stymied previous efforts to hold tech companies

accountable for what they serve to users. Recent court cases have used similar arguments, framing social media to judges and jurors as defective products causing children harm.

Bergman, the attorney whose law center helped to bring social media addiction cases to trial for the first time, can hardly believe the moment we're in now. "If you had told me, told *anybody* in the tech law space in the fall of 2021 that Meta and YouTube would be facing a jury for allegedly harming kids, they'd have said you're crazy," he says. "I felt like we had won the day we started the trial because that was the first time a social media company had ever been held accountable."

Meanwhile we're all still here with our devices, trying to find a relationship with them that adds to our lives rather than diminishing them. "I see connection and disconnection a bit like the tide. There's ebb, and there's flow; they can't exist without each other," says Mariek Vanden Abeele, an associate professor of digital culture at Ghent University in Belgium. "I see [managing] this as kind of a new responsibility that we did not have to consider beforehand."

Meg has been sober from her social media bottom lines for just over five years now. We spoke the day before her birthday, and in years past, she would have spent the entire day online responding to a flood of birthday messages. "There was a fear, for me, of getting [offline]," she says. That fear of missing out would drive her to use Facebook far more than she even wanted to. Now she's swapped FOMO for another outlook that she first heard about in an ITAA meeting: Instead of FOMO, Meg says, "we have the joy of missing out."

Discussions
We're looking to host the most interesting science conversations on the web for this topic.

Allison Parshall is associate editor for mind and brain at *Scientific American* and she writes the weekly online Science Quizzes. As a multimedia journalist, she contributes to *Scientific American*'s podcast *Science Quickly*. Parshall's work has also appeared in *Quanta Magazine* and Inverse. She graduated from New York University's Arthur L. Carter Journalism Institute with a master's degree in science, health and environmental reporting. She has a bachelor's degree in psychology from Georgetown University.

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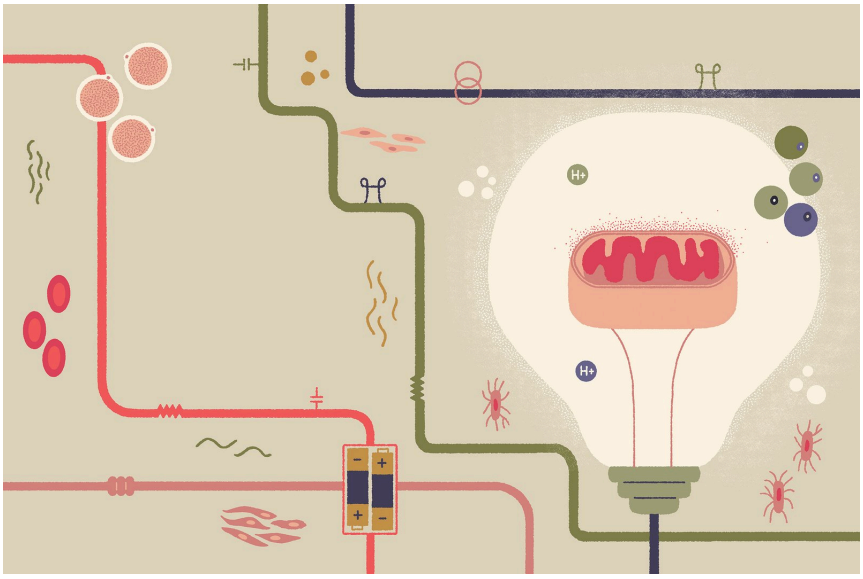
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The provocative new theory that could explain aging and disease

Could disruptions in the flow of energy through mitochondria cause everything from type 2 diabetes to Alzheimer's?

By Rowan Jacobsen edited by Eric Sullivan



Noemi Fabra

As diseases go, type 2 diabetes and epilepsy seem to have little in common. One is a metabolic disorder involving high blood glucose and failing insulin, the other a matter of neurons and electrical pulses. So it is odd that both often respond to fasting. Likewise, depression, obesity, Alzheimer's, heart disease—all are wildly different diseases, yet all are influenced by diet. This shared vulnerability points to metabolism—the way we harvest energy from the food we eat and use it to power our cells.

One of the leading proponents of an energy-centric view of disease is Martin Picard, a biologist who directs the Mitochondrial Psychobiology Group at Columbia University. "Once we see

ourselves as energetic processes," he says, "many things make sense that didn't make sense from a molecular perspective."

Picard and a growing number of colleagues are making the big claim that many chronic diseases we treat as separate failures of separate organs may share one deeper problem: cells struggling to extract and use energy from food. If they are right, a common weakness runs beneath diseases now parceled out among different specialties, and medicine might have spent a century drawing boundaries in the wrong places.

At the heart of this argument lie mitochondria, the microscopic organelles that live in our cells and convert the energy stored in molecules of food into adenosine triphosphate, or ATP, the molecular power pack that drives virtually every reaction in the body, whether it's moving, thinking or healing. We produce our body weight in ATP every day, and we burn just as much.

Any meltdown in our mitochondria will lead to a shortage of ATP, leaving cells with less energy to do their jobs. Picard believes many states of poor health start there, and his once unusual research focus is now mainstream. "Everybody agrees that mitochondrial dysfunction is part of aging biology," says Matt Kaeberlein, former director of the Healthy Aging and Longevity Research Institute at the University of Washington. "It's one of the hallmarks of aging. And I think you can make a pretty reasoned argument that mitochondrial dysfunction can precede and potentially drive many of the other hallmarks of aging."

> "Mitochondrial dysfunction may contribute to the vast majority of common diseases." —Matt Kaeberlein *University of Washington*

Certainly, Kaeberlein says, health is about more than mitochondria. "If you're going to argue that *everything* starts with mitochondrial dysfunction, I think most people would push back against that."

But the idea doesn’t need to explain all of aging to be worth chasing. “Mitochondrial dysfunction may contribute to the vast majority of common diseases,” he says. “Almost any disease you look at, you find a mitochondrial connection.”

That connection has gerontologists and biohackers alike racing to understand what makes mitochondria tick—and what we can do to keep them ticking as long as possible.

Although mitochondria’s central role in our existence is often overlooked, scientists have understood their importance since the 1960s, when British biochemist Peter Mitchell proposed a then radical explanation for how mitochondria capture energy—work that earned him the Nobel Prize in Chemistry in 1978. The process, Mitchell showed, is fundamentally electrical. Electrons stripped from the atoms of our food are drawn toward atoms that will accept them. Among the elements biology has to work with, oxygen exerts the strongest pull.

Animals owe their existence to that attraction. We eat food, break it down into simpler molecules and send those to the mitochondria in our cells, which pry off electrons and let them flow through a string of molecules in their membranes known as the electron transport chain. At each step in this chain, a portion of the electrons’ energy is siphoned off to pump protons across a membrane—Mitchell’s great insight—and the pressure of those protons rushing back through a molecular turbine is what makes ATP. At the end of the chain, oxygen accepts the electrons and, together with protons, becomes water.

As many observers have noted, this flow of electrons through the body resembles the flow of electricity through the power grid. Power plants drive electricity through wires and into our homes, where our devices tap that energy to light our rooms and power our lives. The current has to meet the right amount of resistance in the

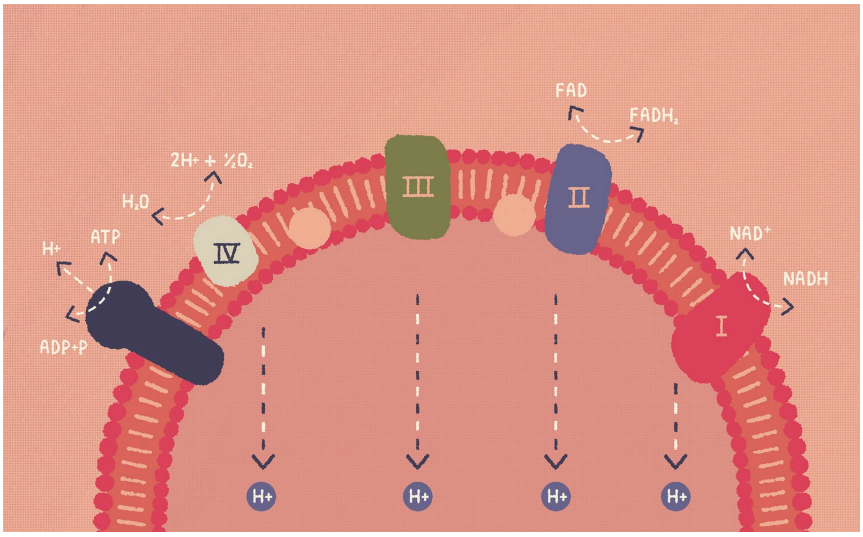
right places. Too little, and the energy races past without doing any work, eventually damaging other components.

Something similar happens in our mitochondria. Under ideal conditions, electrons are slowed in their passage through the electron transport chain just enough that their energy can be transferred. But if the mitochondrial circuits are impaired or if there are simply more electrons from food pouring into the system than the mitochondria can handle, the flow can get bogged down. Electrons can then leak out of the chain, bonding with oxygen in the wrong parts of the mitochondria to form highly reactive free radicals that attack and deform neighboring molecules. These reactive oxygen species, or ROS, can do real damage to mitochondria, cells and organs.

No matter how smoothly electrons flow through mitochondria, a certain amount of ROS is unavoidable in everyday life. The classic free-radical theory of aging, first proposed in the 1950s, held that aging itself resulted from the relentless buildup of ROS in our cells, decade after decade, and the wear and tear this caused. Now we know the body is quite capable of repairing manageable amounts of ROS, which at lower levels can even act as signals, telling a cell to get to work. Only when the oxidative stress gets out of hand do mitochondria get dinged up, but when they do, they move electrons less efficiently, which leads to even more ROS and more damage. The brownouts spread. Eventually cells lose capacity, and the body follows.

Picard has written in this magazine about mitochondria’s role in health. In November 2025 he and Nirosha Murugan, a biophysicist at Wilfrid Laurier University in Ontario, codified this concept in the journal *Cell Metabolism* as the energy resistance principle. If a major cause of chronic disease is simply too many food-derived electrons frying the mitochondrial wires, then a major goal of medicine should be to reduce that overload. “The secret to

longevity," Picard says, "is to find ways to live at optimal energy resistance."



In mitochondria, the electron transport chain helps to convert energy from food into ATP, the molecule that powers our cells.
Noemi Fabra

These disruptions in cellular energy can have very different consequences throughout the body. "The specific pathology depends on where the mitochondrial dysfunction is happening," Kaeberlein says. "If it's in the muscle, you get muscular degeneration, myopathies, things like that. If it's in the heart, you get different types of heart disease." The brain is no exception. "You get things like Parkinson's disease. Certainly mitochondrial dysfunction contributes to other types of dementia, including Alzheimer's." The Alzheimer's link is strong enough that Paula Moreira, who is co-editor in chief of the *Journal of Alzheimer's Disease* and specializes in brain mitochondria, has taken to calling the disease "diabetes of the brain."

The claim that mitochondrial dysfunction is a factor in mental health disorders as well as physical ones is surprising to many people. It's less intuitive to picture energy supply affecting our mind as it does our muscles, but the brain is the most energy-hungry organ in the body. Neurons are closer to Formula One race cars than to family sedans, burning fuel at a furious rate and

threatening to blow up at any moment. If any cells are going to betray signs of energetic stress, it's them.

As mitochondrial dysfunction has become a familiar suspect in disease and aging, efforts to tinker with it now range from a cottage industry of supplements, such as coenzyme Q10, that promise happier, healthier mitochondria, to experimental interventions involving gold nanoparticles. That worries Jonathan Jarry, a science communicator at McGill University's Office for Science and Society in Montreal. "My concern is that people start to believe that this has been successfully translated into the clinic when it hasn't," he says. "So it's one thing to say, 'Hey, here's what we know about this thing that's in our cells. It's really cool. People are studying this.' And it's another to say, 'Hey, we know the problems that this causes, and we know how to fix this, and we can sell you solutions.'"

Kaeberlein agrees that the hacks have thus far disappointed. "None of the data so far looks all that compelling," he says. Of all the supplements suspected to have mito-boosting powers, he says only one is convincing. "The one I would point to is creatine. There's a clear mitochondrial connection, and the biology is very plausible. I think it's a very good supplement. It works in the context of also doing exercise, but there's emerging evidence on creatine and brain function, at least in situations like sleep deprivation or severe cognitive stress. You can buffer the ATP supply of the brain and muscle with creatine."

For now the boring advice has better evidence than the hacks: sleep, exercise and diet.

Sleep is extraordinarily important to mitochondrial health. That may be a big part of what drives our need for it. Neuroscientists at the University of Oxford have recently found direct links between oxidative stress in the brain and the need to sleep. As neurons fire furiously over the course of the day, ROS can accumulate.

Specialized neurons track that buildup, and when a certain threshold is reached, they send a sleep signal to the brain at large. During sleep, neurons go quiet, ROS are cleared, and damaged mitochondria are repaired or recycled.

Exercise also improves the capacity of mitochondria to process energy, in part by stress-testing the system. Any kind of workout is a temporary emergency for the cells involved. Energy demand might rise to 100 times resting levels, which pushes mitochondria and their support systems to the limit, producing a burst of ROS and other stress signals. During recovery, though, the body responds to this challenge so it will be better prepared for the next one, fixing damaged tissue and building more blood vessels and mitochondria. The payoff is a system that can handle more energy without straining the rest of the time. The leg muscles of a marathoner can have twice as many mitochondria per cell as those of a couch potato, spreading the energetic load across much more capacity.

The system also gets better at repair, which may turn out to be a more important part of healthy aging than we've realized. Exercise triggers a process known as autophagy, the cell's salvage system for worn-out parts, including mitochondria. Damaged material gets dismantled; useful components go back into circulation.

Recycling also seems to help explain another old standby: calorie restriction. "Autophagy is an adaptive response to nutrient deprivation," Kaeberlein says. "When you're in nutrient shortage, you don't have the energy to produce a bunch of new complex machinery, so you break down the damaged machinery, and then you can reuse those components. And the most important time to recycle is when you don't have sufficient nutrients to build new machinery."

> These disruptions in cellular energy can cause very different consequences throughout the body.

That efficient scavenging can also clear damaged components before they become a liability. At the same time, reducing the amount of energy entering the system eases the load on mitochondria and reduces oxidative stress. Calorie reduction reliably improves health and lifespan in many organisms, including yeast, worms, flies and rodents. In primates, the results aren't as clear, although long-running studies point to health benefits.

Diet-based interventions have shown promise for the brain as well as the body, a connection physicians made long ago when they noted that fasting could often prevent seizures in people with epilepsy. "Hippocrates knew about it 2,000 years ago," says Christopher Palmer, a Harvard Medical School psychiatrist and author of the 2022 book *Brain Energy*. "In the 1920s physicians put it to the test. And to their shock, it actually worked."

We now know a lot about the mechanisms of action, Palmer says, including "changes in neurotransmitter function, changes in hormone signaling, changes in epigenetics, reduced brain inflammation and changes in mitochondrial function." Those changes seem to be linked to metabolic shifts, among them reductions in the load of glucose, the body's standard fuel, which is produced mostly from carbohydrates. A ketogenic diet can reproduce fasting's metabolic effects by strictly limiting carbohydrates and relying more on fat. When carb intake falls sharply, the liver turns fat into ketones, an alternative fuel for the brain and other tissues that may cause less metabolic stress. Whether this broader metabolic shift can improve psychiatric symptoms is now being put to the test.

In July a team led by researchers at the San Francisco VA Medical Center published the first randomized controlled trial of a ketogenic diet in people with bipolar I disorder and schizophrenia-spectrum disorders. After a month, the ketogenic group showed clear metabolic improvements; the more striking psychiatric and cognitive gains turned up at four months—by which point the diet

had been offered to everyone, eliminating the control-group comparison. The clinical signal remains intriguing but preliminary.

“We’ve known for decades that a ketogenic diet is effective for antiseizure,” Kaeberlein says. “It definitely has effects on brain metabolism that are potent. That’s going to affect mitochondrial dysfunction. So it’s reasonable to think a ketogenic diet will have broader-reaching effects on brain biology that could affect emotion, psychological state, cognitive abilities, all of those things.”

Kaeberlein cautions that it is too early to assume that we understand how these incredibly complex systems work. “The arrows all point in the direction that repairing and preserving mitochondrial function during aging is important for the lifespan and health span benefits we see from these interventions. But because all of these things are multifactorial, it’s hard to know how much is because of their effects on mitochondria versus other things that these interventions are doing.” The wrinkle is that mitochondria live inside the cell and depend on it, so anything that improves their function will improve the cell, but anything that improves the cell will also improve them.

On their face, none of these interventions are revolutionary. The appeal of the energy-centric perspective is that it gives familiar advice—eat lightly, sleep well, get exercise—a common language. “The disease-centric point of view looks at a singular snapshot and then looks backward to see how that applies to health. The energy-resistance framework flips that,” Murugan says. “It reframes health as a dynamic process, and it allows us to look at how different interventions can influence that process. It puts them into context.”

A good example is type 2 diabetes. “From a molecular perspective,” Picard says, “diabetes is excess blood sugar and insulin resistance. So you fix it with drugs. You give insulin to override the resistance.” He reads insulin resistance differently. In some cases, he argues, cells may begin to resist glucose because

their mitochondria are already struggling to process the fuel coming in. In that telling, they need a break, Picard says, not an override.

An energy-centric perspective may also offer early warning signs. Cells under stress release a signaling protein known as GDF15. “It’s the way cells tell other cells, ‘I’m not doing well,’” Picard says. High levels of GDF15, which also rise in pregnancy and illness, trigger feelings of malaise and nausea—a built-in brake on energetically costly activities such as movement and digestion. He thinks the body is then able to conserve energy while confronting the internal challenges. High levels are also among the best predictors of future disease and mortality, which he takes as strong evidence that energetic dysfunction within cells precedes later problems—and may offer us an early opportunity to head them off.

But beyond such practical applications, the biggest appeal of seeing health through the lens of energy may be the new light it shines on the very concept of disease. “Once we see ourselves energetically, it’s going to change how we teach and practice medicine,” Picard says. Rather than fixing parts piecemeal after they break down, we can ask what keeps us all happily humming along in the first place. In a sense, we are all energy benders, channeling an endless flow of what was once sunlight through the labyrinth of our molecules and gracefully shaping it into this thing we call life.

Discussions

We're looking to host the most interesting science conversations on the web for this topic.

Rowan Jacobsen is a journalist and author covering science, nature and the environment. His work has appeared in *The Atlantic*, *Harper's*, *The New York Times* and *MIT Technology Review.*

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https://www.scientificamerican.com/article/the-provocative-new-theory-that-could-explain-aging-and-disease

Why we probably live in a multiverse

The once-fringe quantum Many Worlds theory now seems much more likely

By Philipp Strasberg edited by Clara Moskowitz



Chiara Morra

Quantum mechanics, which recently turned 100 years old, ranks as the most successful theory in the history of physics. Yet on a fundamental level, we still don’t understand what quantum mechanics is saying about how the world works.

Any standard textbook will tell you that the dynamics of quantum mechanics are based on two axioms—mathematical rules that allow us to theoretically compute the outcome of an experiment. Surprisingly, these two axioms are self-contradictory. They present one of nature’s deepest mysteries: the quantum measurement

problem—the fact that the quantum rules describing an object to be measured don't seem to apply to the object doing the measuring. This disturbing contradiction suggests that different physical laws apply to the observer and the observed.

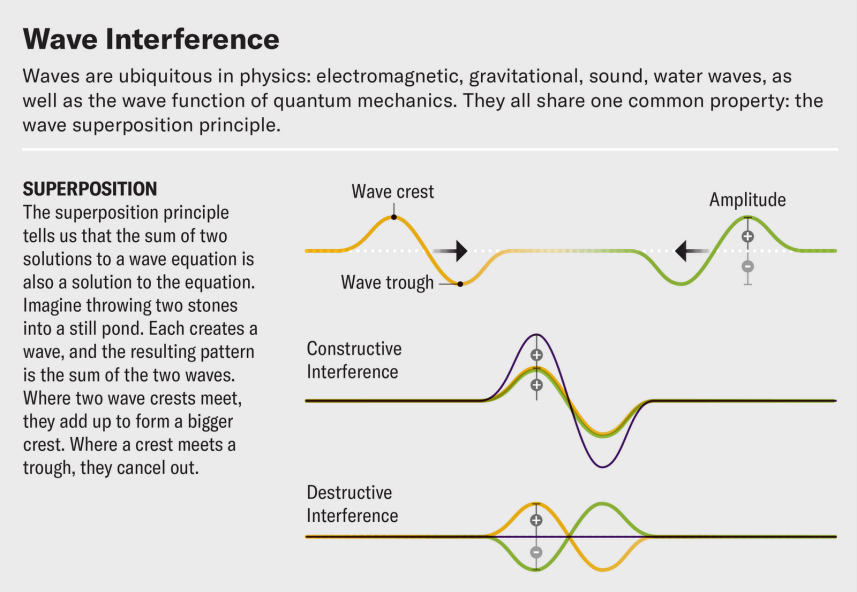
One proposal to resolve the paradox is the Many Worlds interpretation of quantum mechanics. Many Worlds demands that our universe actually be a multiverse. The idea is often considered fringe, even silly. Lately, however, experimental progress has been changing the situation. Scientists have been able to measure quantum effects in larger and more complex systems and over great distances, blurring the line once thought to separate the scale of observer and observed. These signs that quantum rules might cross over into the macroscopic realm are making many physicists reconsider Many Worlds. But what does this notion really mean for the universe we call home?

The first of the two contradictory axioms says quantum systems are described by a wave function called Ψ (or psi, pronounced "sigh"), a mathematical quantity that encodes all the information about a quantum system, such as the position, momentum and spin of all its particles. The wave function evolves over time in a way that's dictated by Schrödinger's equation, named for Austrian physicist Erwin Schrödinger. Like Isaac Newton's law in classical mechanics or Albert Einstein's field equations in general relativity, Schrödinger's equation, which applies to only isolated systems, is the fundamental formula from which every calculation in quantum mechanics begins. In contrast to Newton's and Einstein's laws, however, this equation has a very peculiar property: it is a wave equation, and its solutions must obey the superposition principle.

Superposition is an uncanny aspect of quantum mechanics that allows particles to be in multiple states at once—such as literally here and there simultaneously. This property means that if Ψ1 and Ψ2 are both solutions of Schrödinger's equation—they are two possible states of the quantum system—then the sum Ψ1 + Ψ2 is

also a solution. A key aspect is that if you start with a state in a superposition, it remains in a superposition for all time: superpositions cannot be destroyed.

To appreciate the consequences of superposition, we must understand how waves are added and subtracted. Imagine you are standing in front of a pond with a still surface. Throw a stone into the pond, say, at position P1, and observe the resulting wave pattern. Wait until the surface is still again, then throw in a second stone at another position, say, P2, to watch the resulting wave pattern. Now imagine that you throw the stones simultaneously into the pond at positions P1 and P2. What will you see?



Matthew Twombly

If you watch carefully, you will find that the result is just the sum of the two patterns that each single stone created. If a wave crest from the first stone meets a wave crest from the second, they will combine to form a bigger wave crest, but if a crest and a trough meet, they will cancel each other out. These processes are called constructive and destructive interference, respectively. In this way, multiple solutions to a wave equation can be added together, and the result will still be a solution of the same equation. And it doesn’t stop with two solutions—if you want the wave pattern of three stones, you just combine the three patterns, and so on.

In most cases, wave equations describe an approximation of reality. Light that's sufficiently bright and traveling through matter doesn't obey the superposition principle. Close to shore, water waves break and violate the superposition principle. But Schrödinger's equation obeys the superposition principle without exception.

The second axiom of quantum mechanics is called the measurement postulate. Imagine we want to measure the position of an atom. To make things simple, let's assume that the atom can exist only at two different positions, P1 and P2, and that the wave functions corresponding to these positions are $\Psi 1$ and $\Psi 2$, respectively. If the atom is described by $\Psi 1$, measuring the atom will reveal position P1, and if it is described by $\Psi 2$, our measurement will reveal P2. So far, so good. But if the atom is described by the superposition $\Psi 1 + \Psi 2$, the measurement will randomly select one of the two possible positions as if the other part of the wave function did not exist. It would be as if all the waves created by the stones thrown into the pond suddenly disappeared except for the wave pattern of one particular stone. This phenomenon is called the collapse of the wave function, and the resulting physical picture is known as the Copenhagen interpretation of quantum mechanics.

You might wonder what is so disturbing about this interpretation—after all, it sounds like a coin-flip experiment, in which a result of either heads or tails is unknown until you look at the coin. But the quantum situation is much worse.

Recall that Schrödinger's equation describes the dynamics of atoms and other microscopic particles such as electrons or molecules. Clearly, any tool we use to measure the atom's position is also made up of atoms. We should be allowed to apply Schrödinger's equation not only to the atom we want to measure but also to the detector we use to measure it (and everything else in the laboratory).

> A substantial part, if not all, of the randomness in the world is of quantum origin.

Here's the trouble. We know that superpositions can't be destroyed because Schrödinger's equation obeys the superposition principle. If Ψ1 + Ψ2 describes the initial state of the atom, then after the measurement the entire lab must be in a superposition corresponding to Ψ1 + Ψ2, thus describing the atom at position P1 or P2 and the detector in a state of having measured P1 or P2, respectively. There can be no collapse according to Schrödinger's equation.

So if the measurement axiom is right—and it has been well confirmed empirically—we are forced to accept the following conclusion: the basic physical law, Schrödinger's equation, that we use for the system we measure cannot be applied to the system with which we measure. It's as if there exists a shifty split between the observer and the observed, even though both are made out of the same components of matter.

In 1957 Hugh Everett III, then a Ph.D. student in physics at Princeton University, proposed a radical solution to the quantum measurement problem: drop the measurement axiom as superfluous. What would that mean?

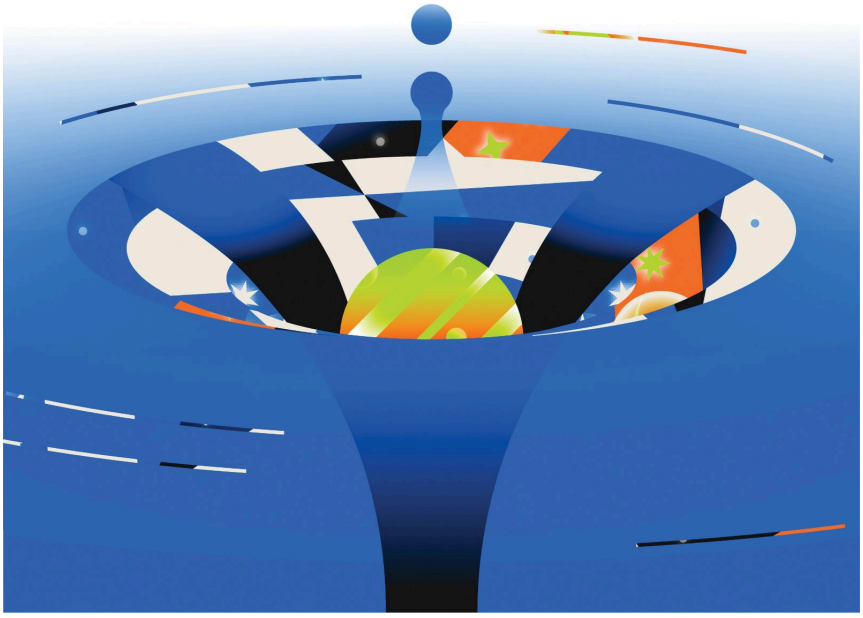
For one, there would no longer be any bizarre divide between the measured system and the measuring system: detectors would be just ordinary physical objects abiding by the superposition principle, just as atoms do. Schrödinger's equation would be universally valid, and the wave function would never collapse.

The consequences of this effect, however, are enormous. Going back to our earlier example, when we measure an atom in a superposition of two positions Ψ1 + Ψ2, the final state after we make the measurement will be a lab that is in a superposition of the atom being in two different positions *plus* a detector in a

superposition of having measured either of the two positions *plus* an experimenter in a superposition of having seen either of the measurement outcomes of the detector. In other words, there would suddenly be two experimenters in the same lab, each of them an exact copy of the experimenter before the measurement except for the tiny difference that they now know the position of the atom: for one of them the atom is at P1, and for the other it is at P2. Yikes!

The situation gets even more bizarre. Theoretical arguments suggest that a substantial part, if not all, of the randomness in the world is of quantum origin. Why? Well, although superpositions can't be destroyed, they can spread from one quantum system to another almost like a virus. We saw this idea already when we noted that the original superposition of the atom resulted in a final superposition of the combined atom-plus-detector-plus-experimenter system.

To see how this spread creates randomness throughout the world, consider an ordinary coin-flip experiment again. There is nothing truly random about whether a coin comes up heads or tails. If we could flip the coin in the exact same way every time using a carefully crafted mechanical device, it would always produce the same outcome, which could be predicted according to Newton's law. But if you try it yourself, you'll quickly notice that it's very hard to control the flight of the coin, and you get unpredictable outcomes. One reason is that a human hand can't flip the coin in exactly the same way every time: tiny variations in the nerve pulses traveling from your brain to your thumb cause variations in your flip strength, which in turn produce different outcomes. But if you could trace things back to the origin of the tiny nerve-pulse variations, you would get to even smaller scales at which quantum effects become relevant: a superposition of different electron states could cause these slightly different nerve pulses. Thus, different coin-flip outcomes could be the result of amplified quantum fluctuations at tiny scales.



Chiara Morra

But if Everett's proposal is true, then each time you flip a coin, you might be creating two copies of yourself: one who sees heads and one who sees tails. The result is a giant cosmic-scale superposition $\Psi 1 + \Psi 2 + \Psi 3 + ...$ that grows like branches on a tree. The universe becomes a superposition of different universes, one for each branch of the wave function. Put differently, the universe is actually a multiverse.

For that reason, Everett's proposal is known as the Many Worlds interpretation of quantum mechanics. It's confusing, however, to call it an "interpretation." The Many Worlds interpretation and the Copenhagen interpretation are not really two different interpretations but two different theories. The former predicts the existence of a multiverse as an inevitable consequence of the superposition principle of Schrödinger's equation. The latter predicts a single universe by denying the universal validity of Schrödinger's equation.

Which theory is right? Is Schrödinger's equation universally valid, or does it apply only to microscopic systems such as atoms?

We don’t know the answer yet, and for a long time it seemed that we might not need to care too much. Many of the pioneering quantum experiments in the early days could be explained with a modified classical theory. Light and matter can be “quantized” if we think of them as composed of small particles (photons or electrons, for instance) that can also behave like waves—a concept known as wave-particle duality. An explicit solution of Schrödinger’s equation was seldom required. Many well-known breakthroughs related to quantum mechanics—Max Planck’s computation of black-body radiation, Einstein’s explanation of the photoelectric effect, Niels Bohr’s idea to explain the spectral lines of the hydrogen atom, and the existence of the electron spin revealed by the work of Otto Stern and Walther Gerlach—preceded the discovery of Schrödinger’s equation.

Later it became clear that all these different effects follow from Schrödinger’s equation. Experimental evidence for the equation, however, remained restricted to the subatomic domain for many decades. Some wondered whether Schrödinger’s equation might be an approximation that works extremely well at the microscopic scale but breaks down at the macroscopic scale, which would mean the measurement postulate of the Copenhagen interpretation is right.

But experimental progress over the past few decades shows that Schrödinger’s equation is not valid only for minuscule particles. The double-slit experiment, for example, is a classic setup where scientists send particles through two gaps in a screen and observe a resulting interference pattern, proving that the particles act as waves. In 1999 a group led by Austrian physicist Anton Zeilinger confirmed that fullerenes, large molecules consisting of 60 carbon atoms, behave just the same. The current record, from 2019, proves molecules with 2,000 atoms show this same quantum property. Meanwhile, a different kind of experiment has demonstrated that clouds of thousands of trapped atoms can be cooled down to low-enough temperatures that exotic quantum mechanical phases of

matter become visible, such as a state called a Bose-Einstein condensate. The wave nature of quantum mechanics as predicted by Schrödinger's equation has even been shown at centimeter scales in superconducting devices.

> Although the multiverse sounds like science fiction, it is a real scientific prediction.

And in a very recent experiment, quantum optical signals in a superposition of different states of light were communicated between South Africa and China over a distance of more than 12,900 kilometers. Because light obeys a superposition principle even under classical physics, this experimental breakthrough doesn't imply that Schrödinger's equation necessarily applies to objects 12,900 kilometers wide. Nevertheless, taking all these results together, researchers who still believe the wave function must collapse at macroscopic scales appear to be walking on thin ice. Schrödinger's equation might be here to stay.

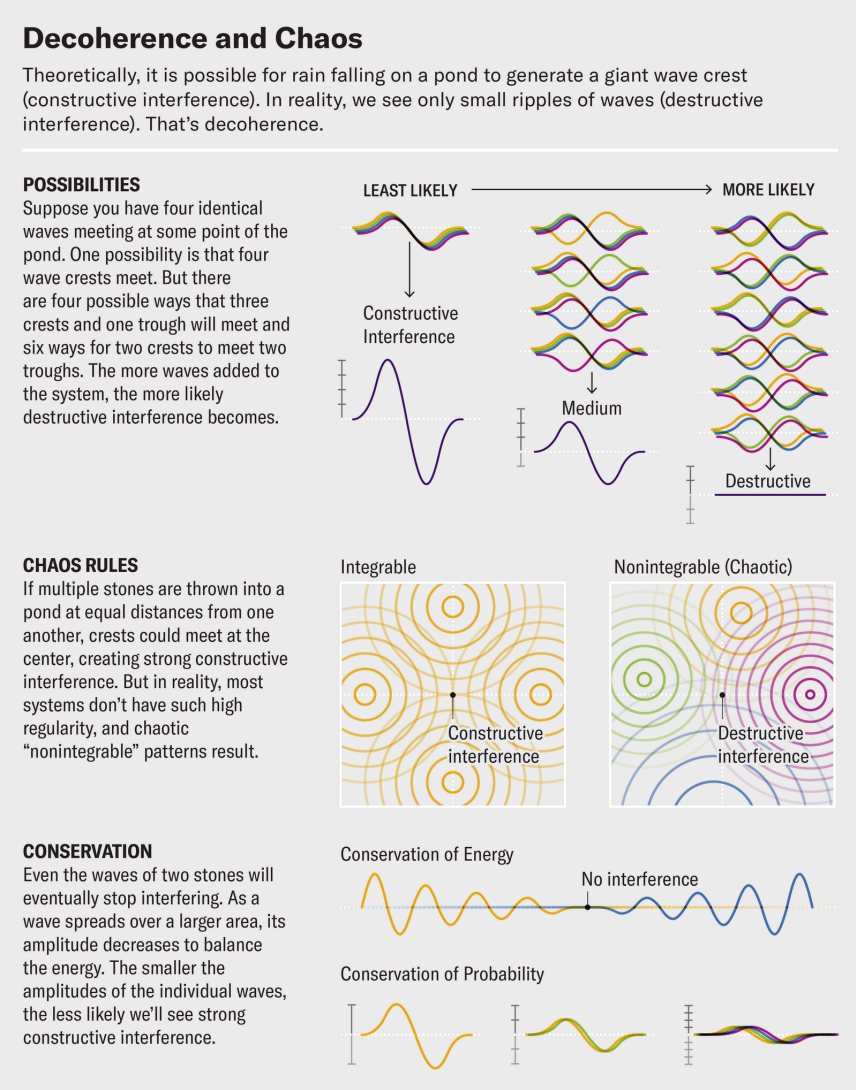
If we assume that Schrödinger's equation applies universally, have we then solved the quantum measurement problem? Unfortunately not, because another issue is lurking around the corner. If everything is described by the wave equation, why do so many things behave like classical particles? Why can't we interact with other branches of the multiverse? If something as simple as a coin flip can create a macroscopic superposition, why don't we see any interference between heads and tails?

The answer is somewhat counterintuitive at first: we actually do see interference effects all the time but only from destructive interference. This is what physicists call decoherence—the inability of many waves to show strong constructive interference.

Note that decoherence is not a new, fundamental law that we add to Schrödinger's equation to obtain classical behavior. It is a consequence of Schrödinger's equation that arises naturally once

there are many interfering waves. In fact, decoherence is a universal phenomenon that we're already familiar with. Think of rain falling on a pond. Even though it is theoretically possible for all the droplets to generate a giant wave crest through constructive interference, in reality we see small ripples everywhere.

One way to understand why decoherence happens is to compare the number of possibilities for constructive and destructive interference. Suppose you have four identical waves meeting at some point of the pond. There is one possibility that only wave crests will meet at this point, and the resulting wave will be four times bigger than the individual ones (constructive interference). There are four possible ways for three crests and one trough to meet, producing a wave that's twice the size of the originals (we could call this medium interference). And there are six possible ways for two crests and two troughs to meet, causing the waves to completely cancel out and produce a flat surface (destructive interference). Repeating this argument for more than four waves reveals that there are many more possibilities for destructive than for constructive interference.



Matthew Twombly

Let's look at the numbers. Water waves caused by, say, 50 stones thrown into a pond have a wavelength of a few centimeters and oscillate with a frequency of around 10 hertz (Hz). In comparison, a quantum mechanical object that is visible to the naked eye contains around 10^{23} particles, each of which has a wavelength less than one billionth of that of a water wave and a frequency a few billion times larger than a water wave's.

Now recall how slow and coarse our human senses are: movies show us only 24 images per second (that is, movies have an image frequency of 24 Hz), and we perceive them as continuous motion. We barely notice that the 0.1-millimeter-wide pixels on our computer displays are discrete. If there are 10^{23} particles interfering with one another on such tiny and fast quantum scales, no human

has a chance to see constructive quantum interference. In fact, numerical simulations of Schrödinger's equation have recently shown that with each additional particle, decoherence becomes roughly twice as strong. This means experiments with only 50 atoms need detectors sensitive to relative deviations of about 0.0000000000001 percent to distinguish quantum from classical effects. That's like trying to measure a 100-meter Olympic race on a femtosecond timescale.

To be fair, these estimates are true for most, but not all, quantum many-particle systems and only if we focus on properties that are easy to measure, such as the coarse and slow stuff humans can perceive. If we precisely tuned a quantum system, we would see that the quantum effects remained. But doing that requires an enormous amount of control and precision—like trying to juggle 50 balls with your eyes closed during a hurricane. This is why it's so incredibly hard to build a working quantum computer.

In our example of the pond, simply comparing the number of possibilities assumes that each stone emits a wave independently. In this case, strong constructive interference could be generated by a precisely tuned pattern of stones. For instance, identical stones thrown in a perfect circle onto the pond would create strong constructive interference when all the waves met in the middle of the circle. But natural systems lack such a high degree of regularity. Real stones will have different masses and shapes, each generating a slightly different wave. Physicists call such systems nonintegrable or chaotic. In these systems, the absence of regularities makes it extremely unlikely that all waves will interfere constructively.

You might wonder how anything interesting can arise from quantum mechanics at all if destructive interference is so prevalent. The answer is that there are some cases where quantum waves interfere constructively and give rise to a pattern familiar from classical physics. Here the analogy with water waves isn't quite

right, but another analogy familiar from everyday life helps. For many years people believed that light was made out of particles flying in straight rays. In most cases, this description is a good approximation because the wave nature of light is revealed only when it interacts with objects whose size is comparable to the wavelength of light (around 500 nanometers).

Under typical circumstances, therefore, light seems to follow a straight line. This effect arises from constructive interference in the direction of the ray and destructive interference outside it, where the light is “dark.” You can picture the quantum multiverse as a collection of light rays flying through a vast, dark space. Each ray describes a distinct macroscopic world. The vast darkness between rays doesn’t correspond to the absence of waves—it arises from destructive interference among many waves.

Although the multiverse sounds like science fiction, it is a real scientific prediction if we assume that Schrödinger’s equation, the fundamental law governing our microscopic world, also applies to the macroscopic world. Right now we can’t say for sure, and we have no perfect test of the question. It’s not clear whether this assumption is true, nor is it clear how to test it experimentally. But there are indirect consequences of the Many Worlds hypothesis that could lend the idea credibility.

For instance, the cosmic microwave background astronomers see throughout space is currently well explained by the idea of inflation, which posits that the early universe was dominated by a quantum field called the inflaton that rapidly stretched space. If physicists can assume that quantum mechanics was applicable to the entire cosmos at early times, shouldn’t it be applicable to the full universe today as well? Or are there other mechanisms coming into play?

Another clue is the fact that the universe appears highly fine-tuned: a slight variation in its fundamental parameters, such as the precise

charge of an electron, the mass of a proton or the strength of the gravitational constant, would make life as we know it impossible. How can we explain the fact that these values seem almost chosen for their convenience for life? A popular answer is known as the anthropic principle: the fact that we are alive means we must find ourselves in a version of the universe compatible with life. We have essentially selected this universe from the array of all possible universes, which may be mostly hostile to life. But what does "select" mean? From which "selection of universes" are we choosing? Here the answer strongly depends on whether there is only one unique universe or a vast collection of them.

Many practical questions ultimately depend on which scales Schrödinger's equation is valid at. Quantum computers' power, for instance, relies on their ability to put large quantum systems composed of many particles into well-controlled superpositions. If that is impossible for systems much larger than atoms, many of humanity's current technological efforts would be wasted.

The question also has consequences for the quest to unify quantum mechanics with general relativity. Finding the right unifying theory strongly depends on which "interpretation" of quantum mechanics is right. Do we need to quantize gravity, or can it be described classically at macroscopic scales?

The Many Worlds interpretation is gaining momentum as a candidate to solve the quantum measurement problem. Despite its seemingly weird prediction that our world is just one part of a multiverse, because of destructive and constructive interference, it is perfectly compatible with our sensation of living in one classical universe. Moreover, it opens possibilities for addressing fundamentally and technologically relevant questions.

Still, many challenging open questions must be answered before the Many Worlds interpretation can be accepted as our best explanation for the measurement problem. One such puzzle

concerns the origin of uncertainty, chance and probability, which are so central to quantum mechanics. What does it mean that an event appears with a certain probability if all possible events are always realized simultaneously? And how do the specific quantum probabilities that we observe in experiments arise in our branch of the multiverse?

Currently not even the strongest proponents of the Many Worlds interpretation agree on how to answer these questions. For now the nature of our universe—or universes—and the quantum rules that guide it are as undecided as the toss of a coin.

Discussions

We're looking to host the most interesting science conversations on the web for this topic.

Philipp Strasberg is a theoretical physicist at the Cantabrian Institute of Physics in Spain, where he studies quantum statistical mechanics, emergent phenomena and foundational questions.

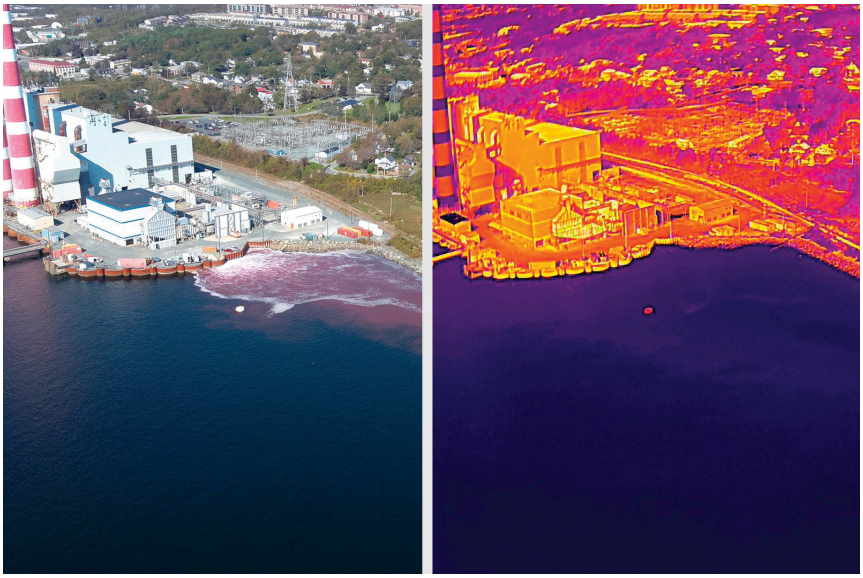
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https://www.scientificamerican.com/article/why-we-probably-live-in-a-quantum-multiverse

Inside the bold, quixotic quest to store planet-warming carbon in the ocean

These start-ups are trying to suck CO_2 from the air and stash it in the sea. Is it working?

By Alec Luhn edited by Eric Sullivan



At a power station on Halifax Harbor, researchers use pink rhodamine dye to trace the path of alkalinity-treated water. A thermal view reveals temperature differences in the harbor.

Ocean Alk-Align Project

On a drizzly morning in March 2025, a bright blue dive boat motored through Halifax Harbor in Nova Scotia toward the Tufts Cove generating station. The boat rumbled past the power plant's three smokestacks, striped red and white like candy canes, to a yellow buoy bobbing in the waves.

Canadian company Planetary Technologies had been adding an alkaline slurry to the generating station's cooling water before it rushed through a concrete tunnel and into the harbor near the buoy. The added alkalinity is meant to shift dissolved carbon dioxide into

more stable forms, drawing more of the gas out of the air. The question was how much.

The captain thrummed the engine to steady the vessel against the incoming tide. Dalhousie University oceanographer Dariia Atamanchuk lowered the team's "carbon cage," a rectangular frame of wires and sensors, into the blue-green waves and checked the water's CO_2 concentration on her laptop. "It is just slightly below atmospheric," she said.

Phytoplankton could have produced that small deficit as they grew. The nearby Sackville River could have delivered low-carbon fresh water from the spring melt. Planetary's alkaline slurry could also have begun to alter the water. "It's hard for us to measure that signal because it's so small," Atamanchuk said.

That ambiguity is the central problem facing marine carbon dioxide removal, or mCDR. Meeting global climate goals will probably require pulling billions of metric tons of CO_2 from the atmosphere annually by mid-century, according to the United Nations. Land-based facilities that do it with fans, heat and chemicals have removed only a few thousand tons of CO_2, at high cost. The ocean already absorbs nearly a third of humanity's emissions, making it an enticing place to try to remove more. But seawater is always moving, and companies cannot simply meter what they claim to remove.

For many of these companies, the business model depends on selling carbon credits, each meant to represent one metric ton of CO_2 removed from the atmosphere. When the removal cannot be measured directly, the credibility of each credit rests on the model used to calculate it.

A few days before the boat trip, I walked through the Tufts Cove generating station with Planetary's lead scientist, Will Burt. On muddy gravel between the smokestacks and the water, a forklift

held what looked like an enormous sack of sand. A worker in a respirator and a hard hat climbed onto scaffolding and began mercilessly beating the sack with a cricket bat to clear a blockage. White powder poured into a round tank below.

The sack held more than a ton of a feedstock made largely of magnesium oxide. Mixed with water in the slurry Planetary uses to dose the plant's outflow, it becomes magnesium hydroxide, the active ingredient in some antacids such as Rolaids. The approach is similar to natural weathering of minerals in rocks, but rather than acting on CO_2 dissolved in tiny raindrops, it converts CO_2 dissolved in the vast ocean into stable forms of carbon. "That's what allows this to be megascalable," Burt said. "That is the same reason it's challenging to measure it."

> The foamy pink water spread across Halifax Harbor; some stayed near the surface and some sank, painting streaks across the water.

A pipe wrapped in orange insulation carried the slurry to a concrete cistern, where it mixed with seawater that had cooled the plant's turbines. I climbed the scaffolding and peered into the outflow. White froth churned as the slightly more alkaline water cascaded into the gray-blue harbor. Once the mixture entered the sea, Burt said, "the carbon removal happens everywhere." Planetary believes ocean alkalinity enhancement could eventually remove billions of tons of CO_2 a year.

The chemistry takes advantage of the ocean's carbonate system. Less than 1 percent of the inorganic carbon in seawater exists as dissolved CO_2; most is held in bicarbonate and carbonate ions. Planetary's slurry shifts more carbon into those stable forms, lowering the CO_2 concentration at the surface. The imbalance allows more CO_2 from the air to dissolve into the water, where much of it can remain as bicarbonate for millennia.

The chemical reaction is only the beginning of the accounting. Air-sea equilibration can continue long after the treated water leaves the outfall. Currents may carry the alkalinized water beyond the area where scientists can sample, and some of it may sink before taking up as much atmospheric CO_2 as the chemistry allows. Measuring, reporting and verifying the removal—MRV, in the field's shorthand—requires researchers to reconstruct what happens after the treated water disappears into the harbor.

In one experiment, Atamanchuk's team released a cloud of rhodamine dye into the Tufts Cove cistern so the researchers could follow the treated water. The foamy pink water spread across Halifax Harbor; some stayed near the surface and some sank, painting streaks across the water. The team collected samples and filmed the plume with an underwater drone to make a three-dimensional map. Most of the dye remained near the surface, although the share changed with the tide and the stratification of the water.



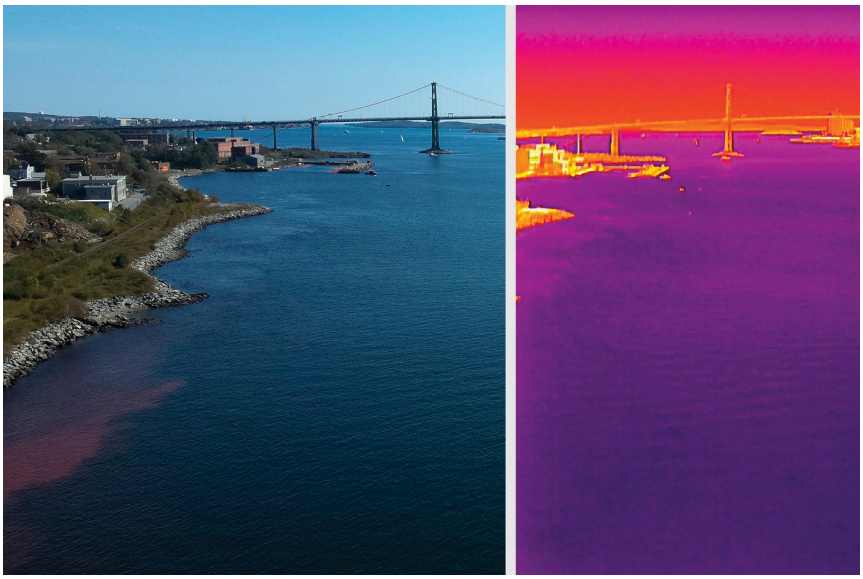
Researchers track how treated water moves with currents and depth—a key step in measuring ocean carbon removal.
Ocean Alk-Align Project

New instruments are beginning to improve the picture. This past April a research team that included Planetary scientists reported the

first field deployment of an autonomous alkalinity analyzer. Positioned 60 meters from the Tufts Cove outfall in 2023, it repeatedly detected Planetary's added alkalinity—although it could not determine how much additional CO_2 ultimately entered from the atmosphere.

For that, the Halifax researchers rely on a high-resolution computer model. While I was at Dalhousie, oceanographer Katja Fennel showed me a simulation of Planetary's trials. Alkaline water appeared as a purple blob on top-down and side views of the harbor. She hit play, and the blob pulsed with each dose, washing up and down the bay with the tide. Purple tendrils snaked toward the Atlantic and dissipated.

Fennel's model calculates how long the treated water remains close enough to the surface to exchange CO_2 with the air and how much the harbor would have absorbed without Planetary's intervention. In the simulation she showed me, the treated water had achieved about 60 percent of its maximum possible carbon removal by the end of the study year. It could continue absorbing CO_2 after that, Fennel said.



The pink tracer follows the shoreline, but tides and water layers reshape its path. Researchers must rely on models to estimate how much carbon it could absorb.
Ocean Alk-Align Project

Researchers at Dalhousie are also watching for ecological effects. They have found a local phytoplankton species and a mussel species that appear to be harmed by heavy exposure to magnesium hydroxide in the lab, but that exposure is higher than what they would face in Tufts Cove. “We have to have monitoring and assessment to make sure that we’re not doing damage,” said Hugh MacIntyre, a Dalhousie professor who previously researched the effects of Planetary’s dosing. “Before all else, do no harm.”

Experiments have already met resistance on shore. Planetary abandoned a proposed project in St. Ives Bay in England after local surfers protested to “keep our sea chemical free”; the company later said a larger rollout was also commercially infeasible. In North Carolina, the U.S. Fish and Wildlife Service warned that an olivine-sand trial by carbon-removal company Vesta could smother organisms on the seafloor, and the project was sharply scaled back. Countries that oversee international rules against dumping at sea identified ocean alkalinity enhancement as one of four techniques requiring priority scrutiny, warning that these approaches could cause widespread or lasting harm and urging that projects outside legitimate scientific research be deferred. The parties have met twice since and still have not settled on how existing treaties should govern the technique.

The field’s history helps to explain that caution. Ocean iron fertilization grew from the idea that iron-rich dust blown across the oceans during past ice ages fed blooms of phytoplankton, helping to lower atmospheric CO_2. Beginning in the 1990s, researchers tested whether adding iron to seawater could reproduce some of that effect. The blooms were easy to trigger, but most experiments could not show how much of the captured carbon sank far enough to remain out of the atmosphere. Added nutrients could also shift food webs or fuel harmful algal outbreaks.

In 2012 American entrepreneur Russ George spread roughly 100 tons of iron in the North Pacific, provoking international criticism

that a commercial venture had begun geoengineering without adequate oversight. The episode helped to turn ocean fertilization into a warning for the broader mCDR field: a visible biological response did not necessarily amount to durable, measurable carbon removal.

> At that rate, removing one billion tons of CO_2 a year would consume almost 8 percent of the electricity produced worldwide.

Other projects have tried to move carbon downward in seaweed. Last year, at the University of California, Santa Barbara, I visited oceanographers Craig Carlson and David Siegel, who were part of a $2.9-million research program studying what becomes of the carbon fixed by kelp. Locals were catching a lunchtime surf as we walked down to the beach. Beyond the breaks stood forests of giant kelp, each capable of growing as much as 50 centimeters a day. Particles of disintegrating kelp drifted downward like falling leaves. About a tenth of that biomass reaches deep water, according to a 2016 *Nature Geoscience* study; the rest is eaten or decomposes on the way down, and the kelp also oozes a dissolved “slime” of carbohydrates and proteins that microbes pull apart. Here, too, the challenge was tracing where the carbon went and how long it stayed there.

Running Tide tried to turn that process into a business. The company, based in Portland, Me., raised $70 million to grow kelp on floating buoys made of waste wood and designed to sink to the deep sea. It claimed to have removed more carbon than any other mCDR company. It shut down in 2024. Running Tide had struggled to grow kelp offshore at scale and shifted toward deployments of wood chips mixed with alkaline substances, said former general manager Kristinn Hróbjartsson. Founder Marty Odlin told me the company had not finished developing sensors and methods to reliably measure key properties such as pH and dissolved inorganic carbon. He blamed the shutdown on weak demand and the cost of

proving removal, including $30 million spent on measurement. “It’s insanely expensive to do MRV in open systems,” he said.

One response is to bring more of the process onshore. At Berth 58 in the Port of Los Angeles, I visited a long barge with yellow railings and a jumble of tanks and pipes threaded with gauges. Gaurav Sant, an engineer at the University of California, Los Angeles, who co-founded mCDR company Equatic, said the young field faced “an issue of trust and confidence.”

Equatic pumps seawater through an electrolyzer, where an electric current passes through a sandwich of plastic and metal plates and separates the water into two streams, one alkaline and one acidic, while also producing oxygen and hydrogen. In the alkaline stream, calcium in the seawater converts CO_2 into calcium carbonate, the durable compound seashells are made of. Magnesium in the stream then transforms CO_2 added to the water in a tank into bicarbonate before returning to the ocean. The Los Angeles pilot uses bottled CO_2 to demonstrate the chemistry; planned plants in Singapore and Quebec are designed to draw it from the air instead. The acidic stream is neutralized with crushed olivine. Because the reactions occur in containers, sensors can measure them. “I can track every gram of carbon dioxide I removed,” Sant said.

Some want the entire field to be held to that standard. In early 2025 Garrett Boudinot, a geochemist who founded alkalinity company Vycarb, argued on the climate news site Carbon Herald that mCDR should be measured directly. “The future of the climate depends on measuring—not modeling—climate solutions,” he wrote.

But that process takes considerable power. Equatic estimated that it takes about 2.5 megawatt-hours per ton of CO_2 removed. At that rate, removing one billion tons a year would consume almost 8 percent of all the electricity produced worldwide. Hydrogen produced along the way could generate electricity in a fuel cell or be sold, potentially offsetting some of that demand. The promise

has already attracted Boeing, which signed a prepurchase option covering 62,000 tons of future carbon removal and 2,100 tons of hydrogen. Open-water approaches may be cheaper to scale and can take advantage of the ocean's enormous surface area. Their claims, however, depend more heavily on models built from observations of only a small part of the moving sea.

Buyers are already putting money behind methods that remain difficult for outsiders to assess. In 2025 mCDR start-up Gigablue sold 200,000 future credits for a method intended to stimulate phytoplankton on mineral particles that later sink. The company has disclosed little about the particles or its supporting data; Chelsey Baker of the U.K.'s National Oceanography Center told me that the public information was too sparse to determine whether the method would durably remove as much carbon as the company claims.

Fennel says measurement alone cannot verify open-ocean removal, because we don't know how the ocean would have evolved without the intervention. Once alkaline water mixes into the sea, modeling becomes unavoidable. Fennel believes the Halifax team may eventually pin its estimate to within 10 percent. For an intervention dispersed through a harbor and out to sea, that may be as close as the ocean gets to a meter reading.

Back on the boat, Atamanchuk's instruments could show only that the surface water held slightly less CO_2 than the air above it. The answer Planetary needed—how much of that difference it could claim—was already moving with the tide.

Discussions

We're looking to host the most interesting science conversations on the web for this topic.

Alec Luhn is a journalist covering climate and the environment whose work has appeared in *New Scientist*, *The Atlantic* and *National Geographic*. He lives in London.

https://www.scientificamerican.com/article/inside-the-bold-quixotic-quest-to-store-planet-warming-carbon-in-the-ocean

How uncertainty breaks the brain

Can our “prediction machine” brains learn to handle a world in chaos?

By Maia Szalavitz edited by Allison Parshall



Van Santen & Bolleurs

Laura Harrison didn’t want to evacuate. The neuroscientist and mother of two had finally gotten her young children settled into bed after their nighttime ritual. But it was the evening of January 7, 2025, and the skies over her home in Sierra Madre, Calif., glowed a haunting orange-gray. Outside, the Santa Ana winds sent debris swirling with powerful gusts.

In nearby Eaton Canyon a spark had already become a roaring fire. Harrison could see the flames advancing down the mountain, but she hadn’t received evacuation orders. She hesitated. She knew which escape routes were still open, but fleeing at the wrong moment could be catastrophic. She had little information about what choices would give her the best odds of getting out safely. Uncertainty shrouded the city along with the smoke.

But the rapidly worsening conditions convinced her to leave. As Harrison drove her family out, she sent photographs to her colleague Galen Buckwalter, who lived nearby, urging him to flee at once. Buckwalter, a research psychologist, became quadriplegic after a teenage diving accident and uses a wheelchair. That night the van he relies on for mobility was in the shop, and he physically couldn't get into his wife's car. He wondered whether he should just shelter in place.

Then fire trucks began wailing their way closer, broadcasting urgent calls to evacuate. Buckwalter told his wife to drive away without him—he would use his motorized wheelchair to escape, hoping to reach a street where an accessible Uber could pick him up. He strapped on a high-visibility vest and fled. "I took off in my chair in this howling wind with ashes all around [and] fire on the hills," he says. For around 20 minutes he motored through choking smoke on almost unrecognizable streets, trying to stay calm. "It was really quite an apocalyptic feeling."

The Eaton fire would ultimately kill at least 19 people and destroy more than 9,000 homes and workplaces. Thankfully Harrison, Buckwalter and their families made it out safely, and their homes survived the blaze. Both scientists work for the Valor Institute for Neuroscience, which is part of a private equity firm. When the Eaton fire ignited, they were, fittingly, in the middle of revising a paper about the connection between the way our brains handle uncertainty and resilience to stress.

Brains evolved to handle uncertainty. They use whatever information they can gather to compute the best course of action for reproductive success and self-preservation—finding food to fuel metabolism, for example, or locating a safe place to sleep. We can never know enough to definitively guarantee our safety, but we can use what information we have to make a best guess. Some neuroscientists now theorize that the brain is a "prediction engine" at its core and that processing uncertainty is its central task.

Evolution prepared us for a world that is uncertain but with hazards that are often foreseeable and, to some extent, calculable. When we know the odds, we can run them and take our best shot at beating them. But the more the odds change on us, the more stressful life becomes—and, research shows, the greater our risk for mental and physical illness.

For many people, the world today feels less predictable than at any other point in their lives. Climate change is driving severe weather and disasters such as the L.A. fires. American norms about politics, science, education and law are being overturned, and artificial intelligence threatens further disruption. Understanding how the brain uses predictions to manage uncertainty may now be more critical than ever—and it may also help explain the causes of mental health disorders that are on the rise in the U.S. and show us how we can foster resilience in ourselves and our children.

In physics, one way to measure uncertainty is entropy—a way of quantifying disorder and randomness. As the bleak second law of thermodynamics states, entropy always increases over time in a closed system. Systems fail; unanticipated difficulties occur; things break or rot. To cope with this reality, philosophical and religious traditions such as Buddhism and Stoicism have long focused on transience, unpredictability, and the inevitability of change and loss.

Being alive allows us to briefly evade the second law. Living creatures dodge entropy for as long as we can by being organized as open systems with permeable membranes. We take in energy from outside to maintain metabolism and to reproduce; we expel disorder through waste. We die when entropy wins by ending—through aging, injury or disease—our body's ability to stay fueled and organized.

According to one popular theory, we defeat chaos for as long as we do because information can bring some temporary order. In the

1940s mathematician Claude Shannon quantified information as the reduction of uncertainty. This founding insight from information theory can be united with physics and biology through the "predictive brain" theory, first proposed in the early 2000s by Karl Friston, a professor of imaging neuroscience at University College London. This theory, also known as active inference or the free energy principle, has become one of the most important frameworks in neuroscience for understanding how the brain works as a whole. From a computational perspective, Friston argues, attempting to reduce uncertainty—basically, working to avoid unpleasant surprises—is the brain's primary goal. "I think uncertainty is at the heart of how we make sense of the world," he says, and reducing it "is a fundamental drive for nearly everything we do."

> "Uncertainty is at the heart of how we make sense of the world. Reducing it is a fundamental drive for nearly all we do." —Karl Friston, *University College London*

Friston's free energy principle mathematically describes how the brain uses the information it obtains from the senses to predict the state of itself, the body and the surrounding environment. According to the theory, we experience the world not through sensory information itself but through these predictions. If, for example, you see a gray blotch on the floor at night and think it's your cat, you might conjure an image of your curled kitty before you realize it's actually a discarded towel.

Our brain makes these predictions both consciously and unconsciously, relying on past sensory and physical experience to run the odds. This may explain why early life experience is literally foundational: to best learn the complex rules of being a functional human, children need a stable and generally predictable environment. We update our projections continuously when sensory information suggests they are inaccurate. Neuroscientists call these inaccuracies "prediction errors," and they are associated

with distinct signals in the brain. The erroneous cat changes seamlessly into the towel it actually is as prediction errors are overridden by new information.

Networks across the brain communicate with one another to create this perception, with information flowing between lower-level sensory regions and higher-level processing networks to accurately represent reality. Friston and his colleagues have found in studies across species that important brain circuits throughout the cortex seem to be wired to match information from the senses with data from memory to make predictions.

What's more, dopamine pathways that were traditionally thought to govern pleasure and punishment also seem to be highly involved in prediction. Multiple studies now show that dopamine cells in these circuits signal prediction errors related to reward. For example, once a rat has been trained to push a lever to get a treat, its dopamine levels will rise at the sight of the lever before the treat is even available. But if it later pushes the lever and gets nothing, those levels will fall below baseline, telling its brain that its prediction was wrong. Dopamine doesn't necessarily signal pleasure here; it just represents the difference between the expected and actual reward, which can be considered the mathematical expression of prediction error. New research has also shown that reducing another signaling chemical, norepinephrine, in certain brain regions can decrease rats' ability to learn from situations where expected rewards are less available.

All of this means a highly unpredictable world is the brain's worst nightmare—and the uncertainty that results can quickly become toxic.

Large prediction errors can be stressful. They signal that something major is wrong, requiring immediate attention and correction. They can activate the hypothalamic-pituitary axis, releasing

glucocorticoid hormones such as cortisol to prepare the body to respond.

Resolving the prediction error typically requires bringing in new information. For example, if I hear a sudden loud noise, I may turn my head to locate the source. Or if I am trying to work and my blood glucose falls lower than expected, my brain attempts to resolve this prediction error about my body's internal state, starting by producing the sensation of hunger. This new information will lead me to seek and eat food, and my glucocorticoid level will increase to focus my attention until I resolve the problem by returning my blood glucose to normal.



Van Santen & Bolleurs

When levels of stress chemicals rise, they put the brain into a mode that uses more energy. In a mildly stressful situation—where, say, you can't eat because you forgot to go shopping and there's no food in your home—this rise is moderate and brief, and it increases learning. You will be less likely to forget to go to the store in the future. But in situations of ongoing uncertainty—such as chronic food instability—the stress is far greater.

Hormones such as glucocorticoids are one way that chronic stress "gets under the skin," according to the late Rockefeller University neuroscientist Bruce McEwen, who pioneered research in this area.

When released under conditions of acute stress, glucocorticoids power fight or flight by shunting metabolic energy away from processes such as digestion and toward the heart, lungs, brain and muscles. These emergency reactions consume energy and cause wear and tear throughout the brain and body.

In the brain, for example, elevated stress hormones can impair learning and memory by flooding the hippocampus with excitatory chemical signals. High stress can also shift the brain's energy resources in ways that make it harder to regulate emotions and control behavior. Such wide-reaching effects may be part of why chronic stress, particularly in early childhood, increases risk for nearly every known mental illness. When people have little relief from stress in a chaotic environment, their brains have difficulty coping with the ongoing prediction errors, Friston, McEwen and neuroscientist Achim Peters argued in a 2017 article.

Chronic stress also increases risk for physical illnesses such as type 2 diabetes, heart attack and stroke in part because of the increased metabolic demands it imposes. A state of "metabolic emergency" can preserve life in the short term, but it can hurt the body in the long term. Metabolism involves constant trade-offs, and when the brain doesn't have the information it needs to stay in balance, things break down—and entropy wins.

If the brain's primary goal is to reduce prediction error, this way of being contains a trap. Wouldn't it be most efficient to withdraw into the most predictable setting possible—an empty, lightless room, perhaps—and stay there? This is called the "dark room" problem.

In the long run, this strategy won't work. You will eventually have to emerge from the small, safe place to which you've retreated. At that point the outside world will have become even more unpredictable because you haven't been gathering new information about it. A more adaptive approach is to seek positive surprises when you can safely do so. Our very nature tends to encourage this

behavior. “We are naturally born curious creatures,” Friston says. We play, solve puzzles and learn new skills even when they bring no obvious, immediate benefit for survival.

Sampling the world through our senses and our interactions with others lets us build rules and patterns for our predictions. But when these rules and patterns suddenly change, the models we built about the world can no longer resolve uncertainty. “We now have to change our world model,” Friston says. “What would that look like? It would effectively look like a grief reaction.”

In these situations, withdrawing into a literal or symbolic dark room—avoiding sources of stress and going to a safe place—can be adaptive in the short term. Friston notes that one theory on the origins of depression suggests that this is its evolutionary purpose: to give us time to consider behavioral changes when a survival or reproductive strategy is failing. If this withdrawal goes on for too long, however, it becomes harmful. Some core symptoms of anxiety and depression, such as avoidance and a lack of motivation, can be seen as the results of a brain stuck in the dark room, according to Friston.

Lisa Feldman Barrett, a professor of psychology at Northeastern University, argues that the brain is not necessarily trying to minimize cost—it’s simply trying to spend energy efficiently. States of uncertainty that cannot be resolved can be particularly inefficient. “In uncertainty, your brain has to assemble many plans and maintain those plans over a period of time,” she says. “And that’s just expensive because there’s more going on. There’s more signaling. There are more chemicals.” Symptoms such as a lack of motivation or pleasure and a feeling of being stuck can be “indicators of being strapped for energy,” Barrett says. When less energy is available to update predictions and you see the world as bleak, you may continue to have that view even when conditions are actually improving.

Alternatively, depression may involve placing too much confidence in outdated forecasts. Once you have decided, for example, that “no one will ever love me” or “everything always gets worse,” your brain may attend more to signals that support those predictions and less to those that don’t. Over time the brain can fall into a rut. An inability to update outdated predictions is one factor that may contribute to mental illness. Another way to describe this situation is as a lack of plasticity or changeability in the brain. Notably, most things known to have antidepressant effects increase brain plasticity—such as exercise, electroconvulsive therapy, selective serotonin-reuptake inhibitors prescribed as antidepressants, and psychedelic drugs.

Anxiety disorders may involve a different type of prediction problem that relates to reduced plasticity. Here you overgeneralize from one bad experience: if you get bitten by one dog, you might fear all dogs; if you fail one test, you may see yourself as a failure. “The more you avoid, the more you think your prediction was accurate,” says Philip Kendall, director of the Child and Adolescent Anxiety Disorders Clinic at Temple University. “‘I can’t go near that dog; it’s going to bite me.’ So [you] don’t go near the dog. [You] don’t get bitten. And you go, ‘See that—the dog would have bitten me, but I’m safe.’ That’s a false confirmation of a prediction, [and it] leads to more avoidance.”

> Managing the stress of uncertainty is like building muscle. You start with small challenges.

Research suggests that having an especially hard time with uncertainty is associated with many anxiety and related disorders, including obsessive-compulsive disorder and generalized anxiety disorder. “We don’t know if it’s appropriate to call [intolerance of uncertainty] a cause or a symptom,” Kendall says. “The good news is that there are treatments that reduce intolerance of uncertainty,” and these treatments can be effective for anxiety disorders.

To treat mental disorders involving faulty predictions about the world, clinicians often use cognitive-behavioral therapy, or CBT, which addresses the intellectual content of the fear: exploring the real probability that the feared negative outcome will happen and instilling confidence in coping skills that can help. In practice, this means helping people to change the way they talk to themselves and to stop avoiding situations that make them nervous. Kendall teaches children and teenagers to break down anxiety-provoking situations, such as finding friends to sit with in the cafeteria or asking someone for a date, into smaller steps and planning what to do if something goes wrong. “When kids do things that they’re afraid of, they don’t fall apart,” he says. “They become proud peacocks.” This practice helps them override the predictions they made based on prior experience, which made avoidance seem like the safer choice. In a study published in 2019, Kendall and his colleagues showed that the more a child’s uncertainty intolerance was reduced by treatment, the more their anxiety fell and their ability to function improved.

Of course, sometimes events are genuinely unpredictable and threatening enough to jeopardize safety. Unpredictability is a hallmark of the circumstances that often produce post-traumatic disorder (PTSD), such as natural disasters, wars, terrorist attacks, vehicle accidents, and sexual or physical violence. These experiences can obliterate people’s ability to predict what will keep them safe. Events that make you feel betrayed or helpless in a life-threatening situation can contribute to the development of PTSD. The sense of continued threat that constitutes the experience of PTSD can be described as a prediction error.

Still, the majority of people faced with traumatic experiences don’t develop psychiatric problems such as PTSD or depression. George Bonanno, a psychologist at Columbia University, explains that many people who have intense symptoms initially—such as nightmares and flashbacks—won’t develop PTSD. These symptoms may reflect the brain’s attempts to integrate and learn

from what happened. “Your brain is trying to make sense of all that information,” Bonanno says. “This is one of these things that the public doesn’t hear about—that intrusive memories [or] even nightmares after something really disturbing or life-threatening are not uncommon, and they’re actually adaptive. But then, for some reason, that turns into PTSD in some people.”

It is surprisingly difficult to predict who will experience PTSD, a phenomenon known to researchers as the resilience paradox. Some features put people at higher risk for developing PTSD or more severe forms of the condition: for example, having suffered multiple previous traumas or having little social support. Determining who will recover quickly is also a challenge. But people who thrive in uncertain and threatening situations tend to have some things in common: traits related to flexibility, Bonnano says. These involve maintaining some degree of optimism about recovery, having confidence in your ability to cope, and seeing the world with what’s known as a challenge-oriented mindset. This is the ability to reframe thoughts and physical signs that could be interpreted as anxiety in a less threatening way.

For instance, if you’re about to give a speech, you can interpret your elevated heart rate as you approach the podium as a sign of fear—or one of excitement about being able to rise to the challenge. The more you can view uncertainty as opportunity rather than threat, the more experience you get successfully managing mild and moderate stressors, and the better you are likely to be at dealing with extreme situations.

In the paper that Harrison, Buckwalter and Friston were editing when the L.A. fires ignited, they describe four potential pathways or “phenotypes” of managing uncertainty. These paths aren’t unalterable; people can adjust them with therapy or other learning experiences. The least resilient is the “fragile” type. Genetic predispositions to sensory overload, which is often experienced by people with autism, attention deficit hyperactivity disorder or

obsessive-compulsive disorder, can put someone on this path early in life. Their existence feels overwhelming and unpredictable, so they create their own dark room by adhering to rigid, repetitive routines that may limit opportunities for learning, social connection and updating of inaccurate predictions. Another on-ramp to the fragile pathway is a childhood of abuse, neglect, or other recurring traumatic experiences. These circumstances, too, create a perception of the world as unpredictable and may lead people to seek escape via alcohol or other drugs.

People with the second phenotype, called durable, might have grown up in a predictable world—for example, a strict religious sect or just a family that experienced little change—and developed only certain coping strategies as a result. They are highly functional in familiar settings but may become fragile or inflexible in new ones. The third possible path is that of resilience. Resilient people experience life's typical ups and downs, but their senses don't overwhelm them. In most cases, they will be able to respond even to potentially traumatic experiences by seeking social support. The final phenotype, confusingly labeled "pro-entropic," describes people who proactively seek information. This makes them even more resilient because regular experience with significant stress allows them to become better at handling ever more extreme challenges.

Although both Harrison and Buckwalter initially denied they needed to evacuate during the L.A. fires—a potentially deadly response—they ultimately got themselves to safety. They used what they'd learned about paths to resilience to guide their choices. "Making the best inferences you can really, really came to the fore for me," Buckwalter says. He knew the light on his motorized chair would help him see in the smoky air, and he grabbed a reflective vest so firefighters rushing to the scene could see him. Keeping himself calm allowed him to navigate his chair on a familiar street made unfamiliar by burning debris and smoke.

Buckwalter says his awareness of the predictive brain allowed him to cope better and regulate his emotional responses so he could do what needed to be done. “This process has been invaluable to me in my life,” Buckwalter says.

After classes resumed at her oldest child’s school, Sierra Madre Elementary, following the fires, Harrison gave a presentation there on the relation between resilience and the predictive brain. Her talk focused on being flexible and creative when responding to stress. First she praised the kids for being “brave superheroes” who had had to deal with a really difficult experience. “You guys were super,” she recalls saying. “You’ve all gotten through this.”

Then Harrison explained how they could get even better at facing future challenges. Stress isn’t always bad, she emphasized, and choosing to change course when something isn’t working is critical to learning. She walked the children through examples of situations when they might not get what they expect, such as not being able to have their favorite breakfast. The idea that they could come up with alternatives on their own, rather than just relying on their parents to fix it, energized the students, especially the youngest ones.

Managing the stress of uncertainty is like building muscle, Harrison says. You start with small challenges, and over time you learn to face larger ones more effectively. It’s important to be curious—to not become too attached to your expectations. “If you expect to be surprised, you can come back more quickly,” she says. “Stress or change in the world is opportunity as well.”

Discussions

We're looking to host the most interesting science conversations on the web for this topic.

Maia Szalavitz is the author of, most recently, *Undoing Drugs: How Harm Reduction Is Changing the Future of Drugs and Addiction* (Hachette Books, 2021). She is a contributing opinion writer for *The New York Times* and author or co-author of seven other books.

https://www.scientificamerican.com/article/how-your-brain-learns-to-cope-with-uncertainty

These extreme cosmic explosions are breaking physics

Astronomers are finally figuring out what powers mysterious gamma-ray bursts

By Robin George Andrews edited by Clara Moskowitz



Nicolò Canova

Astronomers have seen things you wouldn’t believe. Diaphanous auroras on other planets. Collapsing stars off the belt of Orion. The fringes of black holes seemingly on fire. But nothing has ever compared to the almighty force of what one astrophysicist calls “the energy monsters of the universe.”

Every now and then, observatories catch strange flashes lighting up the far-flung reaches of the cosmos. These cataclysmic explosions spray intense radiation and beams of energy through the universe at close to the speed of light. After the initial blast, which spews out gamma rays, a storm of radiation pours into the surrounding galaxy, creating a ferocious afterglow that burns in every part of the electromagnetic spectrum.

Although these events have much in common with various science-fiction superweapons, they are entirely natural eruptions known as gamma-ray bursts, or GRBs. They are by far “the most extreme things in the universe,” says Eric Burns, an astrophysicist at Louisiana State University.

Supernovae, the deaths of the most massive stars, are terrifically energetic. But they’re like firecrackers compared with GRBs, which typically emit energy equal to something like “a million, trillion, trillion times the Hiroshima bomb,” says Brendan O’Connor, an astrophysicist at the University of Maryland, College Park. “In a few seconds they radiate the amount of energy that the sun produces over its entire lifetime.”

These horrifying yet beguiling events might just hold some secrets to understanding the universe. “The physics involved is extremely exotic,” says Charlie Kilpatrick, an astrophysicist at Northwestern University. Certain GRBs produce some of the heaviest elements in the periodic table, including gold. And their extreme, alien physics may help scientists elucidate how the first generation of stars and the first galaxies came to be. Their outpourings of light can be “used as probes of the distant universe,” says astrophysicist Peter Blanchard of the Center for Astrophysics | Harvard & Smithsonian.

Astronomers are puzzled about how GRBs work. In the decades since the objects were first discovered, scientists knitted together a picture that they thought handily explained them. But in the past few years the yarn has begun unwinding. Two recently discovered long-lasting GRBs, which were thought to arise from massive explosions, seem to lack the traditional signs of a blast. Another GRB, by far the brightest one ever observed, did not appear to make any of the expected heavy elements.

And then a seven-hour-long GRB came along—acting like a Death Star whose planet-destroying beam refused to switch off—that defied all explanation. “This is sort of breaking everything we

thought about what produces gamma-ray bursts," says Jillian Rastinejad, an astrophysicist at the University of Maryland, College Park.

Despite detecting around one of these blasts a day, scientists are scrambling for answers. If they can't find a way to make sense of these detonations, they may discover that GRBs are exposing deeper flaws in our understanding of the universe than we realized.

Gamma-ray bursts are a relatively recent discovery. During the cold war, the U.S. wanted to know whether the Soviet Union was sneakily test-firing nuclear weapons. In the 1960s America launched the Vela fleet of satellites, whose sole task was to look at Earth's surface and search for any telltale neutrons, x-rays and gamma rays—extremely high-energy particles—coming from a nuclear explosion.

Worryingly, on July 2, 1967, two Vela satellites recorded something that certainly looked like it came from a nuclear bomb. But on closer inspection, the radiation pattern didn't seem to quite match that of a nuke. Weirder still, it appeared to be coming from space. From July 1969 to July 1972 scientists found 16 of these events buried in the Vela satellite data, all rich in gamma rays. The data, which confounded experts, were initially classified. In 1973, once officials were confident the radiation wasn't coming from a secret, terrifying Soviet weapon, they released the results to the public—but scientists couldn't figure out what was causing the explosions, which they named gamma-ray bursts.

That same year one of the scientists who first uncovered the existence of GRBs, Ray Klebesadel of the clandestine Los Alamos National Laboratory, gave a talk at a meeting of the American Astronomical Society. A journalist from the National Enquirer approached him afterward, asking whether it was possible the GRBs were coming from alien civilizations nuking one another. Charles "Chip" Meegan, a NASA researcher, started to hear about

GRBs from concerned members of the public. “I got a phone call once from a guy who said that he not only knew what was causing the bursts but how to stop them,” he recounted in a 2002 PBS documentary. “And I just said, ‘I’ve got to go to a meeting now.’”

> Scientists may discover that gamma-ray bursts are exposing deeper flaws in our understanding of the universe than we realized.

To discover the truth, space agencies launched observatories to study (among other things) GRBs, including NASA’s Compton Gamma Ray Observatory and the European Space Agency’s Integral observatory. NASA followed up in the 2000s with its Fermi Gamma-Ray Space Telescope and the Swift observatory. Together they found that GRBs are coming not from warring aliens but from cosmic destruction derbies.

By the 2000s scientists had detected thousands of GRBs, and researchers suspected they were being created by two classes of objects. The first was collisions of neutron stars—very dense, rapidly spinning orbs just a dozen or so miles across, made from the left-behind cores of massive stars that had died. The second was collapsing supermassive stars, which end their lives as supernovae. Both events are thought to generate black holes, at least most of the time.

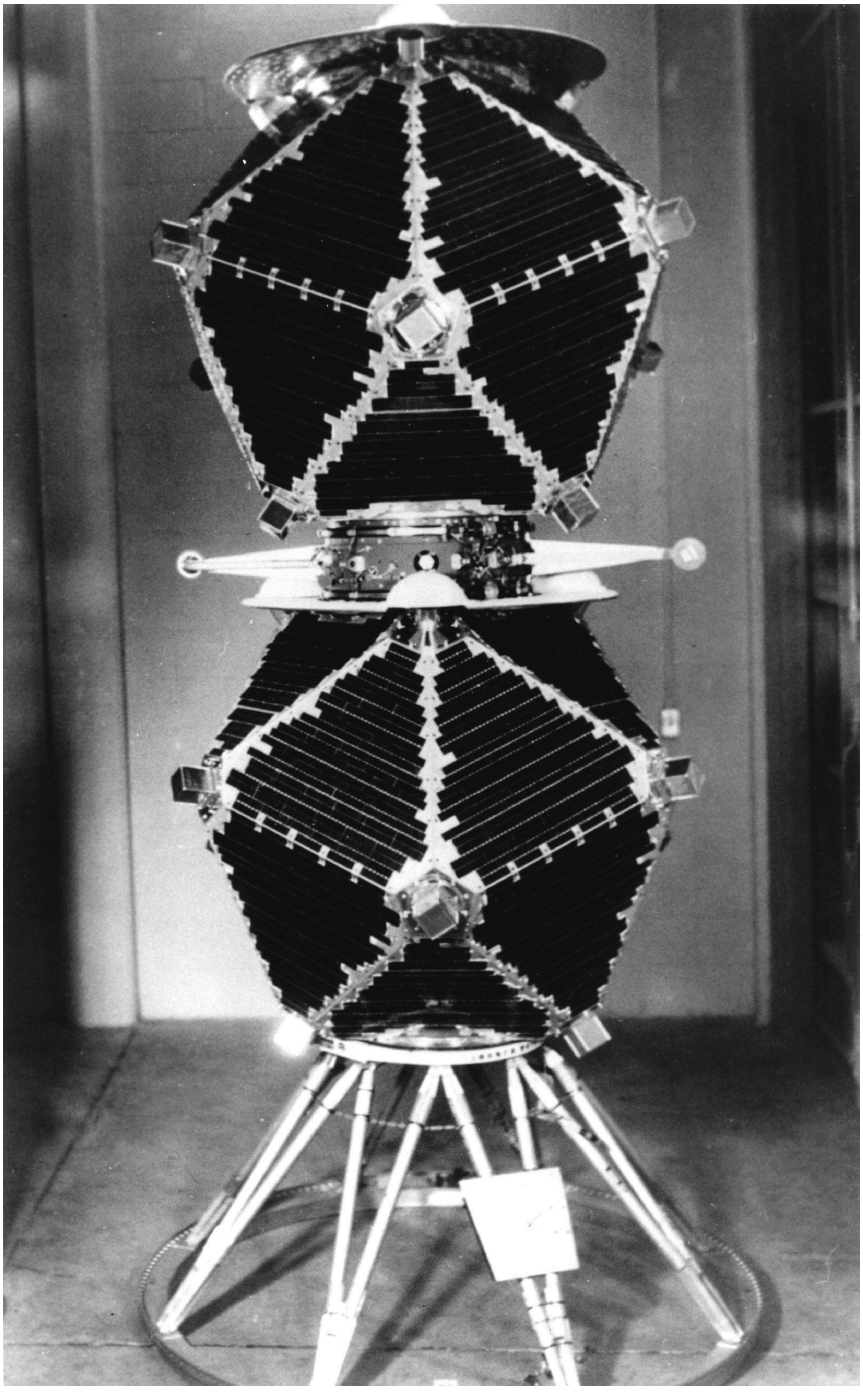
GRBs coming from these monstrosities aren’t firing gamma rays out in all directions—they are mainly shooting unfathomably powerful jets of radiation in concentrated beams. “It’s really, really hot plasma and particles moving out at basically the speed of light,” Burns says. The composition of these jets isn’t crystal clear, but scientists suspect electrons, positrons (the antimatter versions of electrons) and perhaps protons are pulled along swirling magnetic lines before being sent out from the black hole’s poles. “You get two jets,” O’Connor says, “but we only see one.”

If one of the jets happens to be pointing toward Earth, that's when we detect a GRB. In most cases, we probably miss them entirely because the beams are pointed elsewhere. And when one does head toward us, it's usually coming from billions of light-years away, so our planet is hit by only a modicum of this annihilative radiation. When the jets fly out, they smash into the interstellar matter around them, causing a menagerie of radiation—from x-rays and visible light to radio waves—to stream into space, creating an eerie afterglow that can persist for hours, months or even years.

By the 2000s scientists had split the explosions into two broad categories: short-duration GRBs (let's call them shorties), which lasted for no more than two seconds, and long-duration GRBs (longies), which could endure for tens of seconds. Longies, which make up the majority of GRBs, are associated with collapsing supermassive stars, whereas shorties are linked to crashes between two neutron stars.

The difference between shorties and longies comes down to mass. In both cases—when neutron stars merge and when supermassive stars collapse—stellar matter is ejected into space as a black hole is born. And in both cases, that material eventually falls onto the black hole, and some gets launched out of both its poles as jets, causing the phenomenon we call a GRB. But collapsing supermassive stars are, as the name suggests, behemoths: they have plenty of mass, which allows their beams to fire for a while. Neutron stars have less mass, so when they crash together, the resulting GRB is briefer. "The amount of mass accreted onto the black hole is really the driving factor in how long the jet is going for," Rastinejad says.

But even when GRBs last just a few milliseconds, they are, in those moments, "the most energetic thing in the universe," says Yvette Cendes, a physicist and astronomer at the University of Oregon. And that does make them superweapons—just not the kind commanded by any galactic empires.



The U.S. launched the Vela satellites in the 1960s to search for signs of nuclear detonations. They found the first indications of gamma-ray bursts.
U.S. Department of Energy

Scientists have previously speculated that a GRB would vaporize anything caught in its death rays up to 200 light-years from the source. Fortunately, the explosions seem to take place in very remote galaxies undergoing intensive star formation and not—as far as we have seen—in our more mature Milky Way. But if Earth did somehow get supremely unlucky, and we were directly hit by a

jet even a few thousand light-years away, our situation would be grim.

Brian Thomas, a physicist and astronomer at Washburn University in Kansas, investigated what might happen. Earth wouldn't completely melt. But at this distance the jet's energy wouldn't have dissipated much during its voyage through space. It would crash into our planet's atmosphere and break apart oxygen and nitrogen molecules, causing a flood of nitrogen oxides to form. These compounds quickly destroy the stratospheric ozone, the gas that filters out harmful ultraviolet radiation from the sun. For approximately a decade, Thomas estimates, Earth's surface would face a far stronger UV bombardment than it does today. Organisms not resistant to this onslaught—anything from microbes and crops to whales and humans—would experience an almost violently severe sunburn. The event might beget an ecological collapse or perhaps a mass extinction event. "It's going to be bad," Thomas says, putting it mildly. But the chances of such a strike are so vanishingly small that we need not worry. Instead scientists see GRBs not as destroyers but as ways to clue in to the evolution of the universe.

Gamma-ray bursts could be the key, in fact, to understanding the origins of the building blocks of the cosmos.

The big bang flooded the universe with hydrogen and helium, plenty of which fused together within stars to create many heavier elements right up to iron. But that's where stars' output stops. Forging the rest of the elements took something much more extreme.

Many of the heaviest elements arise only in environments where lots of free neutrons crash into atomic nuclei at great velocities. This is called the rapid neutron-capture process, or the r-process for short. About half the elements heavier than iron are thought to be created through the r-process—including materials such as gold,

platinum, thorium, uranium and iodine. “A huge fraction of the elements in the periodic table are produced by the r-process,” Blanchard says. “We find them on Earth. But they weren’t formed here.”

Iodine is an essential element for animals: it is a crucial component of thyroid hormones, which regulate metabolism. Uranium is one of the key elements giving Earth’s geological heart its heat, allowing everything from plate tectonics to volcanism to happen. Gold is a precious commodity in plenty of cultures, as well as in electronics. The r-process elements, then, help to “make life what it looks like,” Rastinejad says.

Neutron star mergers and collapsing supermassive stars are both perfect settings for the r-process. That means studying GRBs may tell us about how these elements originated. Their afterglows are like “lightbulbs,” Kilpatrick says, illuminating the faraway, ancient galaxies they emerged from, even those that existed just a few hundred million years after the big bang.

In 2017 an extraordinary event provided the smoking gun for this theory of element creation. That year scientists running the U.S.-based Laser Interferometer Gravitational-Wave Observatory (LIGO) experiment and Virgo, a similar facility based in Italy, were looking for ripples in the fabric of spacetime created by clashes between celestial titans. On August 17 the Klaxons rang: something in NGC 4993, a galaxy 130 million light-years away in the constellation Hydra, was throwing out intense gravitational waves. By measuring those waves, scientists determined the mass of the two objects involved, which revealed their identity: colliding neutron stars. Shortly afterward NASA’s Fermi space telescope detected a burst of gamma rays coming from the same spot in the night sky. They named it GRB 170817A. GRBs are designated by streams of numbers and letters like this one. For the sake of narrative ease, let’s give it a nickname: Lightbulb.

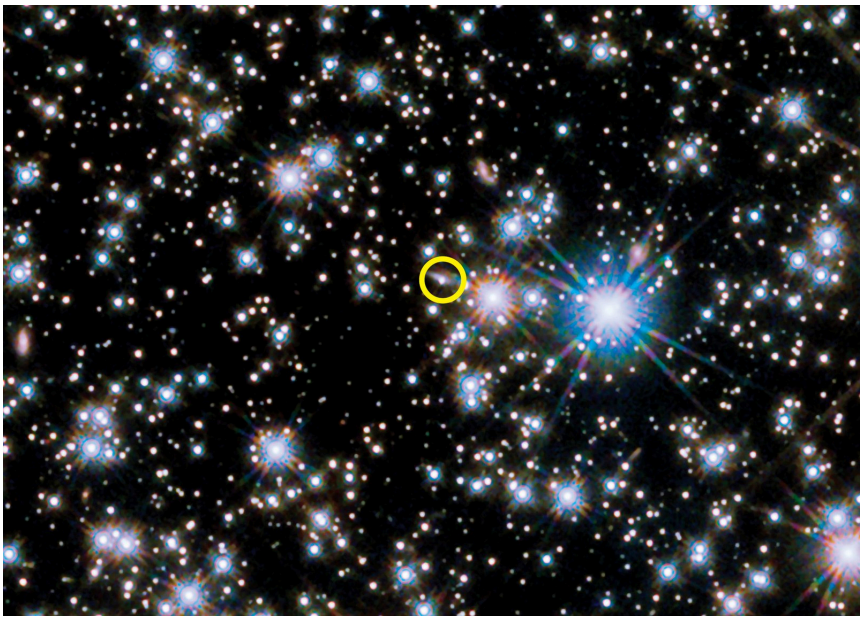
Lightbulb was also accompanied by a kilonova, a bright surge of radiation involving oodles of neutrons—the very kind that could kick-start the r-process and forge all those heavy elements. One calculation suggested Lightbulb made so much gold and platinum that the shiny pile would be 10 times as massive as Earth. The observation seemed to cap off the tidy understanding of GRBs and the two processes that create them. In 2021, back when Rastinejad was a fairly novice Ph.D. student, she was told that the story of GRBs "had, for all intents and purposes, played out." But that December a new light glimmered in the heavens. The universe had just thrown astronomers a curveball.

The new firestorm, NAMED GRB 211211A—detected by Fermi and Swift in a galaxy 1.1 bil-lion light-years away—lasted for 50 seconds. This event was a classic long-duration GRB caused by a supernova, right?

Wrong. When Rastinejad and her colleagues used telescopes in Hawaii and Arizona to look at the source, they expected to find the brilliant glow of a dying star. Instead they saw space soaked in a blood-red hue. "We were really shocked," Rastinejad says. It appeared as though the bluer photons of an explosion were being absorbed by a huge cloud of electrons, then reemitted as red light. Such a cloud is typically linked to an eruption of heavy elements—those created during a neutron star collision and its resulting kilonova. "Holy moly, this is crazy," Rastinejad recalls thinking at the time. "We all got very excited."

Their research, published in 2022, confirmed that there was no supernova and no collapsing supermassive star. Even now nobody can confidently explain what happened. It looked like two neutron stars collided and produced a rather long GRB. But how could two tiny neutron stars provide enough mass to fuel a 50-second-long pair of jets? "People will debate for a long time as to what actually produced that," O'Connor says. "We know something weird happened." But what?

A kilonova doesn’t need two neutron stars to form. Only one is required if a black hole is also involved. At the time, Rastinejad and her colleagues wondered whether a black hole’s gravity had ripped up a companion neutron star, as if unseen godly hands had pulled it apart like dough. This action would have triggered a kilonova while also liberating a lot of material, stuff that would take its time falling onto the black hole and fueling longer jets.



The Brightest of All Time (BOAT) gamma-ray burst, which erupted in October 2022, is circled in this image from the Hubble Space Telescope.
NASA, ESA, CSA, STScI, A. Levan/Radboud University Nijmegen; Image processing: Gladys Kober

But lately Rastinejad’s team has been working with a new theory. The scientists suspect that two neutron stars did collide but that that wasn’t all: a huge disk of stellar matter also formed around the merger, which meant there was a lengthy chain of fuel spiraling downward and powering the jets for a longer period. “That’s, at least in my mind, the best model we have,” says Kilpatrick, a member of the team.

Others, including Burns, aren’t convinced. Ejected material falls onto a dense neutron star in around one one-hundredth of a second. The jet should always be brief. Scientists saw a kilonova, so most of them suspect at least one neutron star was involved. But “I think the second object is a white dwarf,” Burns says. A white dwarf is

the dead heart of a modestly sized star, such as our sun, that has since been extinguished. Being considerably larger than a neutron star, “a white dwarf takes much longer to eat.” That, he says, could produce a longer GRB. Suffice it to say, GRB 211211A—let’s call it Dynamite for blowing up everything we thought we knew—left everyone bewildered.

Then, in 2023, Fermi detected another event—one that lasted for 200 seconds and also lacked a supernova. GRB 230307A, or Bonfire, “is basically a twin” of Dynamite, O’Connor says. By this time the James Webb Space Telescope (JWST) had been launched. As soon as the blast was spotted, astronomers pointed the telescope at Bonfire and, using its infrared scope, detected light emitted by tellurium, an r-process element—making it very likely that Bonfire involved neutron stars and, somehow, made a long-duration GRB.

Both events revealed that the old models of long- and short-duration GRBs had holes in them. And definitive explanations for Bonfire and Dynamite aren’t forthcoming. During both, the LIGO and Virgo gravitational-wave detectors were down for maintenance. If they had been on, the identity of the second colliding objects, whatever they were, might have been revealed. “It was a total bummer,” Rastinejad says.

This astronomical misfortune, though, was mitigated by an event sandwiched between Dynamite and Bonfire: GRB 221009A, the Olympic champion of GRBs. On October 9, 2022, something exploded 2.4 billion light-years away from Earth. Despite the distance, its jet sucker-punched our planet. For more than 10 minutes we were hammered with gamma rays. The upper atmosphere recoiled so violently that radio communications were disrupted. Lightning detectors in India lit up because of it. And various gamma-ray observatories in orbit were saturated and blinded by the torrent of incoming radiation.

The event's jaw-dropping radiance stunned astronomers, who think that a GRB of this strength strikes Earth only once every 10,000 years. Even its afterglow was so absurdly incandescent that JWST was able to locate the source supernova only several months later. "It was mind-blowing," Kilpatrick says. It was the most intrinsically energetic event ever seen.

We don't need to conjure up a nickname for GRB 221009A. Astronomers did it themselves, calling it the Brightest of All Time, or the BOAT. And its origin is pretty clear. Eons ago a star up to 30 times the mass of our sun collapsed and perished, forming a black hole. Aside from its brightness, everything else about the BOAT fits the model of a typical long-duration GRB. That, along with its proximity, has allowed scientists to test out another theory about GRBs.

R-process elements certainly come from neutron star mergers. But these mergers are far too rare to explain the abundance of heavier elements in the universe. Could collapsing supermassive stars forge the remainder? The BOAT, scientists thought, would reveal the answers. So they trained JWST on the blast, hoping to see evidence for at least some of those heavier elements. "We didn't see any," Blanchard says—not a smidge. That seems to leave astronomers with a serious accounting problem. They would love to see several more BOAT-like events close to home to test out this elemental theory on more than one occasion. "If these all happened in the Milky Way, we would have all of the answers," O'Connor says. "If we didn't die."

By last year astronomers knew they were in trouble. "We know that our classification scheme has been wrong for 30 years," O'Connor says, "or simplified." That's when another event, GRB 250702B, revealed itself in all its relentless, savage glory.

On July 2, 2025, a burst with energy equivalent to 1,000 suns blazing for 10 billion years lit up Fermi's detectors, heading from a

dusty galaxy about eight billion light-years from Earth. It was potent but not as bright as the BOAT. But whereas the BOAT lasted for 10 minutes—quite persistent for a GRB—this event endured for at least seven hours, easily earning the new record. Let's call this bringer of light Lucifer. And its duration wasn't all that was notable about it. Lucifer also seemed to pulse—to dim, then reignite—three times. "You can't have a star explode three times in the space of a day," Rastinejad says.

> "We know that our classification scheme has been wrong for 30 years—or simplified." —Brendan O'Connor *University of Maryland, College Park*

Astronomers suggested a range of scenarios that could explain the three pulses, including one in which the jet is wobbling. From Earth's perspective, we'd see it go "on" and "off" as it beamed right at us, then listed ever so slightly away before coming back into view again, says Eliza Neights, a graduate student at the George Washington University. Other GRBs are also thought to pulse, so this feature of Lucifer hasn't thrown too many astronomers for a loop. Its duration, though, was utterly perplexing. "It was flagged immediately as being something strange," Neights says. "For the first month or so following it, we had no idea what it was. There were all these crazy hypotheses thrown out there."

One way to increase the duration of a GRB's jets would be to slow down the collapsing star's ingestion of ejected matter. To do that, you could spin the object faster, which would make it harder for the eventual black hole to hastily gobble up its entire meal. But spin the collapsing star around too fast, and it "will literally just fall apart," Burns says, preventing the rest of the cataclysm. That limitation nixes this option: nothing could be spinning fast enough to explain the jet's duration.

Astronomers were left with the troublesome task of pairing up two weird objects, smushing them together and hoping to get Lucifer as

a result. Neutron stars certainly don't work. An elephantine star becoming a black hole doesn't explain it either. "There are not many things that could look like this," O'Connor says. There's even a chance it's not a GRB at all, traditionally speaking. A study led by Neights and Burns describes what could be a potentially winning duo. The first part is a minuscule black hole, roughly three times the mass of the sun and only about 10 miles across. The second is a similarly lightweight star. And billions of years ago, long before our solar system existed, they might have been locked in a deadly dance.

The black hole is so small that it wouldn't engulf the star in the blink of an eye. Instead it would peel off its hydrogen fuel, leaving behind a remnant helium core. Nothing especially ruinous happens during this theft—until the black hole plunges into the burgled star. There "it starts rapidly consuming material and shredding it from inside," Neights says. The stellar core implodes into the black hole, and a GRB fueled by a vast amount of material from all around it is born. When the jets fire out of its poles, they puncture through the ailing star's shell, causing it to explode as a supernova. The process is baroque—complicated and messy. "That's the best explanation we've been able to find," Neights says. It might be the correct one. But Lucifer remains a puzzle with no clear answers.

Cendes says she wouldn't bet on anything right now, but it's "something wonky with a black hole." It's hard to prove any theories, she says, "when you have a sample size of only one." Lucifer is, for now, a one-of-a-kind event. Perhaps it always will be. Either way, along with Dynamite, Bonfire and the BOAT, Lucifer demonstrates that our comprehension of GRBs isn't as solid as we once thought.

Are scientists hindered by a weak grasp of the physics driving GRBs? Do they not have enough advanced space observatories seeking out fainter, perhaps stranger GRBs? Or are astronomers hampered by nothing more than chance in that they just haven't

seen the particular GRBs that will allow them to solve their mysteries? “It’s all three,” O’Connor says.

The energy monsters of the universe will probably remain somewhat unknowable well into the future. But astronomers will persist in their quest to understand them—not just because they are curious but because GRBs are perfect for impatient scientists. The universe is rife with galaxies, stars and worlds that change very little during an astronomer’s brief time on Earth.

“So much of the universe is happening on these billion-year timescales,” Rastinejad says. But nearly every night we see a GRB’s furious ignition somewhere far from home. “I think it’s very fun and very cool to be working with things that are happening on human timescales,” she says. Blink, and you’ll miss them.

Robin George Andrews is a volcanologist and science writer based in London. His most recent book is *How To Kill An Asteroid* (W.W. Norton, 2024).

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https://www.scientificamerican.com/article/these-extreme-cosmic-explosions-are-breaking-physics

The Science of Health

- **What actually makes ultraprocessed foods unhealthy?**
  Scientists can’t figure out why ultraprocessed foods are unhealthy for us—and new studies suggest that the “processing” isn’t the problem

Scientists can’t figure out what makes ultraprocessed foods unhealthy

Scientists can’t figure out why ultraprocessed foods are unhealthy for us—and new studies suggest that the “processing” isn’t the problem

By Lori Youmshajekian edited by Allison Parshall



Jay Bendt

For breakfast, instant oatmeal. For lunch, a turkey sandwich on whole wheat bread. For dinner, a pasta bake with mozzarella cheese, and for dessert, low-fat yogurt and dried fruits. This diet is quite healthy and meets federal dietary guidelines—yet almost all of the calories come from ultraprocessed foods.

In countries such as the U.S. and the U.K., ultraprocessed foods make up at least half of the average diet. Research from the past few years has linked eating a lot of them to heart disease, diabetes and early death. Those findings have put the foods on everyone’s radar—including federal policymakers in the U.S., who put out new dietary guidelines this year telling the public to avoid highly

processed foods. The idea is that the processing itself might have effects on the food that make it easier for us to eat much more of it without noticing and for our body to digest it quickly or even that certain added ingredients such as preservatives and emulsifiers might be affecting our gut and how we handle blood glucose levels.

But is the problem really the processing, or could it be the kinds of nutrients these highly processed foods tend to contain? Ultraprocessed foods are often laden with ingredients we're told to avoid, and new findings are adding to the evidence that it's the nutrient profile—such as low fiber content and high amounts of sugar and saturated fat—more than the processing that affects our health in the long run. This could explain the existence of "healthy" ultraprocessed foods, such as yogurt and whole-grain breads, and suggests that classifying foods based on processing is trickier than it seems because it could steer people away from foods that are possibly doing them no harm.

To find out whether all ultraprocessed foods are equally bad for us, a team at the National Institutes of Health looked at national dietary survey data from more than 5,000 people. The researchers identified foods already labeled as "ultraprocessed," according to a common food-classification system called Nova, and they checked whether they were also high in the things dietary guidelines tell us to cut back on: added sugar, salt, saturated fat, refined grains and alcohol.

Unsurprisingly, about 90 percent of the calories that people consumed from ultraprocessed foods came from items that were high in nutrients to avoid. The researchers linked these same foods to higher body mass index and body fat, elevated insulin levels and increased inflammation.

But notably, the ultraprocessed foods that contained fewer nutrients to avoid, such as canned beans and vegetables, low-sugar yogurt, diet sodas and low-fat products, weren't correlated with the same

adverse health indicators. This result suggests that the health effects linked to ultraprocessed foods might derive from their nutrition profile—although the observational nature of the data makes it difficult to identify what's really behind the effects. The preliminary findings were presented at the American Society for Nutrition's July meeting.

Leah Lipsky, the study's lead author and a nutrition epidemiologist at the NIH, says the research doesn't mean people should eat a diet composed of only ultraprocessed foods with a better nutritional profile. "That's not quite what we're trying to get at," she says. Instead it suggests that the "ultraprocessed" framing isn't the best place to begin to understand a food's health effects. Lipsky also notes that some food manufacturers are reformulating ultraprocessed foods to market them as more healthful choices. But this approach misses the point, she says. "We really want to try to encourage people to eat more foods straight from the produce section" rather than relying on the "ultraprocessed versus nonultraprocessed" distinction.

Other large studies have also found that the link between ultraprocessed foods and disease risk is being driven by specific types of foods within the category. In particular, processed meats and cheese, as well as sweetened drinks, pose more of a risk, whereas foods such as breads, cereals and plant-based alternatives don't have the same associations with poor health. In 2023 a team at the U.S. Department of Agriculture built a menu that met federal dietary guidelines even though almost all of the calories came from ultraprocessed foods, adding weight to the idea that not all these foods are necessarily "bad."

Yet some evidence suggests the processing still matters. In 2019 another team at the NIH reported on 20 adults who ate ultraprocessed and unprocessed diets with food options carefully matched for calories, sugar, fat, salt and fiber. On the ultraprocessed diet, people ate roughly 500 extra calories a day and

gained weight. In a similar study reported last year, researchers in England gave 50 people two different diets, both designed to meet the U.K. healthy-eating guidance but with one consisting of ultraprocessed foods and the other of minimally processed ones. Both caused weight loss, but people lost twice as many pounds on average on the minimally processed diet.

These studies, though short in duration, all point in the same direction, says Marion Nestle, professor emerita of nutrition at New York University: people have better health outcomes when they eat less ultraprocessed diets. Ultraprocessing physically alters the structure of some food molecules, she explains. Corn chips, for example, start as whole corn kernels, which get cooked in an alkaline solution to loosen the hull and ground into wet dough that is then formed into shapes and fried. This process eliminates the physical structure of the corn and creates something resembling a predigested food—which the gut digests faster than it would a whole food, Nestle says.

Whether it is how soft and easy to chew these foods are or their tasty combinations of fat, salt and sugar, “something about [ultraprocessed foods] encourages people to eat more without realizing it,” Nestle says.

The processing and the nutrient makeup may both matter. David Julian McClements, a food scientist at the University of Massachusetts Amherst, says the foods that end up in the ultraprocessed group tend to be the ones nutritionists have been warning about for decades anyway—those that have been broadly but imprecisely referred to as “junk food.” He thinks the processing does have real effects, but they’re “much smaller than the [nutritional]-composition effects.”

The experts don’t think it’s realistic for most people to stop consuming ultraprocessed foods altogether. “Most people haven’t got the time, the energy, the money or the inclination” to cook from

scratch, McClements says. His suggestion is for scientists to work out which processed foods cause harm and why, redesign those specifically and then test whether the redesign actually helps—although such trials are slow and expensive.

Nestle’s advice to us is simpler. “Eating less junk food is good health advice,” she says.

Lori Youmshajekian is a science journalist who reports on consumer health, environmental issues and scientific misconduct. She holds a master’s degree in science journalism from New York University and has written for *National Geographic*, *Wired* and *Retraction Watch*, among other outlets.

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https://www.scientificamerican.com/article/what-actually-makes-ultraprocessed-foods-unhealthy

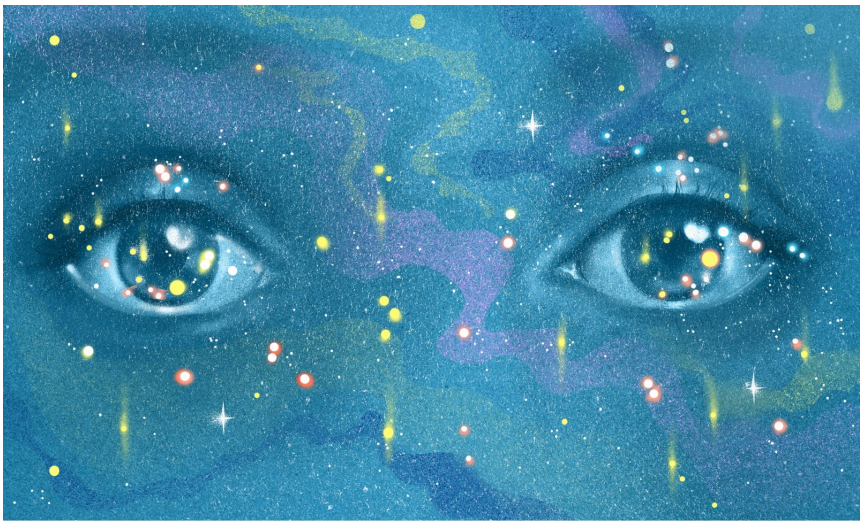
Meter

- **Poem: ‘Life in Bilateral Wonder’**
  Science in meter and verse

Poem: ‘Life in Bilateral Wonder’

Science in meter and verse

By Gurupreet K. Khalsa edited by Dava Sobel & Clara Moskowitz



Masha Foya

“*My favorite number is 2 ... 2 is kind of responsible for the concept of whether things are even or odd.*”
—Melanie Matchett Wood, American mathematician

Two—the first even number, prime, balancing two ones like weights on a meter stick rocking on a fulcrum. We balance on two feet, and if we wobble we can stretch out our two arms on two sides and focus on a point with our two eyes.

> Information gobbled
> sorting itself like black
> and white marble tiles
> in the foyer

Two eyes, either side of the nose, nicely central on my face, not too high or too low, not too spread out, binocular vision, and what

would my vision be like if I were a flounder flouncing along sandy bottoms of bay shallows, both eyes perched on the top half of my flattened body? Does the flounder roll his eyes in exasperation because his eyes don’t line up with his puckered mouth?

Two nevers
multiplied
equals always

A shape cannot have only two sides—even a slice of bread, which, when dropped, lands on the floor buttered side down (symbolic of life’s frustrations and setbacks, perhaps the flounder can relate), has a third dimension, and if I say it twice it’s a bi-mention.

Bicameralization:
“It takes all the running you can do
to keep in the same place,”
says the Red Queen

Binary bifold, life’s bilateralism. Bihydrate; that is, drink two glasses of water, which is a liquid not unlike the primordial quark-gluon plasma ooze except *that* is created at a temperature of four trillion degrees and I wonder how they do that because wouldn’t the laboratory explode and even two glasses of water would not be enough to quench such a fire. So the quark-gluon plasma is said to be a liquid, and don’t get me started on entanglement entropy, which appears to be a circumstance where not only is everything like a knotted tangle of yarn but is also disaggregating faster than you can say upsilon (which has two components of bottom quarks and anti-bottom quarks—of course, a thoroughly dichotomous situation). And the flounder, most decidedly pro-bottom, flutters.

I think I’ll take
a walk in spacetime
wiggle my fingers
among the strings

Nothing odd about that.

This poem resembles a haibun—a Japanese literary form that combines prose with haiku.

Gurupreet K. Khalsa, a current resident of Alabama, considers connections, space, time, reality, illusion and possibility. She holds a Ph.D. in instructional design and is a part-time instructor in graduate education programs. Her work has appeared in multiple print and online journals.

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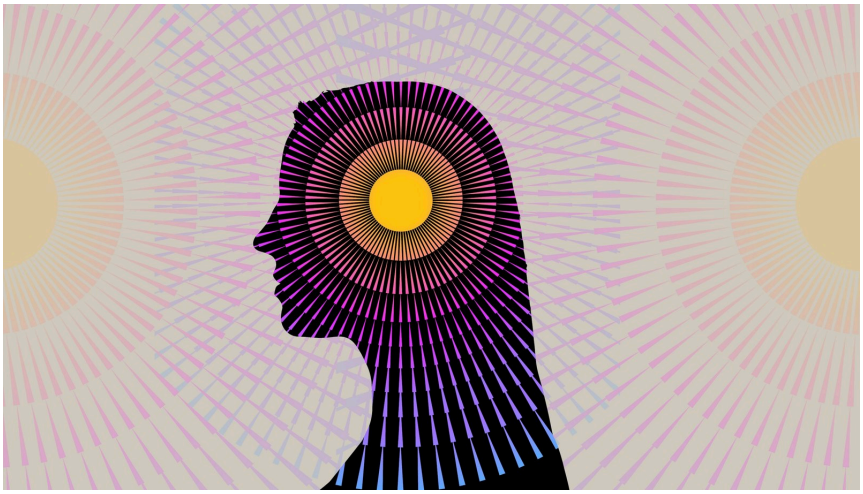
Mind Matters

- **How psilocybin alters consciousness to create a sense of ‘oneness’**
  Psychedelics can create a new “hidden order” in brain activity that could explain how these drugs alter our sense of self

How psilocybin alters consciousness to create a sense of ‘oneness’

Psychedelics can create a new “hidden order” in brain activity that could explain how these drugs alter our sense of self

By Allison Parshall edited by Claire Cameron



GeorgePeters/Getty Images; *Scientific American*

Taking psychedelics can dissolve the boundary between you and the rest of the world, leaving a profound feeling of “oneness.” Now scientists have revealed that specific changes to the brain may cause this altered sense of consciousness. The results, published in *Nature*, show how taking psilocybin—an active ingredient in magic mushrooms—causes brain networks that typically process the self and the outside world separately to become integrated.

“It reframes our understanding of the psychedelic experience by providing a biological basis for some of the meaning and coherence people report,” says lead author Devon Stoliker, a computational neuroscientist at Monash University in Melbourne, Australia.

During a psilocybin trip, brain activity becomes more random or chaotic. Networks of neurons that usually fire together become

desynchronized, and new connections can form. This shake-up of the brain's existing patterns of connectivity may contribute to psilocybin's potential therapeutic effect for certain mental health conditions. But the experience is also paradoxical, Stoliker says: "If the brain becomes more disorganized, then why do some people have highly meaningful and coherent experiences?"

In their study, Stoliker and his colleagues purported to find a "hidden order" in this chaos. Curiously, this order emerged in people whose conscious experience had been most strongly altered by psilocybin.

The researchers gave 62 healthy adults with no psychedelic experience a dose of psilocybin, then used functional magnetic resonance imaging to scan their brains in varying contexts: resting, meditating and listening to music—all with their eyes closed—and while watching a movie. Many participants experienced a blissful state of oneness, which the researchers call "embeddedness." One participant reported, "I melded with the MRI machine, floor, walls, air. It was like nothing existed except for me or me as part of something bigger."

The scientists measured how synchronized different brain networks were with one another. They reported that systems responsible for processing sensory input from the outside world became less connected with the rest of the brain, whereas those responsible for higher-level processing became more connected. "Broadly, [this result] suggests a shift in the balance between internally generated and sensory processing," Stoliker says.

Next, the researchers used machine learning to identify patterns hiding in the data. Specifically, they analyzed individual participants' brain activity while the trip evolved and as the participants did different tasks. People who had less profound experiences had widely variable brain activity—even if they were engaged in the same actions as their peers. But for those who felt a

strong sense of oneness, key dimensions of their brain activity tended to look the same across the group, forming distinct patterns for each activity they performed.

In other words, Stoliker explains, “network activity became more similar when participants experienced less separation between self and world.” The findings offer a neurobiological explanation for why the context of a psychedelic trip—where and when it happens or what someone does during it—can so greatly influence the experience.

“Context sensitivity is a double-edged sword,” says Nico Dosenbach, a neurologist at Washington University in St. Louis, who was not involved in the study. The results highlight the importance of having a “controlled, positive setting, which would make therapy more expensive and difficult to deliver—but not unrealistic,” he says. Computational neuroscientist Adeel Razi, the study’s senior author, notes that using machine learning to study individual brains during a trip could help reveal why the same drug produces such different experiences in different people.

Investigators are currently split on whether the apparent benefits of psychedelics stem from the conscious experience of tripping or from the underlying changes in brain connectivity, explains Lorenzo Pasquini, a neuroscientist at the University of California, San Francisco, who wasn’t involved in the study. The new work doesn’t answer that question, he says, but it hints at the importance of people’s conscious experience.

On a more philosophical level, the findings suggest that the division we typically perceive between ourselves and the world is actively managed by the brain, Stoliker says. “Rather than psychedelics simply creating an illusion that the self and the world are one,” he says, “they may temporarily reduce the brain processes that normally maintain this separation.”

Perhaps the illusion is not the sense of oneness but our perceived separation from the world to begin with. “Psychedelics may offer a temporary glimpse of a bigger picture,” Stoliker says.

Allison Parshall is associate editor for mind and brain at *Scientific American* and she writes the weekly online Science Quizzes. As a multimedia journalist, she contributes to *Scientific American*'s podcast *Science Quickly*. Parshall's work has also appeared in *Quanta Magazine* and Inverse. She graduated from New York University's Arthur L. Carter Journalism Institute with a master's degree in science, health and environmental reporting. She has a bachelor's degree in psychology from Georgetown University.

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https://www.scientificamerican.com/article/how-psilocybin-alters-consciousness-to-create-a-sense-of-oneness

The Science of War

- **How a fusion start-up could help maintain the U.S. nuclear arsenal**
  The U.S. hasn’t detonated a nuclear weapon since 1992. Now a fusion start-up whose co-founders include former national lab scientists could help conduct experiments that make sure the stockpile works

How do you test a nuclear arsenal without blowing it up?

The U.S. hasn't detonated a nuclear weapon since 1992. Now a fusion start-up whose co-founders include former national lab scientists could help conduct experiments that make sure the stockpile works

By Sarah Scoles edited by Eric Sullivan



Richard Mia

A new Albuquerque mixed-use neighborhood called Mesa del Sol, near the airport, is still under construction. But it's coming together: earth-toned homes have popped up; rock-based snake sculptures line the medians. There's also a road called Pacific Fusion Drive—and it now leads toward a construction site where a company of the same name broke ground on a $1-billion fusion research and manufacturing campus on a hot, dusty August day.

The bigwigs in attendance—New Mexico's governor and senators, Albuquerque's mayor, and leaders from the nuclear establishment,

among others—escaped the heat and dust inside a fancy air-conditioned tent with a carpeted floor. They spent the morning talking about what Pacific Fusion could do for New Mexico, the clean-energy sector and nuclear-weapons research.

Early in the day's events, National Nuclear Security Administration head Brandon Williams stepped up to the mic, two pieces of heavy machinery holding a U.S. flag behind him. "I represent the NNSA," he said. "We're about 70 percent of the budget of the Department of Energy. Let me just tell you, the NNSA is not sending anybody to Mars, and we're not reading your text messages." It was not to be confused, in other words, with NASA or the National Security Agency.

The NNSA maintains and is currently modernizing the U.S.'s nuclear stockpile. The U.S. hasn't blown up a nuclear weapon in an explosive test since 1992. Instead weapons scientists rely on smaller-scale experiments and sophisticated simulations to understand how the weapons should behave. Now, under a public-private partnership agreement with Pacific Fusion, some of that nuclear-security work could happen at a commercial fusion facility.

At national laboratories such as Sandia, Lawrence Livermore and Los Alamos, scientists use enormous machines—for instance, a Sandia fusion facility called the Z Machine, just a few miles from that eerily cool and undusty tent, and the National Ignition Facility (NIF) at Lawrence Livermore in California—to mimic aspects of bombs. The results help scientists keep nuclear weapons safe and functional without having to set one off.

The co-founder of Pacific Fusion, Keith LeChien, used to work at both those labs and on those machines. While there, he saw a way to improve on their technology: do fusion bigger, better and more efficiently. He started Pacific Fusion to build that idea out.

According to the memorandum of understanding (MOU) that Williams was there to sign on behalf of the federal government, some nuclear-security research may take place at Pacific Fusion's future facility. "It's the foundation of investment from our national labs and our government that makes today possible," Williams said to the audience. He called up LeChien so he, too, could sign. "You go first," Williams said. "Give me a second to back out." People looking on, some still munching on mini quiche and green-chile breakfast potatoes, laughed.

Publicly, Pacific Fusion talks mostly about its future ability to supply electricity, a promise the fusion industry has made for years. But the technology is also useful for national security, and those applications don't require the same level of power-generating fusion success that the electrical grid does. "With atomic science, there will always be dual use, and that is also true for what Pacific Fusion wants to do," Williams told me in a cozy media-only section of the event.

> Pacific Fusion aims to create a fusion reaction an order of magnitude more powerful than those at the national labs.

In modern nuclear weapons, conventional explosives set off a fission reaction—the splitting of big atoms—whose intensity then ignites a fusion reaction that combines light atoms. Understanding exactly how fusion works, in the grittiest detail, is therefore of extreme interest to nuclear-weapons scientists. So, too, is the way objects react when a fusion reaction booms nearby. Can, say, a circuit board that's essential to one nuke survive the detonation of another, nearby nuke? "Systems have to be able to withstand high pressure, high temperature, all with energy released in just a very short time," Sandia director Laura McGill said during a panel on Pacific Fusion's defense applications.

Figuring out whether those systems can survive is a big part of what Sandia's Z Machine and Lawrence Livermore's NIF were

built for. They are both “pulsed-power” instruments: they store, compress and release a huge amount of energy in a short period, sending it toward a tiny container of fusion fuel. That pulse crushes the fuel, driving its atoms to unite and release their own huge amount of energy, whose nature and effects scientists can study.

LeChien and many Pacific Fusion employees have been on the government side of this work (Los Alamos’s director Thom Mason, for instance, told me he first heard of the company when it started hiring his staff). LeChien started working on the Z Machine as an intern in 2001. He went on to NNSA headquarters, where he ran a fusion program that oversaw Z and NIF, and then he moved to Livermore. During his time working at these federal entities, he co-invented a piece of equipment, the impedance-matched Marx generator, that could potentially improve their fusion-igniting processes.

At Z, storing up the initial charge takes a few minutes, and the energy pulse then has to be temporarily compressed several times so that when it smashes against the fuel, it packs a huge, quick punch. Pacific Fusion’s system, at least in theory, charges on the scale of seconds and then requires only one compression.

The company hopes that on the very ground where the fancy tent flapped, it can demonstrate a fusion reaction that releases more energy than goes into it, an achievement called net facility gain. It’s something that hasn’t ever actually been achieved. Pacific Fusion also aims to create a fusion reaction an order of magnitude more powerful than those at the national lab facilities. This process, which it calls “high yield,” “has interesting use cases for different defense applications,” LeChien says.

NNSA is being encouraged to pursue partnerships like this one. Other parts of government already contract out rocket launches, cloud storage, satellite intelligence and artificial-intelligence analysis to the tech sector. “We don’t move fast enough, and we

don’t possess all of the innovation,” Williams says of his agency. “We do provide a fantastic foundation, but we really need the innovation of private industry to move things quickly.” And that’s important right now, he adds, because China is fast building up its fusion capabilities.

For now, much of this work is still hypothetical; Pacific Fusion’s campus, of which the demonstration facility is part, is currently under construction at the end of Pacific Fusion Drive.

Near the end of the event in August, ceremonial shovels stuck into the earth shone in the sun behind Williams. “What you’re seeing today in this MOU is an acknowledgment that the foundational work of atomic science is no longer exclusively the purview of a state government,” he said. “There’s a first-place prize for deterrence: first place is freedom and prosperity. And there’s a second-place prize for deterrence. It’s called annihilation and subjugation.”

Sarah Scoles is a Colorado-based science journalist and a contributing editor at *Scientific American*. Her newest book is *Countdown: The Blinding Future of Nuclear Weapons* (Bold Type Books, 2024).

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https://www.scientificamerican.com/article/how-a-fusion-start-up-could-help-maintain-the-u-s-nuclear-arsenal

Science Crossword

- **Science crossword: Scroll bars**
  Play this crossword inspired by the November 2026 issue of Scientific American

Science Crossword

By Ella Dershowitz

This crossword is inspired by the most recent issue of Scientific American. *Read it here. Print readers, check your answers by selecting "Assist" above and then "Reveal Grid" or by selecting "Print" and then "Solution."*

We'd love to hear from you! E-mail us at games@sciam.com to share your experience.

Ella Dershowitz is a crossword puzzle constructor who has been published in the *New York Times*, the *Los Angeles Times*, the *Wall Street Journal* and AV Club Xwords (AVCX), among other outlets. Besides wrangling tiny boxes for puzzles, she is an actress who has appeared in tiny boxes on film and television, as well as in slightly larger boxes at off-Broadway productions in New York City, theaters in the Bay Area and those throughout the Northeast. She is currently playing the title role in an off-Broadway production of *Mrs. Stern Wanders the Prussian State Library* through January 19, 2025.

https://www.scientificamerican.com/article/science-crossword-scroll-bars

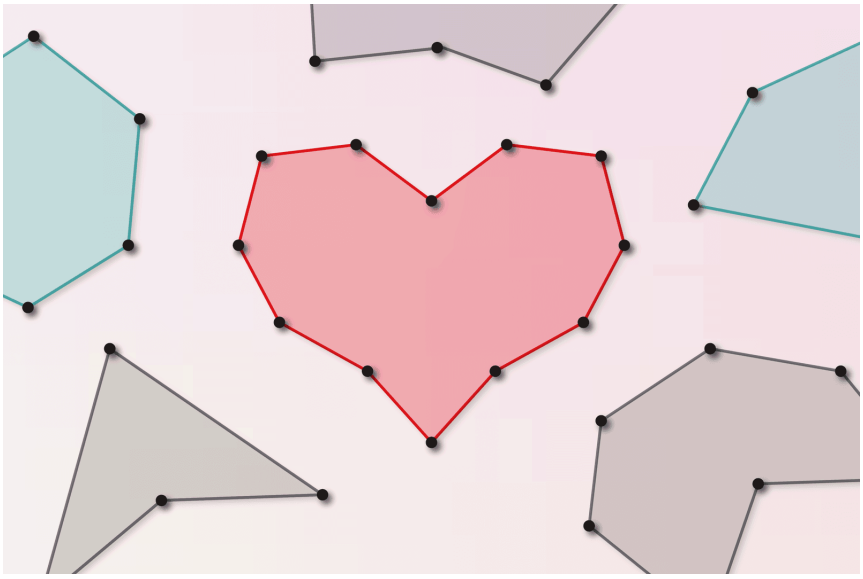
Math

- **How the ‘happy ending problem’ launched a new branch of math—and a romance**
  A puzzle about dots on a page led to one of the most profound areas of modern math

How the ‘happy ending problem’ launched a new branch of math—and a romance

A puzzle about dots on a page led to one of the most profound areas of modern math

By Jack Murtagh edited by Jeanna Bryner

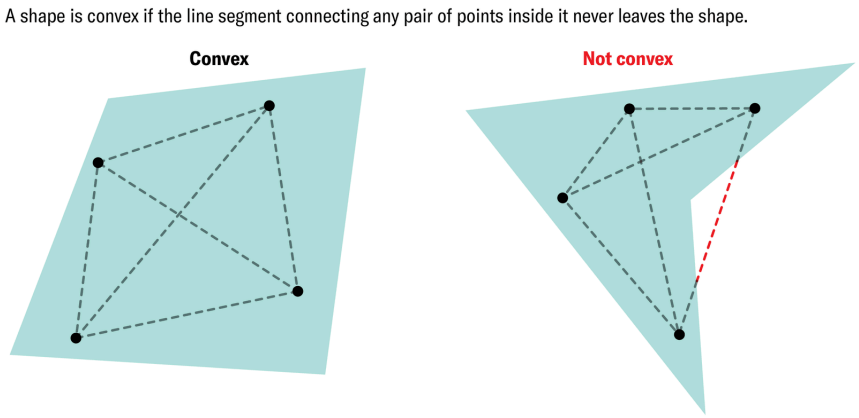


Amanda Montañez

In the early 1930s a group of bright university students held gatherings in Budapest to discuss math. Among them was Paul Erdős, an eccentric prodigy who would go on to become the most prolific mathematician in history. Other attendees included mathematicians George Szekeres and Esther Klein. One day Klein presented a puzzle to her friends: a deceptively simple question about dots scattered across a page. Unbeknownst to them, the puzzle would help launch a new branch of modern math about order among chaos and spark a romance that would last the rest of Klein’s life.

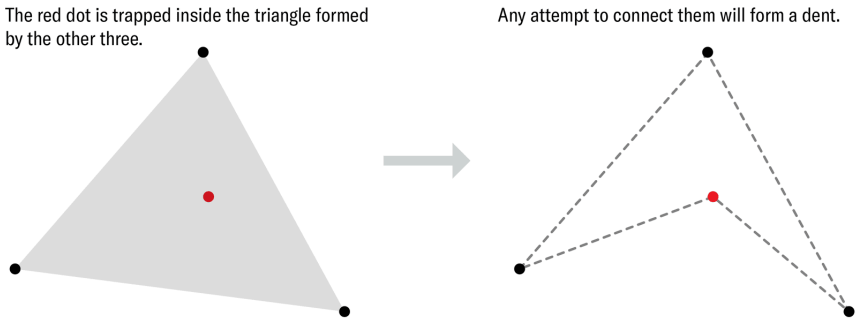
Here’s what Klein presented to her pals: speckle a page with five dots placed wherever you like, as long as no three dots fall along

the same straight line. Will four of those dots always form the corners of a four-sided shape with no “dents” or crossed sides? In mathematical terms, must some four of the five dots form a convex quadrilateral?



Amanda Montañez

With only four dots, it’s clearer that a convex quadrilateral might elude you, depending on where the dots fall. But the five-dot case changes things.

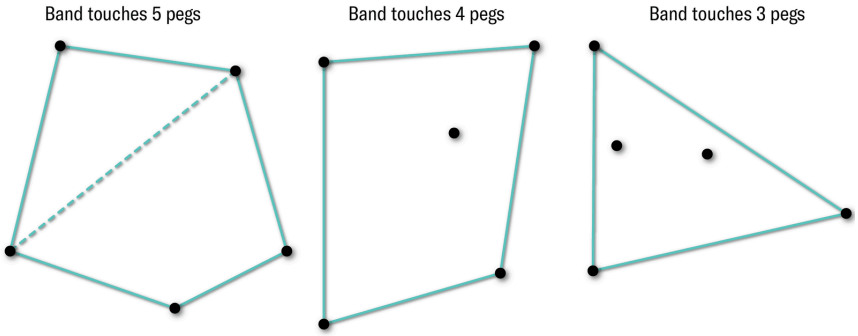


Amanda Montañez

Grab a pencil and test it for yourself. Place five dots at random, with no three dots on the same line, and hunt for a four-cornered, dentless shape. Then try to be clever by attempting to intentionally arrange the dots so that they deny you such a shape. You will fail every time. No matter how you place five dots, four of them will always form a convex quadrilateral. Klein proved this with an elegant argument.

Imagine the dots protrude from the page to form pegs. If you stretch a rubber band taut around the outermost pegs so that all of them are surrounded, it will trace a convex boundary because rubber bands without slack don't form spontaneous dents. There are three cases to consider: the band touches five pegs, the band touches four pegs with one trapped inside, or the band touches three pegs with two trapped inside.

It can't touch only two pegs unless all five lie in a straight line, which the rules forbid. If the band touches four pegs, then we've already found our convex quadrilateral (four sides) with those four corner points. If it touches five pegs, forming a convex pentagon, then you can lift the band off of any one of the pegs, and it will snap to the remaining four to form a convex quadrilateral. The tricky case occurs when the band touches only three points, forming a triangle with two interior points.

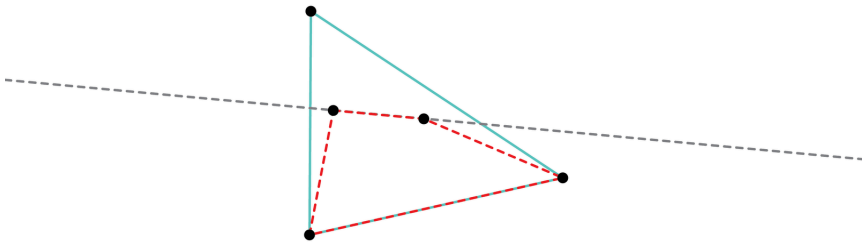


Amanda Montañez

Does this three-point configuration always admit a convex quadrilateral? It does. Draw a straight line through the two interior points and extend it across the page. That line slices the plane into two sections, one of which will contain two corners of the outer triangle; see the bottom half of the graphic below).

Those two corners and the two interior dots always form a convex quadrilateral together. Why? For four dots, the only way to fail is if one is trapped inside the other three, unable to participate in the boundary. But none of these four dots can get trapped. The two triangle corners are among the farthest-out points of the entire

cluster, so nothing can box them in. And if you take those two corners and form a triangle with either of the other dots, you'll notice the fourth dot always sits outside of it, unenclosed.



Amanda Montañez

After impressing her friends with this puzzle and its neat resolution, Klein did what mathematicians do: she wondered how it generalizes. How many points would you need to guarantee, say, a convex pentagon, a convex hexagon or a convex 100-gon? Or perhaps no amount of points would be guaranteed to produce these larger shapes. Could it be that you could arrange any number of points while avoiding the creation of a convex hexagon?

Erdős and Szekeres settled part of the mystery in 1935. They proved that for any number of sides you choose, some finite number of points is always enough to ensure that shape: gather that amount of dots, with no three in a line, and a convex polygon of the desired size is guaranteed. Enough dots will force a convex hexagon. The same goes for a convex 100-gon—you just need more dots. With the correct number of dots, no arrangement, however cunning, can dodge these shapes forever.

Erdős dubbed this the "happy ending problem" because somewhere in the course of trading geometric drawings, Klein and Szekeres fell in love. They married in 1937 and enjoyed a storied life together. As a Jewish couple facing the rise of Nazi persecution in Europe, they fled, spending the war years as refugees in Shanghai, where their first child was born. In 1948 they resettled in Australia and built a life steeped in math: Szekeres worked as a professor and became president of the Australian Mathematical Society while

Klein taught math at Macquarie University in Sydney and ran extracurricular sessions where she posed geometric problems for students, just as she had done with her friends in her youth.

The legacy of the happy ending problem extends beyond romance. It helped to seed an active branch of modern math known as Ramsey theory. While writing their paper, Erdős and Szekeres discovered a theorem from a young British polymath named Frank Ramsey, who had already done groundbreaking work in philosophy, economics and math by his untimely death at age 26. Ramsey wrote in more abstract terms, but his celebrated theorem is best framed this way: Imagine a cocktail party where some attendees are acquaintances and some are strangers. For a suitably large party, will there always be some group of people who are either all mutual acquaintances or all mutual strangers? Ramsey proved that, yes, with enough invitees, it becomes impossible to handcraft a guest list that avoids cliques of acquaintances or strangers. To ensure a bigger clique, you simply need a bigger party.

How much bigger is a thornier question that remains open today, but Erdős was the first person to make significant progress on it. Ramsey's theorem and the happy ending problem share a common insight: once systems grow large enough, pockets of order become unavoidable. That's the central thread of Ramsey theory. You can scatter points or invite guests as haphazardly as you want, and yet tidily arranged clusters will still emerge. It's a vibrant branch of research with roots reaching into many disparate areas of math. Erdős was the central figure in the field throughout his career, and his work on the happy ending problem helped to kick that off.

Ramsey theory is profound, but it comes with a signature frustration. Proofs in the field tend to guarantee that order must appear somewhere at some point without revealing where it hides or exactly how large a structure must grow to guarantee it. For the happy ending problem, Klein showed that five points guarantee a

quadrilateral, and Erdős and Szekeres reported that nine points necessitate a pentagon, but eight can still avoid one; concrete quantifications stalled there for 70 years. The next number was not found until 2006. Szekeres, still circling the problem that had introduced him to his wife more than seven decades earlier, teamed up with Australian mathematical physicist Lindsay Peters and marshaled an extensive computer search to confirm that 17 points necessitate a convex hexagon, whereas 16 can still avoid one. The paper was published after Szekeres's death.

The discovery matched a conjecture that Erdős and Szekeres had put forth as young men. They predicted that $2^{(n-2)} + 1$ points is the minimum needed to force an *n*-sided convex polygon. The quadrilateral, pentagonal and hexagonal cases all obey this formula, but no larger shapes have been confirmed. Erdős left behind a $500 cash prize to anyone who proves that it holds in general. The prize remains unclaimed, although in 2016 mathematician Andrew Suk, then at the University of Illinois Chicago, came tantalizingly close.

Klein and Szekeres both died in 2005, within an hour of each other. They left behind two children and a mathematical legacy worthy of the name "happy ending."

Jack Murtagh is a freelance math writer and puzzle creator. He writes a column on mathematical curiosities for *Scientific American* and creates daily puzzles for the Morning Brew newsletter. He holds a Ph.D. in theoretical computer science from Harvard University. Follow him on X @JackPMurtagh

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https://www.scientificamerican.com/article/how-the-happy-ending-problem-launched-a-new-branch-of-math-and-a-romance

Q&A

- **Nobel laureate Omar Yaghi on how he fell in love with chemistry**
  The Nobel laureate shares how artificial intelligence will change the way science is done

Omar Yaghi

The Nobel laureate shares how artificial intelligence will change the way science is done

By Jeanna Bryner



Justin Sullivan/Getty Images

Omar Yaghi is a professor of chemistry at the University of California, Berkeley, and a joint winner of the 2025 Nobel Prize in Chemistry for his pioneering work on metal-organic frameworks (MOFs). Yaghi is also founding director of the Berkeley Global Science Institute, which is aimed at creating research centers in developing countries. He is co-director of the Kavli Energy NanoSciences Institute, the California Research Alliance by BASF and the Bakar Institute of Digital Materials for the Planet. *Scientific American* spoke to Yaghi about the state of science today, the coming artificial-intelligence revolution, and how researchers can prepare.

An edited transcript of the interview follows.

How would you describe the current state of American science?

I think that American science, the culture of innovation, has always been strong because of something the U.S. system has that other systems don't: the mentoring between a scientist and the students. This mentoring passes the knowledge from one generation to the next. The mentor also gives all they have, all they know, all their experiences, to the student with no reservations. This is how innovation gets passed on.

What fuels this innovation are top thinkers and researchers who are well funded by our government. For many, many years our funding was very competitive; if you worked hard and you were doing good research, you would get funding. The current state is not so encouraging because of the cutting back on grants and support of science by the very agencies that many university researchers rely on.

I think one problem is that science is becoming very expensive to carry out, so society is demanding some answers as to what this cost is leading to. And we often emphasize that "I have a product that could be deployed, that could be commercialized." We forget that our best products are these new young scientists who think in a different way, who can solve problems in ways that nobody else can. The human element of this—human education and the human ability to solve problems and to discover new things that change the world—is not as emphasized as the actual product that might result. And I think that's missing the point of why universities are there in the first place.

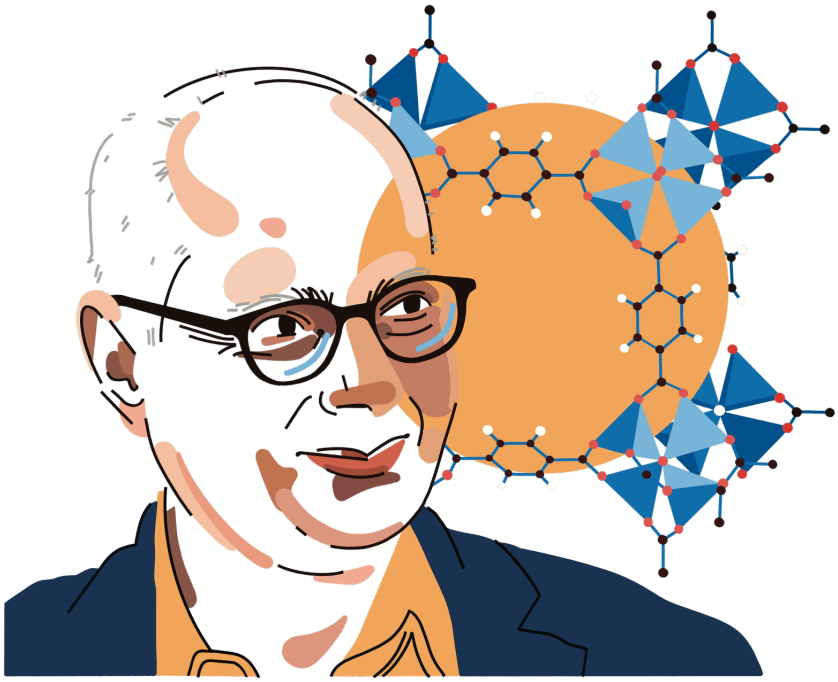
> "I think the love for things like [science] comes from the most humble and unexceptional circumstances." —Omar Yaghi
> *U.C. Berkeley*

What needs to change in American science?

Scientists need to engage in this artificial-intelligence revolution. We need to engage in these AI models to make them more useful, not just in speeding up tasks in the laboratory but also in suggesting questions that we normally wouldn't ask. That needs to happen as a matter of survival of the advanced research system in the U.S. We need to be engaged in this important AI revolution so that we can modernize our labs and be commensurate with how society is proceeding and the rate at which it's moving forward. That will be revolutionary for recruitment of the best minds among those young scholars because now they are going into a modern lab where they don't have to slow down.

What gives you optimism right now?

In a large society like the U.S., you can always find enough young people interested in doing research and stimulating their minds by doing the hard work. And not everybody has to be able to do that, just enough to form a nucleus of great thinkers who want to study and advance the frontiers. What worries me is that we are not preparing for what I think is an AI revolution. We need to reinvent how research is being done to account for the fact that AI can speed up discoveries and potentially carry out a lot of the tasks that are consuming our time and resources. In our labs, things proceed very slowly, and there is no reason for that, given that society is operating at the speed of light, whether it is trade or aviation or even [getting information more through] online resources and LLMs [large language models] than from some of their university classes.



Shideh Ghandeharizadeh

If you were given a $100-million grant that didn’t require any preliminary data or guaranteed milestones, what high-risk project would you start?

I would target the development of autonomous labs, where a researcher can sit at a terminal next to a box that will do the chemistry and instruct the robot on what to do, and the robot goes off and finds the conditions under which to make a new material, then characterizes the material and gives you data.

What was a eureka moment when you realized a big idea was going to work?

I think when I was talking to one of my students back in the mid-1990s. He had observed the formation of a beautiful crystal that looked like diamond, and he had taken this crystal out of its original solution, and it turned from looking like a diamond into looking like a white powder. The student deemed this observation unimportant because it meant that whatever compound was behind this change was not interesting, and it was not stable.

I said to the student, “Analyze it without taking it out of its original solution so that it doesn’t lose its diamondlike character.” When he analyzed that, we discovered that it is what we called MOF-5, the very compound that the Nobel Committee [for Chemistry] cited and the very compound that broke all records of porosity and started the field of MOFs. We looked at each other, and I knew we had done something significant.

He was at the Nobel celebration and reminded me of what I said when we saw the structure: “This is Nobel-worthy.”

What’s a favorite story from your early career?

When I was a kid, the way I fell in love with chemistry was through chemistry drawings, molecular drawings. I didn’t know they were molecules. But when I was 10 years old, I saw those drawings in a library, and I thought, “Wow, this is very interesting.” I thought I had discovered drawings that nobody had seen before. So it got me interested in the drawings. And then later I learned they were molecules, and the molecules are all around us. I became really, really interested in chemistry, and nothing since has turned me away from chemistry. I think the love for things like that comes from the most humble circumstances and the most unexceptional circumstances. It was not an intellectual moment by any means. It was more like, “Oh, I see these things look strange, but something is drawing me to them.” And I felt, deep down, that maybe this is a secret I can keep, as if nobody has seen them before.

In retrospect, when I think deeply about it, maybe I was looking for structure in my life. My life was chaotic. My humble home with many kids was quite chaotic, and we were not a well-to-do family. Maybe I saw that as a structure in my life, maybe something that I can think about away from the day-to-day circumstances.

How has AI changed your own research?

My field, reticular chemistry, is about making structures from molecular building blocks. We make porous materials that have applications in extracting water out of the air to make drinking water, taking carbon dioxide out of the air to make clean air, or storing hydrogen to deliver clean energy. The fact that they can be designed precisely on the atomic and molecular level makes them extremely important in customizing materials for very specific applications.

Two or three years ago a colleague and a young student in my lab decided that AI would be very important in furthering the field because you can get to your answers much faster. It's easy to make these materials but not so easy to crystallize them. So we've discovered that just by using ChatGPT, we could speed up the crystallization from years to weeks. Crystallization is the ability of the building units to come together in an ordered fashion based on the conditions under which you assemble them. That's a trial-and-error process of trying to find the right conditions under which all the pores are homogeneously distributed and the material can work cohesively and repeatedly for, in the case of water harvesting, many, many years.

Another example in my lab is that we are adapting machine-learning algorithms to a specific class of materials. Students who are using this machine-learning algorithm that has been adapted to our chemistry will discover twice as many new compounds compared with students not using it. So the answer to the question of "How did AI transform my field?": it's a revolution.

Jeanna Bryner is editor in chief of *Scientific American*. Previously she was editor in chief of Live Science and, prior to that, an editor at Scholastic's *Science World* magazine. Bryner has an English degree from Salisbury University, a master's degree in biogeochemistry and environmental sciences from the University of Maryland and a graduate science journalism degree from New York University. She has worked as a biologist in Florida, where she monitored wetlands and did field surveys for endangered species, including the gorgeous Florida Scrub Jay. She also received an ocean sciences journalism fellowship from the Woods Hole Oceanographic Institution. She is a firm believer that science is for everyone and that just about everything can be viewed through the lens of science.

https://www.scientificamerican.com/article/omar-yaghi

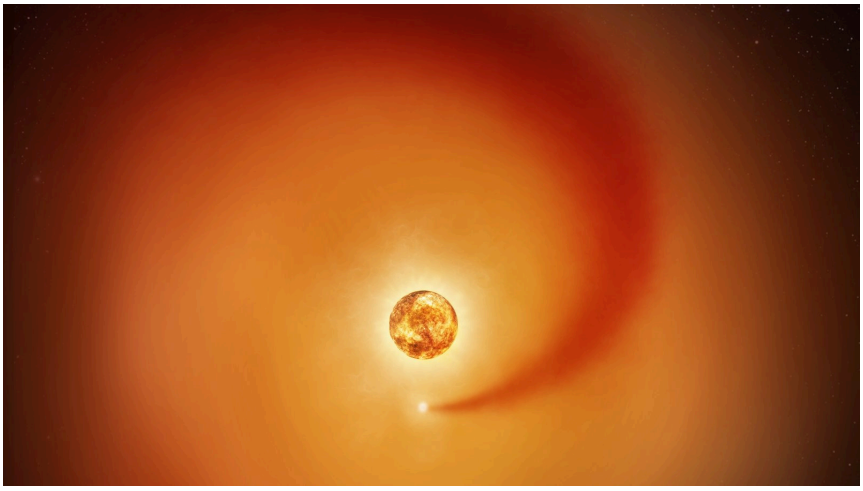
The Universe

- **When will Betelgeuse go supernova?**
  Betelgeuse, the sky’s most famous red supergiant, is well studied but poorly understood

What happens when Betelgeuse blows its top?

Betelgeuse, the sky's most famous red supergiant, is well studied but poorly understood

By Phil Plait edited by Lee Billings & Clara Moskowitz



This artist's concept shows the red supergiant star Betelgeuse and a companion star orbiting in the supergiant's tenuous outer atmosphere.

NASA, ESA, Elizabeth Wheatley (STScI); Science: Andrea Dupree (CfA)

Betelgeuse has always been something of a mystery. And by "mystery," I mean a source of endless fun, fascination and frustration for astronomers.

Betelgeuse is one of the brightest stars in the sky, marking the right shoulder of the constellation Orion. It's a red supergiant, a massive star with more than a dozen times the sun's heft, that is blasting out many tens of thousands of times as much light as our own star. It's also dying, already well on its way to becoming a supernova.

It's one of the most studied stars in the sky yet still puzzlingly difficult to understand.

Some two millennia ago Chinese astronomers reported Betelgeuse as being yellow. Has it changed color over the centuries? What does that mean for what its internal structure is and how it emits light? The star dimmed tremendously in late 2019, not returning to its typical brightness until early 2020, which caused much consternation among stargazers. It took years to show that the star had eructated out a huge cloud of opaque dust that blocked its light until the cloud dissipated enough to let Betelgeuse shine through once more. Could that happen again?

Astronomers have long known that Betelgeuse has an unexplained six-year-long periodic fluctuation in its brightness in addition to its less dramatic 400-day cycle. They suspected the cause was a second star—a "Betel buddy," if you will—orbiting the supergiant; the system dims when the companion star is behind the supergiant and hidden from our view. Astronomers observing Betelgeuse in 2024 found hints of a second star, and now a paper published in the journal *Astronomy and Astrophysics* strongly supports its existence. This stellar companion is apparently a luminous star that has three times the mass of the sun but is so close to the far brighter supergiant that its light had been hidden in the glare all this time. Confirming this mystery object's identity is pretty straightforward: observe Betelgeuse in 2027, when the object is half an orbit farther along compared with its first detection. If it appears on the other side of the star from where it was in 2024, it's a binary companion.

If so, it orbits the red supergiant (now called Betelgeuse A) at a minimum distance of about 1.3 billion kilometers, roughly the distance Saturn is from the sun. Even so, from the perspective of this companion star—Betelgeuse B—the supergiant is so swollen that it occupies half the sky. Red supergiants are awesome beasts.

> There is one thing we know for sure about Betelgeuse A: someday it will go supernova in one of the most titanic explosions.

Despite its mysteries, there is one thing we know for sure about Betelgeuse A: someday it will go supernova in one of the most titanic explosions this universe can generate. We don't know when, precisely—that's another mystery. It could be tonight, but most signs point toward it happening more like 100,000 years from now.

This cataclysmic end for the star is inevitable. In the core of Betelgeuse A, thermonuclear processes are at work: almost unthinkable heat and pressure cause helium nuclei to be merrily fused into carbon nuclei. Eventually that carbon will begin to fuse into neon, then the neon will fuse into magnesium, and so on up the periodic table until iron builds up in the core. Iron doesn't release energy when it fuses; it actually steals energy that is otherwise needed to support the colossal mass of the star around it. This theft rug-pulls the entire star, causing the core to collapse. Fantastic temperatures—upward of 100 billion degrees Celsius—will be achieved, triggering a detonation that destroys the star: a supernova.

The onslaught of energy is almost beyond comprehension. Within a few moments the nascent supernova will release as much energy as the sun will over its entire lifetime. The event will be so luminous that it could be detected halfway across the observable universe.

Astronomers still debate the effects this explosion will have on Earth—probably minimal ones, given its distance from us. But it's easy to think that for a star literally right next to it, the event would be catastrophic. Stars are sturdily constructed, however, and even being right next to a supernova doesn't guarantee destruction. It won't be fun, but it won't be a death sentence, either.

Interestingly, there has been very little research on the consequences inflicted on a wide-binary companion during such an event. What follows are the results from my personal back-of-the-envelope calculations—enough to give us an idea of what lies ahead for Betelgeuse B.

Right away we do know companion stars can survive: there is a class of objects called runaway stars, which are screaming through space at hundreds of kilometers per second, far faster than a typical star. There are a few ways these stars can get kicked up to these speeds, but one is to orbit a star that goes supernova. The sudden loss of mass means the strength of the primary star's gravity drops suddenly, causing the secondary companion to get flung out into space at dizzying speeds. We know these stars exist, and they were probably in tighter orbits around their supernova progenitor star than Betelgeuse B is. They survived the explosion, so that gives us confidence Betelgeuse B will make it out, too.

But it won't do so unscathed. The sudden influx of heat will be massive, though temporary and unlikely to have lingering effects. There will also be a lot of matter thrown out by the exploding star, but it will thin rapidly as the material expands, and the amount that physically impacts Betelgeuse B will be a tiny fraction compared with the star's own mass. Of course, that material will be moving at several thousand kilometers per second. So the physical blow will still be enormous—not enough to disrupt the star (again, stars are sturdy), but it will certainly cause tremendous heating and most likely strip some mass off the star. A stellar body slam will do that.

Amazingly, all this energy from the supernova is only a small fraction (about 1 percent) of the total that's emitted. Most is in the form of subatomic particles called neutrinos, which carry little energy individually but are generated in vast numbers when a star explodes. Neutrinos are notoriously unaffected by normal matter—most pass right through it. Still, so many are made in a stellar explosion (something like 10^{58}, a number so huge that it made the hair on the back of my neck stand up just to type it out) that, at the distance of Betelgeuse B from the now former Betelgeuse A, a human would get a near-fatal dose of radiation in just neutrinos (aside from the other extremely fatal effects). The amount of neutrino energy absorbed by the companion star, however, would

be tiny compared with the amount of energy the star emits on its own, so again, little effect is likely.

After the explosion, Betelgeuse A will probably leave behind a superdense neutron star with an incredibly strong magnetic field. This will certainly affect Betelgeuse B but how will depend on the exact circumstances of the explosion and the characteristics of the neutron star, which, at this time, are difficult, if not impossible, to predict.

So in the end, a star like Betelgeuse B will survive, even as close as it is to the hellish fury of a supernova, but the exact outcomes are still not clear.

Our uncertainty about this star and its potential fate are a reminder of just how much about this star system we don't know—despite its obvious relative proximity to us; even its distance isn't nailed down. One of the most attention-grabbing stars in the sky still keeps its secrets close—and that is one of the main reasons astronomers have so much fun studying it.

Phil Plait is a professional astronomer and science communicator in Virginia. His column for *Scientific American*, The Universe, covers all things space. He writes the *Bad Astronomy Newsletter*. Follow him online.

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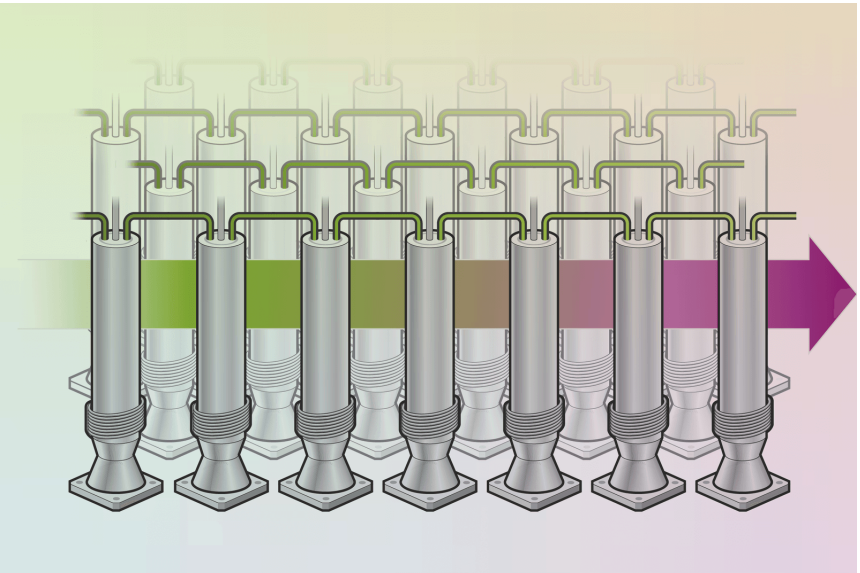
Graphic Science

- **A graphic explainer of uranium enrichment**
  Inside the process of isolating uranium-235

See how uranium enrichment really works

Inside the process of isolating uranium-235

By Clara Moskowitz & Nick Bockelman edited by Clara Moskowitz & Amanda Montañez



Nick Bockelman

Uranium enrichment is a contentious topic these days. This process of purifying uranium to increase the concentration of the isotope uranium-235 is important for nuclear power plants, but it is also crucial for building nuclear weapons.

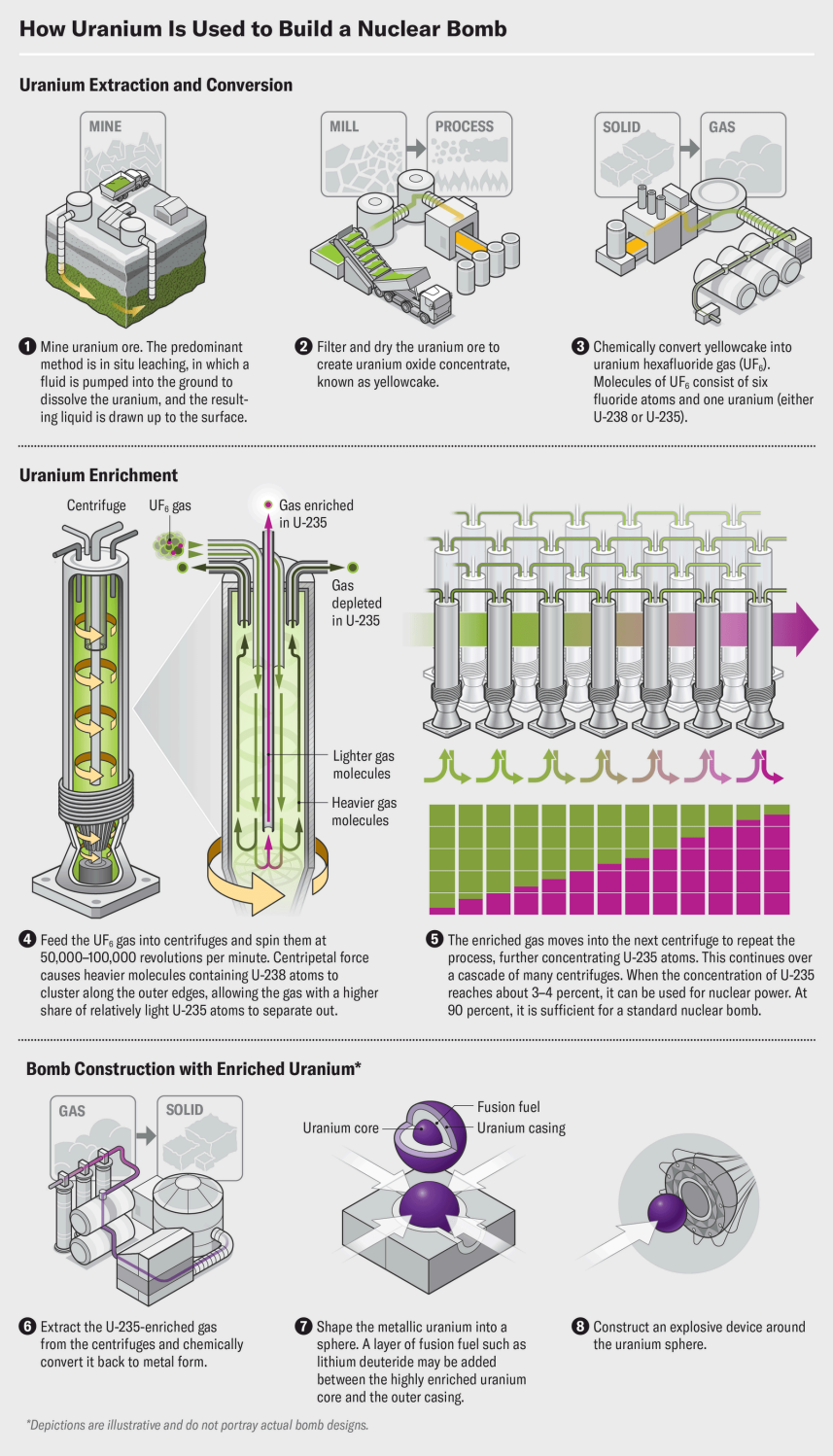
The ongoing war in Iran has centered on nuclear issues, including enrichment—Iranian officials have resisted any peace-deal terms that restrict the country's enrichment program. And a proposed agreement between the U.S. and Saudi Arabia would break with precedent by allowing enrichment in that country for the first time —worrying nuclear nonproliferation advocates.

Both nuclear power and nuclear bombs require a lighter, rarer kind of uranium, U-235, rather than the more common, heavier U-238. (The difference is in the number of neutrons per atom—the usual

variety has three more.) Both isotopes are found in nature, but the smaller one is less stable and more likely to break apart when hit by an errant neutron. This quality makes it “fissile,” meaning it can easily split and prompt a chain reaction: one uranium atom breaks apart, releasing energy and neutrons, which hit other atoms and break them apart, and so on.

Power plants moderate this reaction, so they require only 3 to 4 percent U-235. Weapons, on the other hand, need a much higher concentration—90 percent—to spark an uncontrolled reaction.

But how, exactly, does enrichment work, and why is it dangerous?



Nick Bockelman

Clara Moskowitz is chief of reporters at *Scientific American*, where she covers astronomy, space, physics and mathematics. She has been at *Scientific American* for more than a decade; previously she worked at Space.com. Moskowitz has reported live from rocket launches, space shuttle liftoffs and landings, suborbital spaceflight training, mountaintop observatories, and more. She has a bachelor's degree in astronomy and physics from Wesleyan University and a graduate degree in science communication from the University of California, Santa Cruz.

Nick Bockelman is an illustrator and graphic designer.

https://www.scientificamerican.com/article/a-graphic-explainer-of-uranium-enrichment

History

- **November 2026: Science history from 50, 100 and 150 years ago**
  In 1926 we wrote about how there was a good chance Mars had both plants and animals. And in 1876 we probed the duplicity of the artificial butter business

What was *SciAm* writing about 50, 100 and 150 years ago? Life on Mars and justice for crabs

In 1926 we wrote about how there was a good chance Mars had both plants and animals. And in 1876 we probed the duplicity of the artificial butter business

By Jeanna Bryner edited by Jeanna Bryner



1876, Family of Wood Bugs: “This illustration shows Pentatoma grisea [now called *Elasmucha grisea*], a type of wood bug. These insects can be seen on the boughs of trees, the young following their parent like chickens after a hen.”

Scientific American, Vol. 35, No. 19; November 4, 1876

1976

Urban Trees

“How can trees survive in an urban environment? The human artifacts we call cities first appeared less than 10,000 years ago. The ancestors of today’s trees are some 20,000 times older than the oldest cities. As trees have been exposed to more than 100 million

years of selective pressure to adapt to natural environments, it borders on the biologically miraculous that any tree can occupy an environmental niche as hostile as a city street.

“The fact that urban trees manage to survive is certainly not the result of rapid evolutionary adaptations. Neither the 200 years that have seen the rise of the industrial metropolis nor the 100 centuries of coexistence between trees and cities of any kind is a period long enough to have let selective pressures sort out the genes needed to give rise to a city-proof tree. It is man’s intervention, not nature’s, that has kept trees in the city.”

1926

Life on Mars Update

“Within the last few years new and powerful methods of observation have been applied to Mars. A series of impor-tant investigations speaks strongly in favor of the theory of Martian conditions developed by visual observers. The determinations of temperature make it practically certain that the polar caps are really composed of frozen water. The spectroscopic observations confirm this, and the presence of oxygen makes it appear more likely than not that vegetation exists on the planet.

“If vegetable life exists on the planet, as seems fairly probable, animal life may exist. The latter could not do so otherwise. But we appear to be far from having any unquestioned evidence that it does.”

As of the summer of 2026, there is no evidence of vegetation or animal life on Mars.

Classifying the Arthropod

“It would be an aid to nature study if the term ‘arthropod’ were as commonly used, as familiarly accepted and as well understood as the words ‘insect’ or ‘spider,’ ‘reptile’ or ‘bird.’ The arthropod group includes animals that are without a backbone or any bones at all, strictly speaking, their bodies being encased in a ringed, exterior tegument and having legs and, sometimes, wings. In this great division is included the crabs, lobsters and shrimps as one class, the spiders, mites and scorpions as another, the millepedes and centipedes as the least numerous group and the insects as the greatest.”

1876

Artificial Butter

“For some time, there has been a prevalent impression that if the manufacture of artificial butter has not died out, at least no product of this description is now industrially made which has any standing in the market. The facts, however, are altogether against any such conclusion, for quite recently no less than 50 artificial butter factories were counted in this city, and large quantities of artificial butter are sold to the customer as genuine butter.

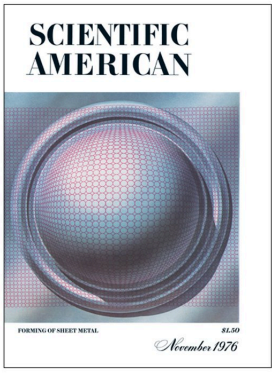
“There is a duplicity in this business which is reprehensible. But if people cannot distinguish the artificial from the natural product and if the former is more healthful, it would be difficult to prove any damages save to the moral sense to all of the community.”

Twinkling Stars

“The scintillation of stars, and its close connection with changes of weather, has much interested many scientists, and recently spectroscopic researchers have found a connection between scintillation and various meteorological elements. Their chief results are as follows: The intensity of scintillation increases invariably with the occurrence or approach of rainy weather and

with the increase of tension of vapor in the air on one side, as well as the increase of pressure and decrease of temperature on the other.

“Thus, a continued investigation of scintillation would be of great service, not only for the prevision of weather but also for the general study of meteorology.”







Jeanna Bryner is editor in chief of *Scientific American*. Previously she was editor in chief of Live Science and, prior to that, an editor at Scholastic's *Science World* magazine. Bryner has an English degree from Salisbury University, a master’s degree in biogeochemistry and environmental sciences from the University of Maryland and a graduate science journalism degree from New York University. She has worked as a biologist in Florida, where she monitored wetlands and did field surveys for endangered species, including the gorgeous Florida Scrub Jay. She also received an ocean sciences journalism fellowship from the Woods Hole Oceanographic Institution. She is a firm believer that science is for everyone and that just about everything can be viewed through the lens of science.

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https://www.scientificamerican.com/article/november-2026-science-history-from-50-100-and-150-years-ago

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